FASTLEARN EXAMINER

8 – 9

TOPICAL REVISION

INTEGRATED SCIENCE

NEW SYLLABUS PAST EXAM QUESTIONS

2011 – 2016

Mwiya Namakando

Cover design by **Nelson Nkhoma**

©2017 FASTLEARN PUBLISHERS

TABLE OF CONTENTS 2011 - 2016

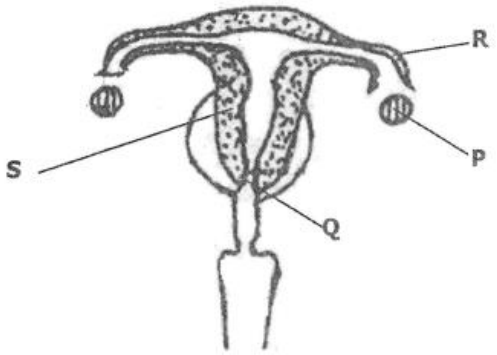
		Questions	Answers
1	**HUMAN REPRODUCTIVE SYSTEM AND PUBERTY**		
	1.1 – PAPER 1	4	95
	1.2 – PAPER 2	6	95
2	**NUTRITION**		
	1.1 – PAPER 1	8	96
	1.2 – PAPER 2	9	96
3	**WATER, AIR AND LAND POLLUTION**		
	1.1 – PAPER 1	10	96
	1.2 – PAPER 2	10	97
4	**PLANT & ANIMAL CELLS**		
	1.1 – PAPER 1	11	97
	1.2 – PAPER 2	13	97
5	**PLANT GROWTH AND NUTRIENTS**		
	1.1 – PAPER 1	14	98
6	**COMPOSITION OF MATTER**		
	1.1 – PAPER 1	14	98
7	**PHYSICAL CHANGE OF STATE**		
	1.1 – PAPER 1	15	98
	1.2 – PAPER 2	16	98
8	**MIXTURES**		
	1.1 – PAPER 1	18	99
	1.2 – PAPER 2	20	99
9	**MASS, WEIGHT, VOLUME & DENSITY**		
	1.1 – PAPER 1	23	100
	1.2 – PAPER 2	25	101
10	**HEAT TRANSFER & EXPANSION OF SUBSTANCES**		
	1.1 – PAPER 1	27	102
	1.2 – PAPER 2	30	102
11	**REFLECTION AND REFRACTION OF LIGHT**		
	1.1 – PAPER 1	33	103
	1.2 – PAPER 2	35	104
12	**COMPOSITION OF AIR**		
	1.1 – PAPER 1	36	104
	1.2 – PAPER 2	38	105
13	**CIRCULATORY SYSTEM**		
	1.1 – PAPER 1	40	106
	1.2 – PAPER 2	41	106
14	**RESPIRATORY SYSTEM**		
	1.1 – PAPER 1	42	106
	1.2 – PAPER 2	44	107
15	**SEXUALLY TRANSMITTED INFECTIONS (STIS)**		
	1.1 – PAPER 1	45	107

		Questions	Answers
16	**CYCLES IN THE BIOSPHERE**		
	1.1 – PAPER 1	45	107
	1.2 – PAPER 2.....	46	108
17	**WATER MANAGEMENT**		
	1.1 – PAPER 1	47	108
	1.2 – PAPER 2.....	48	108
18	**CONSERVATION OF ANIMALS AND PLANTS**		
	1.1 – PAPER 1	48	108
	1.2 – PAPER 2	50	109
19	**PHOTOSYNTHESIS**		
	1.1 – PAPER 1	51	109
	1.2 – PAPER 2	55	109
20	**TRANSPIRATION**		
	1.1 – PAPER 1	58	110
	1.2 – PAPER 2.....	59	111
21	**CHEMICAL REACTION**		
	1.1 – PAPER 1	61	111
	1.2 – PAPER 2	64	112
22	**LIGHT AND ITS NATURE**		
	1.1 – PAPER 1	68	113
	1.2 – PAPER 2.....	72	116
23	**ELECTRIC CURRENT AND VOLTAGE IN CIRCUIT**		
	1.1 – PAPER 1	73	116
	1.2 – PAPER 2.....	80	117
24	**PRESSURE**		
	1.1 – PAPER 1	83	121
	1.2 – PAPER 2.....	83	121
25	**ENERGY AND ITS CONSERVATION**		
	1.1 – PAPER 1	84	122
	1.2 – PAPER 2.....	86	122
26	**COMMUNICATION**		
	1.1 – PAPER 1	87	123
	1.2 – PAPER 2	90	123
27	**PRACTICAL EXAMS (PAPER 3)**		
	1.3 – PAPER 3	92	N/A

GRADE 8 TOPICS

1.1 HUMAN REPRODUCTIVE SYSTEM AND PUBERTY – PAPER 1

1 Below is a diagram of the female reproductive system.



Which letter identifies the organ called cervix? [2016.P1.Q1]

A P B Q C R D S

2 Why is it important for one to observe personal hygiene of their reproductive organs? [2016.P1.Q2]

A Allow the organs to grow gradually. B Avoid the organs getting infections.

C Prevent sexually transmitted diseases. D Help prevent HIV transmission.

3 Below is a list of activities that take place in the process of fertilisation and embryo development over a period of time.

1. fusion of sperm and ovum 2. sexual intercourse

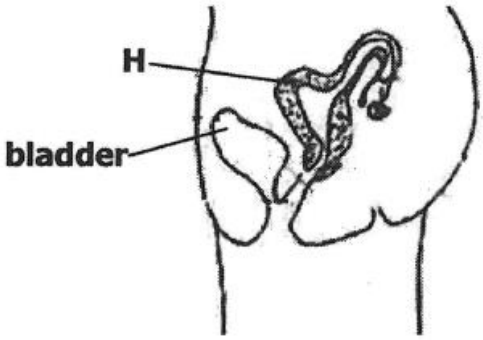
3. formation of zygote 4. attachment of zygote to uterus

Which of the following shows the correct order of how these activities happen? [2016.P1.Q3]

A 2 -> 1 -> 3 -> 4 B 1 -> 3 -> 4 -> 2

C 2 -> 1 -> 4 -> 3 D 1 -> 4 -> 2 -> 3

4 The diagram below shows the side view of the female reproductive organs.



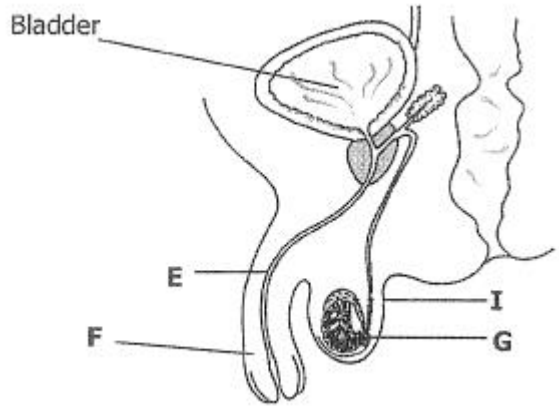
Identify structure H and its corresponding function. [2016.P1.Q4]

	Structure H	**Function**
A	Placenta	- Supplying food and oxygen to foetus
B	Oviduct	- Passage of the ovum from ovary
C	Uterus	- Organ where embryo develops
D	Ovary	- To produce female sex cells

5 Which of the following is a natural sign of puberty that occurs only in boys? [2015.P1.Q1]

A Growth of pubic hair B Deepening of the voice
C Menstruation begins D Pimples appear on theface

6 The diagram below shows the side view of the male reproductive organs.



What is the function of structure I? [2015.P1.Q2]

A Producing sperms B Producing ova
C Protecting the testes D Storing male sex cells

7 A pregnant woman is 120 days pregnant and would like to find the approximate number of remaining days in an average gestation period. Calculate the number of remaining days for her before she delivers. Which of the following shows the correct answer? [2015.P1.Q3]

A 40 days B 160 days C 280 days D 320 days

8 A breast feeding mother produces milk for her baby. This is called ..... [2013.P1.Q18]

A lactating. B weaning. C dieting. D teething.

9 The following are female reproductive organs.

1. Uterus 2. Oviduct 3. Vagina 4. Cervix

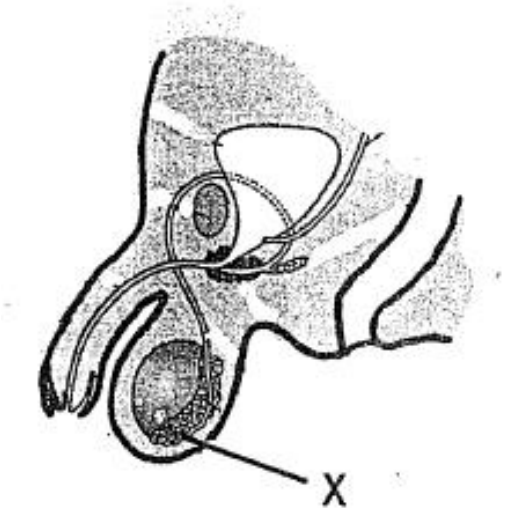
Using the numbers, which of the following best shows the correct order of the movements of the sperm before it finally fertilizes the ovum. [2013.P1.Q19]

A 4 --> 2 --> 3 --> 1 B 3 --> 4 --> 1 --> 2
C 3 --> 1 --> 4 --> 2 D 2 --> 4 --> 1 --> 3

10 A woman gave birth on 28 October 2011. If this was a normal full term delivery, when was she likely to have fallen pregnant? [2012.P1.Q18]

A January, 2011 B February, 2011
C May, 2011 D June, 2011

11 The diagram below shows the male reproductive organs.



What is the name and function of structure X? [2011.P1.Q18]

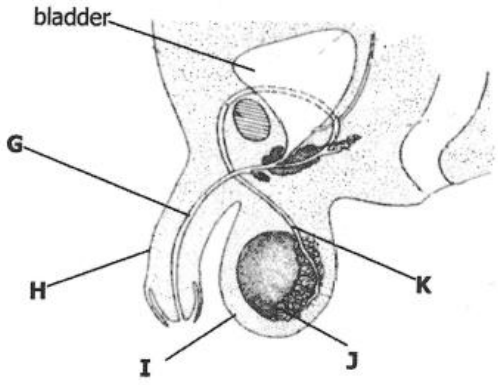
	NAME	FUNCTION
A	Sperm duct	making of sperms
B	Epididymis	storage of sperms
C	Sperm duct	storage of sperms
D	Epididymis	making of sperms

12 In which of the following organs does implantation normally take place? [2011.P1.Q19]

A Ovary B Oviduct C Uterus D Vagina

1.2 HUMAN REPRODUCTIVE SYSTEM AND PUBERTY – PAPER 2

1 THE HUMAN REPRODUCTIVE SYSTEM AND PUBERTY [2016.P2.Q1]



The diagram below shows the side view of the male reproductive organs.

(a) Identify the parts marked G and I. [2016.P2.Q1a]
(b) State the function of part 3. [2016.P2.Q1b]
(c) Explain what would happen if part K was to be tied. [2016.P2.Q1c]
(d) State two signs of puberty in boys. [2016.P2.Q1d]

2 MAN [2013.P2.Q6]

Study the diagrams below showing the female and male reproductive organs.

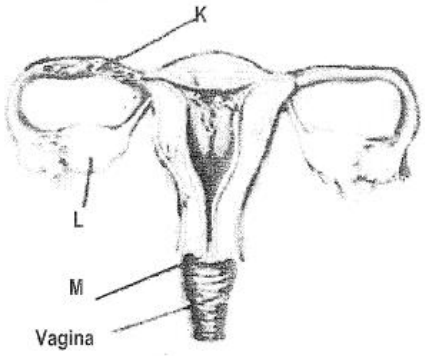


DIAGRAM A

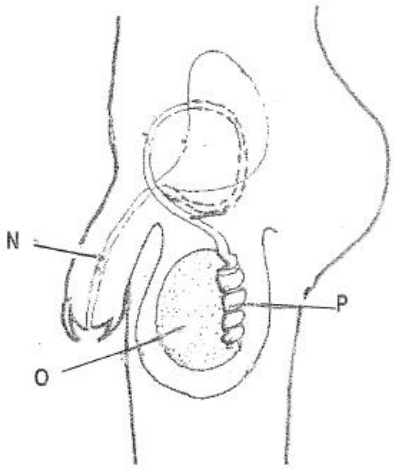


DIAGRAM B

(a) Identify the organs labelled M and P. [2013.P2.Q6a]

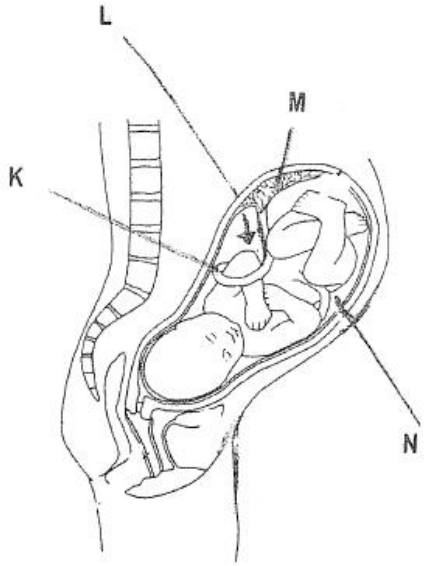
(b) What are the reproductive functions of parts marked L and N? [2013.P2.Q6b]

(c) On the diagram, which letter shows where fertilisation takes place? [2013.P2.Q6c]

(d) How long is the gestation period in human beings? [2013.P2.Q6d]

3 HUMAN REPRODUCTION [2010.P2.Q5]

Study the diagram showing a stage in the foetus development.



(a) Identify the parts labelled K and L. [2010.P2.Q5a]

(b) Describe the functions of liquid marked N. [2010.P2.Q5b]

(c) State one substance other than gas which moves in the direction of the arrow. [2010.P2.Q5c]

(d) In which labelled part does gaseous exchange occur between mother and foetus? [2010.P2.Q5d]

(e) Explain why the mother at this stage requires a special diet which includes proteins and carbohydrates. [2010.P2.Q5e]

2.1 NUTRITION – PAPER 1

1 State the food type that you would prescribe for a manual worker. Food rich in ... [2016.P1.Q5]

A fats. B proteins. C minerals. D carbohydrates.

2 Which of the following information may not be obtained from the children' clinic card? [2016.P1.Q6]

A Number of siblings. B Number of visits to the clinic.
C Age of the child's mother. D Age of the child.

3 The table below shows symptoms and nutritional deficiency diseases.

	Symptom	Deficiency disease
A	Poor vision in light	Kwashiokor
B	Swollen stomachs and thin limbs	Marasmus
C	Poor healing of wounds	Rickets
D	Unhealthy bones and teeth	Scurvy

Which one correctly shows the symptom and corresponding deficiency disease? [2015.P1.Q4]

4 Study the food nutritive substances below.

1. Bread, rice, potatoes, mealie meal
2. Butter, cheese, port, doughnuts
3. Beef, chicken, beans, liver
4. Calcium, magnesium, potassium, sodium

Which of the following best describes carbohydrates and proteins? [2015.P1.Q5]

	Carbohydrates	**Proteins**
A	1	3
B	2	3
C	1	2
D	3	4

5 A breast feeding mother produces milk for her baby. This is called ..... [2013.P1.Q18]

A lactating. B weaning. C dieting. D teething.

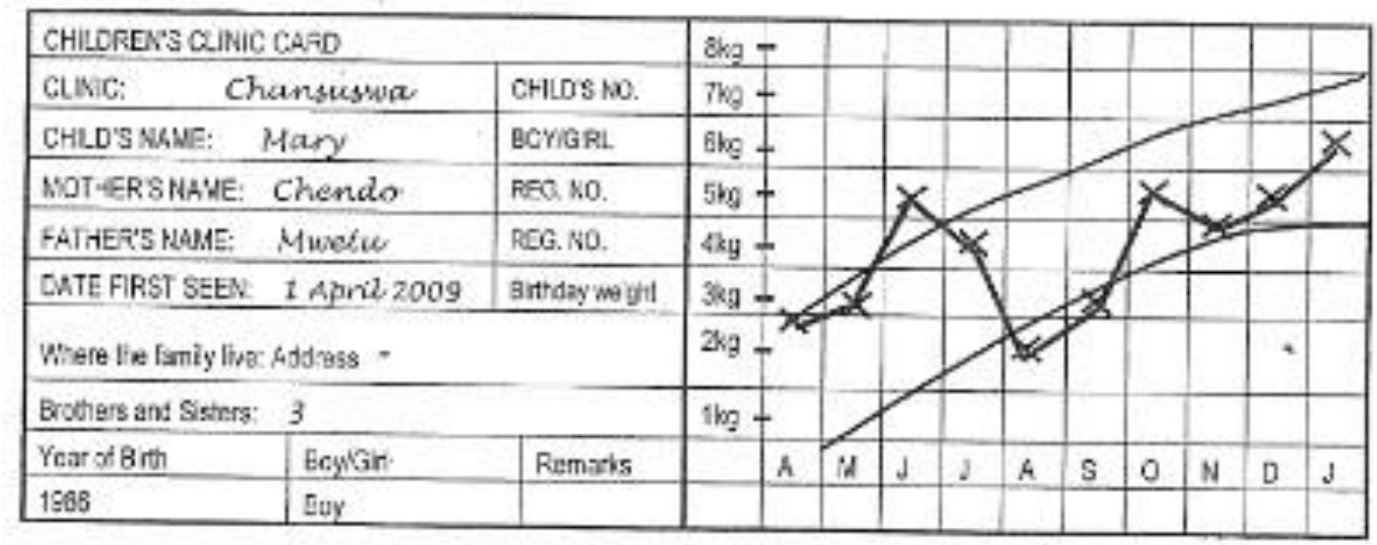
6 A human being at complete rest uses only 4.0 kilojoules of energy per minute. A gram of glucose has an energy value of 16.0 kilojoules of energy. How long will 4.0g of glucose last as a source of energy during complete rest? [2012.P1.Q4]

A 4 minutes B 12 minutes C 16 minutes D 64 minutes

7 Which of the following diseases can be prevented or controlled by giving a child a balanced diet? [2012.P1.Q19]

A Kwashiorkor, marasmus, anaemia. B Kwashiokor, poliomyletes, gum bleeding
C Rickets, poliomyletes, anaemia. D Rickets, marasmus, gum bleeding.

8 The diagram below shows part of an under five clinic card. [2011.P1.Q20]



CHILDREN'S CLINIC CARD		
CLINIC: Chansuswa	CHILD'S NO.	
CHILD'S NAME: Mary	BOY/GIRL	
MOTHER'S NAME: Chendo	REG. NO.	
FATHER'S NAME: Mwelu	REG. NO.	
DATE FIRST SEEN: 1 April 2009	Birthday weight	
Where the family live: Address -		
Brothers and Sisters: 3		
Year of Birth	Boy/Girl	Remarks
1998	Boy	

In which month was the child over-weight?

A April B June C August D October

2.2 NUTRITION – PAPER 2

1 HEALTH

Dietary needs vary from person to person. [2015.P2.Q2]

(a) Name one class of nutrients in a balanced diet necessary for;

(i) Proper skin formation, [2015.P2.Q2a(i)]

(ii) Providing energy to do work. [2015.P2.Q2a(ii)]

(b) Suggest a reason why a pregnant woman needs more of the following in her diet;

(i) Proteins [2015.P2.Q2b(i)]

(ii) Iron [2015.P2.Q2b(ii)]

(c) Explain why mothers are advised to take their children monthly to the under five clinics. [2015.P2.Q2c]

3.1 WATER, AIR AND LAND POLLUTION – PAPER 1

1 Which pair shows correctly causes of pollution to the environment? [2016.P1.Q7]

	Caused by human	**Caused by nature**
A	burning fossil fuels	volcanic eruptions
B	forest fires	industrial emission
C	wind erosion	incinerators
D	pollen dispersal	waste from mining plants

2 Which of the following statements best describes pollution? Pollution is the addition of..... [2015.P1.Q6]

A harmful gases in air. B contaminating substances in water.
C inorganic fertilizers in soil. D harmful substances to the environment.

3 Which of the following is not a common type of pollution in rural areas of Zambia? [2015.P1.Q7]

A Water pollution. B Soil pollution.
C Noise pollution. D Land pollution.

4 Which of the following is not a way of reducing negative effects of inorganic fertilizers in a maize field? [2015.P1.Q10]

A Using composite manure B Keeping the soil aerated
C Adding agricultural line to the soil D Rotating the crop with rice crop

5 Which of the following is the main pollutant that causes acid rain? [2012.P1.Q37]

A Dust particles B Carbon monoxide
C Sulphur dioxide D Smoke particles

6 Which of the following is an effect of pollution by carbon dioxide on the environment? [2011.P1.Q39]

A Increased rainfall B Increased air temperature
C Prolonged night time D Prolonged illnesses

3.2 WATER, AIR AND LAND POLLUTION – PAPER 2

1 MAN'S INTERACTION WITH THE ENVIRONMENT

The list below shows some natural resources used by human beings in their day to day life.

TREES	COPPER	WATER	COAL	IRON

(a) From the list above, choose ;

(i) one diminishing resource which is renewable. [2011.P2.Q10a(i)]

(ii) one diminishing resource which is not renewable. [2011.P2.Q10a(ii)]

(b) Which one of the resources causes air pollution when it is being used? [2011.P2.Q10b]

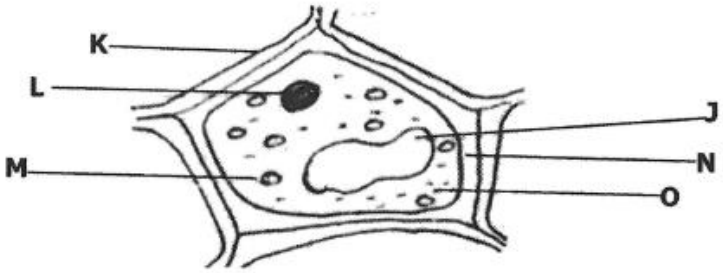
(c) Which one of the resources reduces air pollution caused by the use of the resource named in (b) above? [2011.P2.Q10c]

(d) Which one of the resources causes land pollution when extracting it? [2011.P2.Q10d]

(e) Suggest one way of reducing the diminishing resources. [2011.P2.Q10e]

4.1 PLANT & ANIMAL CELLS – PAPER 1

1 Below is a diagram of a plant cell as seen under a microscope.



Which labelled structure represents the protoplasm? [2016.P1.Q8]

A L and 0 B L and J C N and M D K and M

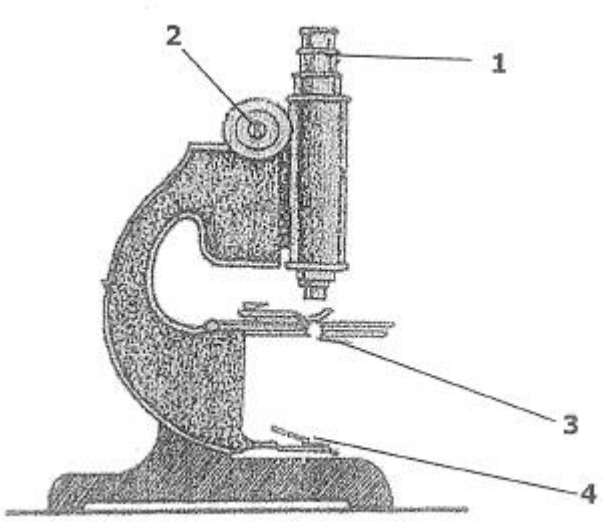
2 The list below shows functions of parts of cells.

1. Protection of cell
2. Selectively permeable
3. Controls all cell activities
4. Contains chlorophyll

Which functions are only of parts found in a plant cell? [2016.P1.Q10]

A 1 and 2 B 1 and 4 C 3 and 2 D 3 and 4

3 The diagram below shows a microscope.



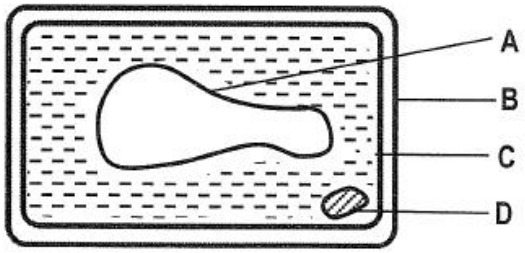
Which part is adjusted to obtain a clear image? [2015.P1.Q8]

A 1 B 2 C 3 D 4

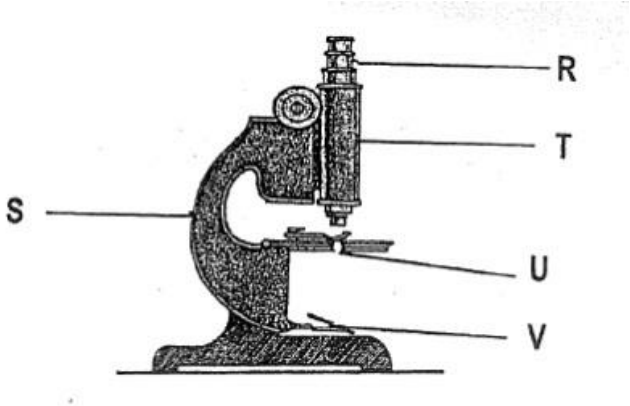
4 Which of the following shows the correct structure and comparison between plant and animal cells? [2015.P1.Q11]

	Structure	Plant Cell	Animal cell
A	Cytoplasm	Present	Absent
B	Cell membrane	Absent	Present
C	Cell wall	Present	Absent
D	Chloroplasts	Absent	Present

5 Which labelled structure in the diagram of a cell is not selectively permeable? [2013.P1.Q8]



6 The diagram below shows a simple microscope.

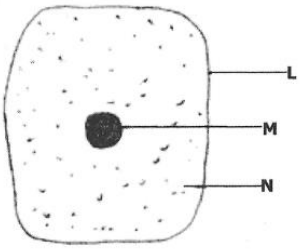


Which letters can be used to trace the path taken by light when a specimen is being viewed through the microscope? [2011.P1.Q8]

A R → S → U → V
B R → T → U → V
C V → U → T → R
D V → S → T → R

4.2 PLANT & ANIMAL CELLS – PAPER 2

1 ANIMAL CELL. The diagram below shows an animal cell. [2016.P2.Q2]

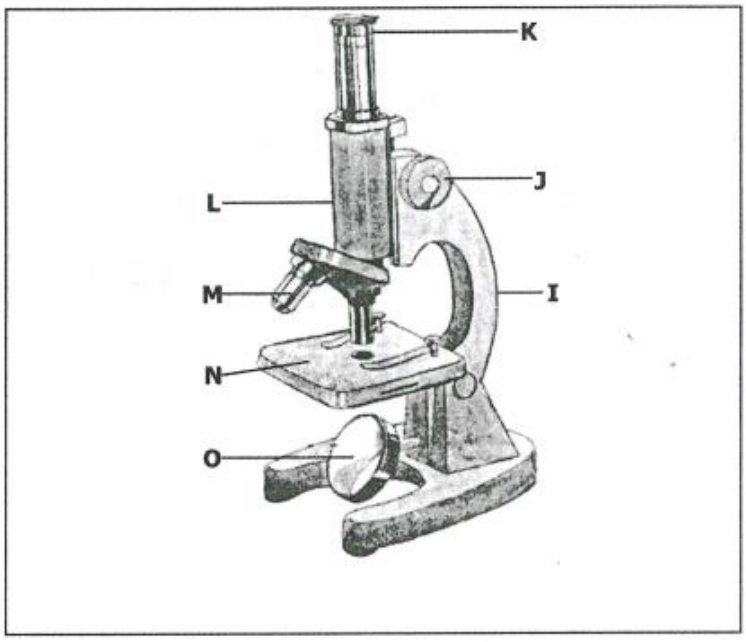


Animal Cell

(a) Which letters on the diagram represents the nucleus and cytoplasm? [2016.P2.Q2a]
(b) State the name of the part of an animal cell which selects what goes in and out of the cell. [2016.P2.Q2b]
(c) What two parts of a plant cell are not found in animal cells? [2016.P2.Q2c]
(d) What instrument is used to observe the structure of a cell? [2016.P2.Q2d]

2 PLANTS AND ANIMALS

The diagram below shows parts of a microscope. [2015.P2.Q1]

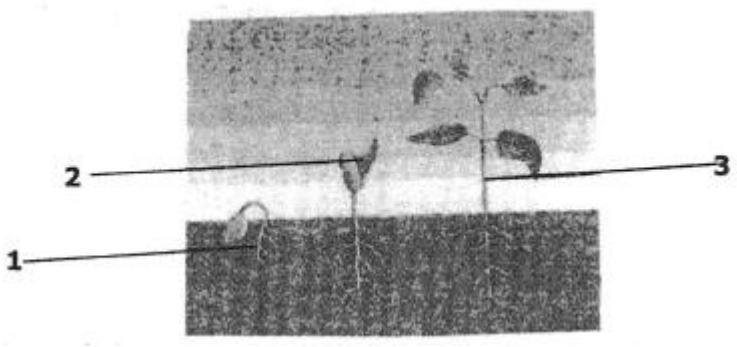


(a) Which two labelled parts contain lenses which help to show the image of the specimen? [2015.P2.Q1a]
(b) What name is given to the lenses in the parts identified in question (a) above? [2015.P2.Q1b]
(c) Using the letters on diagram, identify the labelled part on which a person has to put his eye in order to see the image of the specimen. [2015.P2.Q1c]
(d) Describe the function of the labelled parts;
(i) J [2015.P2.Q1d(i)]
(ii) O [2015.P2.Q1d(ii)]

5.1 PLANT GROWTH AND NUTRIENTS – PAPER 1

1 The diagram below shows stages of plant growth.

Which regions of plant growth are shown by the numbered parts in the diagram? [2016.P1.Q9]



	Shoot	Root	Stem
A	1	2	3
B	3	1	2
C	2	1	3
D	2	3	1

2 A pupil noticed that the plants in their garden had a yellow colour on the edges of leaves. The element that could be lacking in the plant is... [2013.P1.Q7]

A potassium. B phosphorous. C nitrogen. D magnesium.

3 Apart from nitrogen, which other element, when lacking in the soil, will result in poor growth of a plant? [2011.P1.Q9]

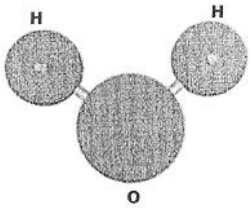
A Calcium B Chlorine C Phosphorus D Sulphur

6.1 COMPOSITION OF MATTER – PAPER 1

1 Which list correctly identifies the symbols of common atoms? [2016.P1.Q11]

	Iron	Copper	Hydrogen	Carbon
A	Fe	C	H	Cu
B	Cu	Fe	H_2O	CO_2
C	Fe	Cu	H	C
D	Cu	C	H_2	Cu

2 The diagram below shows a model of a molecule.



The molecule is of... [2015.P1.Q12]

A water. B hydrogen. C copper. D nitrogen.

7.1 PHYSICAL CHANGE OF STATE – PAPER 1

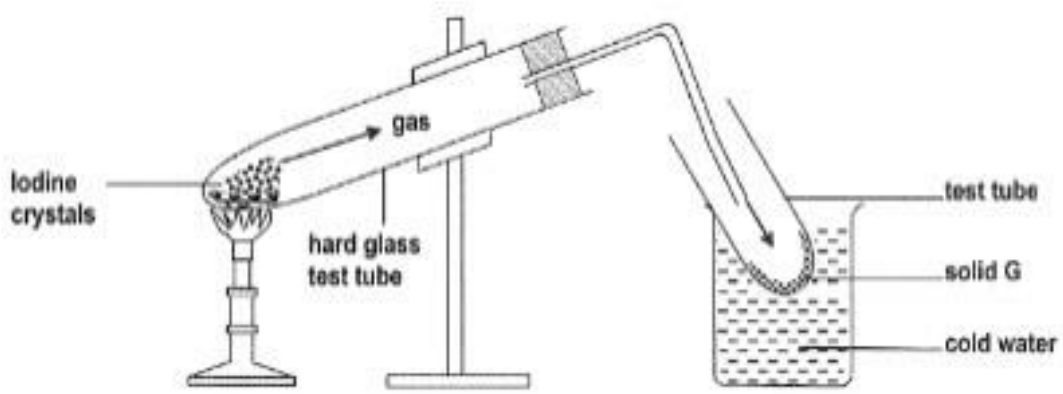
1 If the substance A boils at 98°C and melts at -100°C, determine its state when the temperature is -130°C. [2016.P1.Q12]

A Solid B Liquid C Gas and Solid D Liquid and Gas

2 Which of the following is not true about the freezing point of water? The freezing point of water is ... [2015.P1.Q13]

A also its point of melting.
B when water loses heat energy and changes into solid.
C also its boiling point.
D 0°C under normal conditions at sea level.

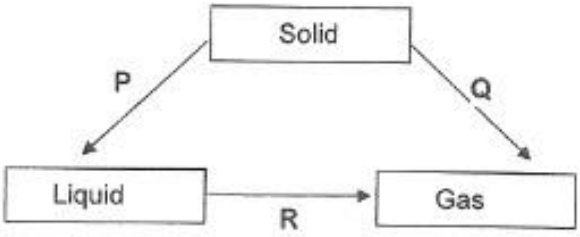
3 Study the diagram below showing an experiment on change of state.



What is the process by which solid G is formed in the test tube? [2013.P1.Q2]

A Condensation B Freezing C Sublimation D Vaporisation

4 The diagram below represents the physical changes between the three states of matter. [2012.P1.Q1]



Which of the following shows the correct names of the processes labelled P, Q and R?

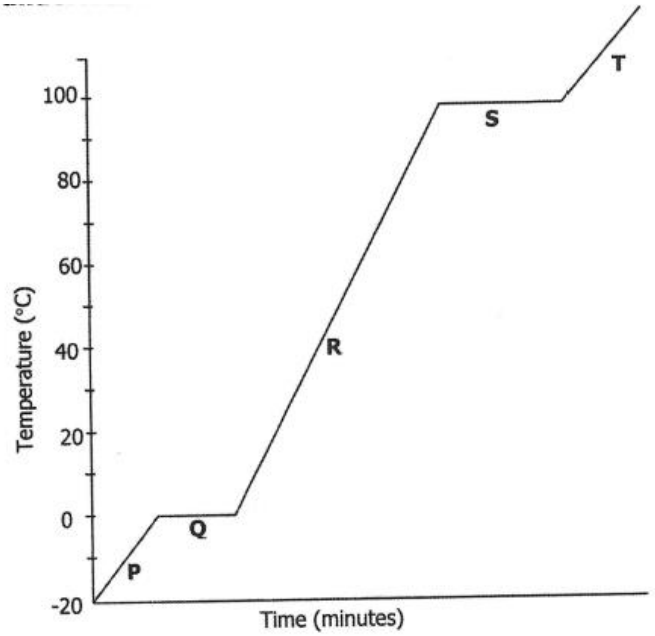
	P	Q	R
A	Sublimation	Melting	Freezing
B	Freezing	Condensation	Evaporation
C	Melting	Condensation	Evaporation
D	Melting	Sublimation	Evaporation

5 State the substances that can only exist in two states of matter. [2011.P1.Q1]

A Water and carbon dioxide. B Carbon dioxide and iodine.
C Iodine and water. D Water, carbon dioxide and iodine.

7.2 PHYSICAL CHANGE OF STATE – PAPER 2

1 CHANGES OF STATE [2016.P2.Q5]



(a) Which letters on the graph represents; the

(i) melting point [2016.P2.Q5a(i)]

(ii) boiling point [2016.P2.Q5a(ii)]

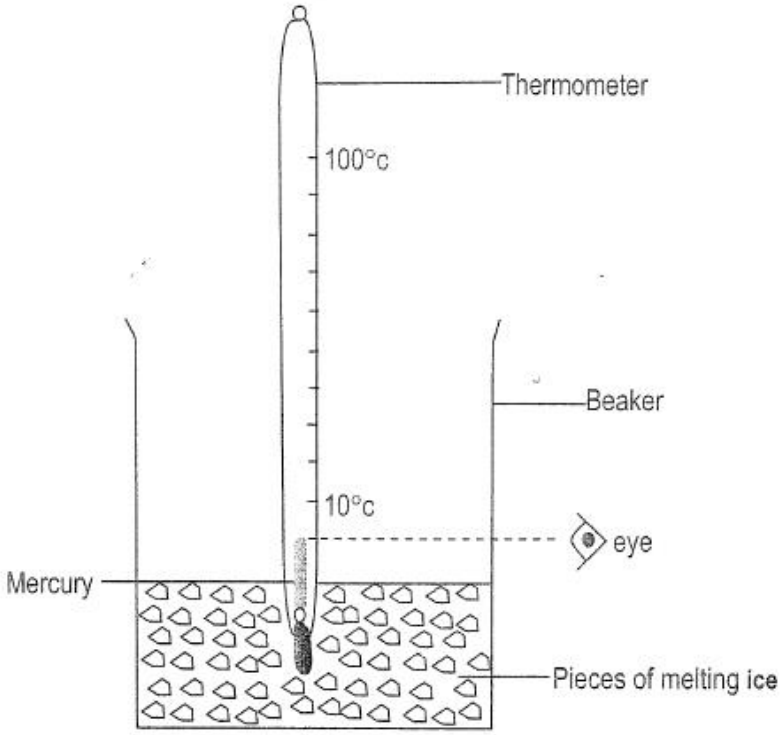
(b) Give a reason for your answer in (a) above. [2016.P2.Q5b]

(c) Identify the stage in which the substance X would be in liquid state only. [2016.P2.Q5c]

(d) At which stage would substance X exist in two different states of solid and liquid at the same time? [2016.P2.Q5d]

2 **WATER** [2013.P2.Q1]

The diagram below shows an experiment on changes of state.



(a) What is the reading on the thermometer? [2013.P2.Q1a]

(b) What must be done to the melting ice for it to completely change to the next state of matter? [2013.P2.Q1b]

(c) Describe the change of state that the ice will undergo in question (b) above? [2013.P2.Q1c]

(d) Give one reason why liquid mercury is used in thermometers. [2013.P2.Q1d]

(e) The temperature at which ice changes its state is called . . . [2013.P2.Q1e]

(f) Some substances can change from solid state into gaseous state without becoming a liquid. Such type of change is called ... [2013.P2.Q1f]

3 **WATER** [2010.P2.Q1]

The table below is a summary of the properties of matter.

State of Matter	**Volume**	**Shape and Size**
Solid	Fixed	Fixed
Liquid	Fixed	(i)
Gas	(ii)	Takes shape of container

(a) Complete the table by filling in the blank spaces (i) and (ii). [2010.P2.Q1a]

(b) Using a named liquid, give one example of a physical change. [2010.P2.Q1b]

8.1 MIXTURES – PAPER 1

1 Which of the following methods of separating mixtures show a correct sequence that can be used to obtain pure salt from sea-water? [2016.P1.Q13]

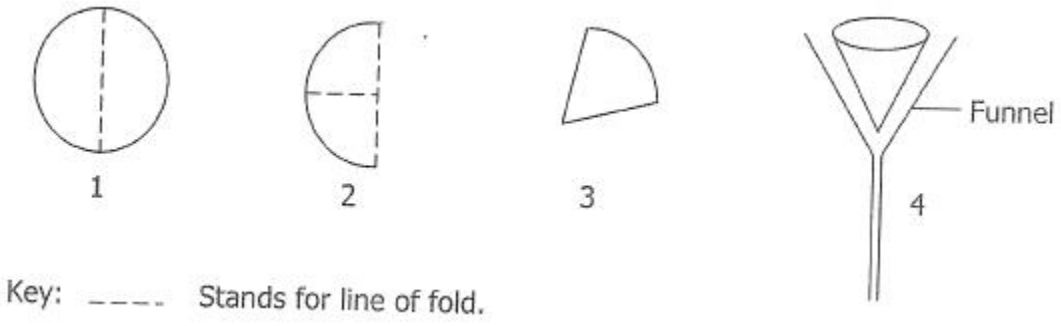
A Evaporation -> Sublimation -> Distillation

B Filtration -> Evaporation -> Crystallisation

C Evaporation -> Chromatography-> Crystallisation

D Filtration -> Chromatography -> Sublimation

2 The following are ways in which a filter paper is folded before placing it on a funnel during filtration method as shown in diagram 4.

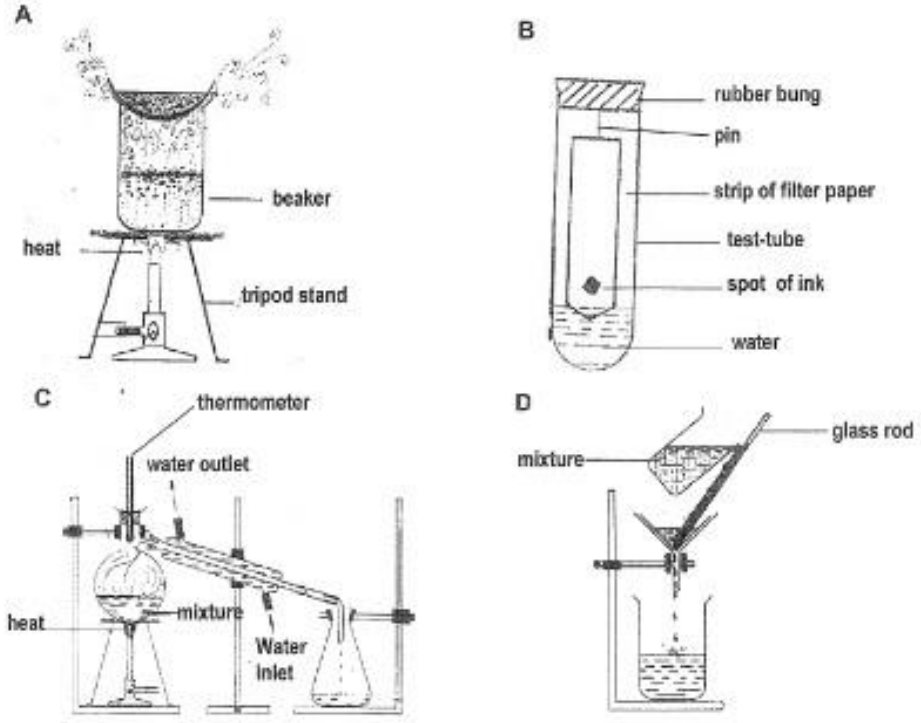


Which of the following correctly shows the order of folding of the filter paper? [2015.P1.Q14]

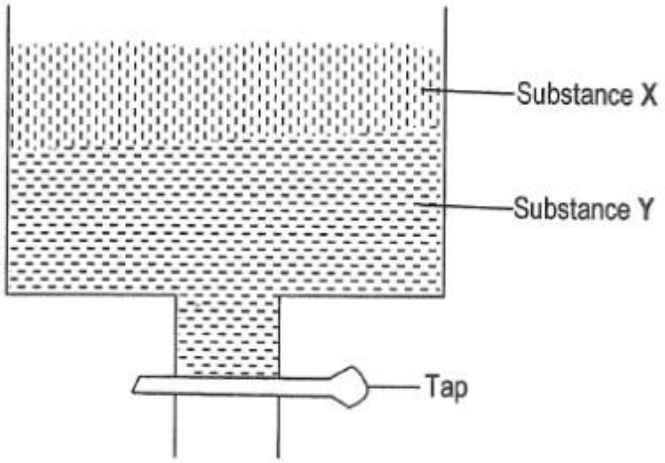
A 1 2 3 B 2 3 1

C 3 2 1 D 2 1 3

3 Which separation technique results in the formation of a distillate? [2013.P1.Q36]



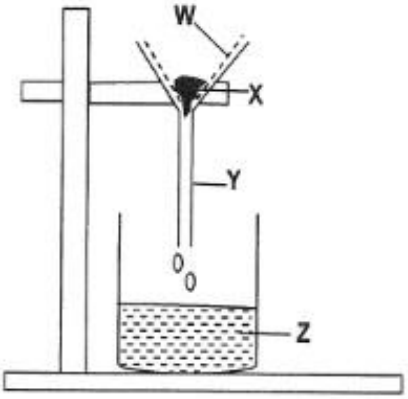
4 The diagram below shows a separation technique used to separate two substances.



The separation of the two substances is made possible because ..... [2012.P1.Q33]

A Substance X is immiscible with Substance Y.
B Substance X is miscible with Substance Y.
C Substance Y is able to flow while Substance X is not.
D Substance X is able to flow while Substance Y is not.

5 Below is the apparatus for filtration.



Which of the following order correctly labels **W, X, Y** and **Z**?

	W	X	Y	Z
A	residue	filtrate	funnel	filter
B	funnel	residue	filter	filtrate
C	filter	residue	funnel	filtrate
D	filter	filtrate	residue	funnel

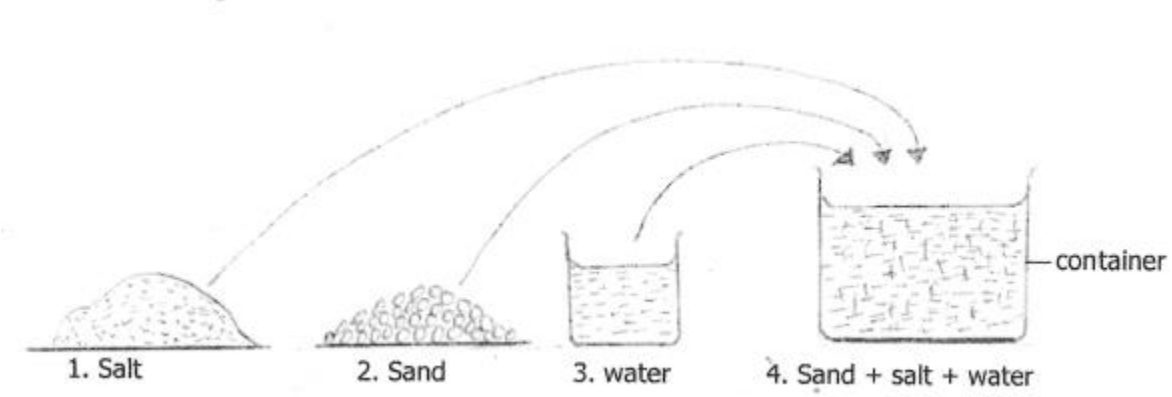
6 A separation technique which involves dissolved substances separated by using their different rates of movement on a medium is known as ..... [2011.P1.Q34]

A fractional distillation. B simple distillation.
C filtration. D chromatography.

8.2 MIXTURES – PAPER 2

1 SEPARATION TECHNIQUES

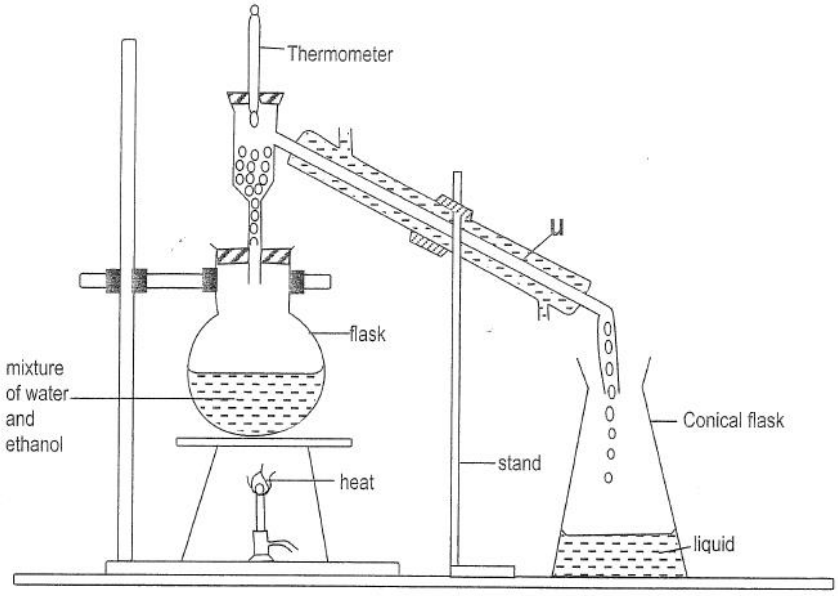
The diagram below shows three different substances salt, sand and water put together in one container. [2015.P2.Q4]



(a) What was formed when the salt, sand and water were put together into one container? [2015.P2.Q4a]
(b) What happened to the following when they were put with the water in the container?
  (i) salt [2015.P2.Q4b(i)]
  (ii) sand [2015.P2.Q4b(ii)]
(c) Explain the method that can be used to separate each of the following from the water in the container.
  (i) sand, [2015.P2.Q4c(i)]
  (ii) salt. [2015.P2.Q4c(ii)]
(d) What method could be used to separate two miscible liquids? [2015.P2.Q4d]

2 QUESTION 9 – SEPARATION TECHNIQUE [2013.P2.Q9]

The diagram below shows the experiment on separation of a mixture.

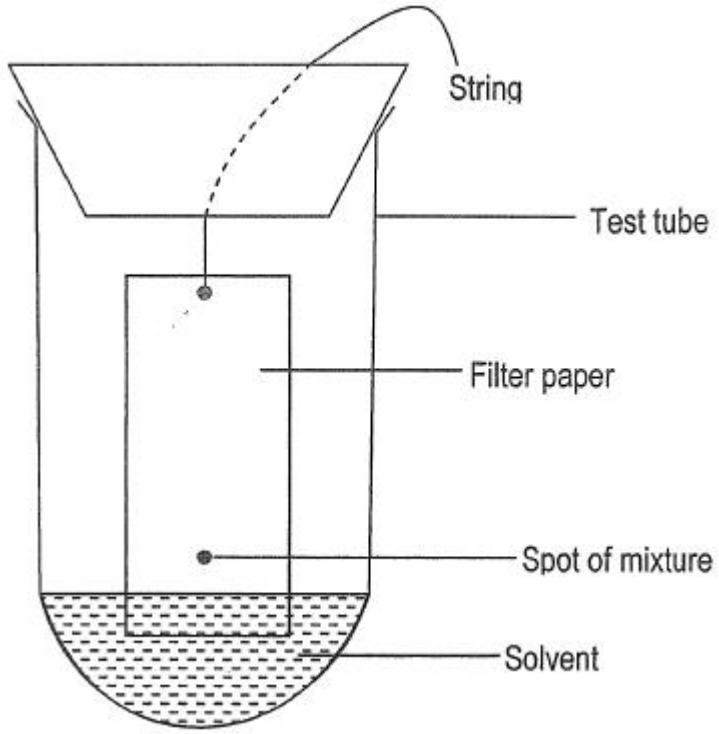


(a) What is the name of the separation process shown in the diagram? [2013.P2.Q9a]
(b) State two properties of the mixture that allows the separation to take place. [2013.P2.Q9b]

(c) What is the function of the apparatus labelled U? [2013.P2.Q9c]
(d) Name the first part of the mixture that will be collected in the conical flask. [2013.P2.Q9d]
(e) Suggest one industrial use of the separation technique shown in the diagram. [2013.P2.Q9e]

3 QUESTION 7 – MATTER [2012.P2.Q7]

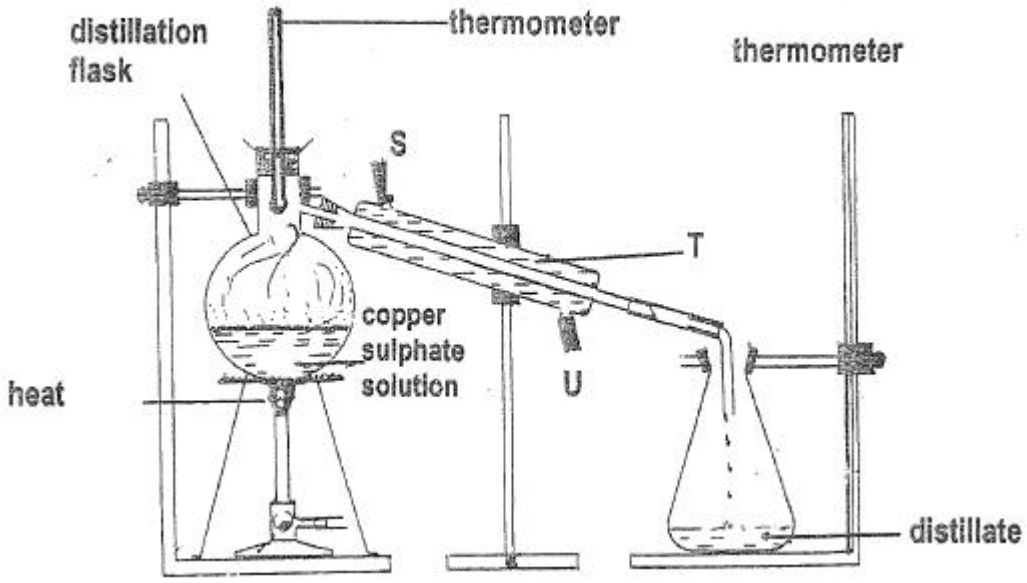
Study the diagram showing an experiment on separation techniques.



(a) (i) Mention one mixture which can be separated by the technique shown above. [2012.P2.Q7a(i)]
(ii) Suggest a solvent for the mixture you have mentioned in a (i) above. [2012.P2.Q7a(ii)]
(b) What property of the filter paper makes it possible for it to be used in the separation technique? [2012.P2.Q7b] [1]
(c) What name is given to the band of colours that will be formed on the filter paper after separation? [2012.P2.Q7c]
(d) State one industrial application of this separation technique. [2012.P2.Q7d]
(e) Which separation technique will be the most suitable to separate salt from its solution? [2012.P2.Q7e]

4 MATTER (SEPARATION TECHNIQUES) [2010.P2.Q9]

Study the diagram showing an arrangement of apparatus used in the separation of copper sulphate solution.



(a) Name the apparatus marked T. [2010.P2.Q9a]

(b) Identify the process shown above being used to separate the copper sulphate solution. [2010.P2.Q9b]

(c) State the function of the apparatus labelled T. [2010.P2.Q9c]

(d) Which letter shows the point at which cold water enters apparatus labelled T in order for it to work effectively. [2010.P2.Q9d]

(e) Suggest one additional piece of apparatus which could be added to the set in order for the separation of a mixture of water and ethanol to occur. [2010.P2.Q9e]

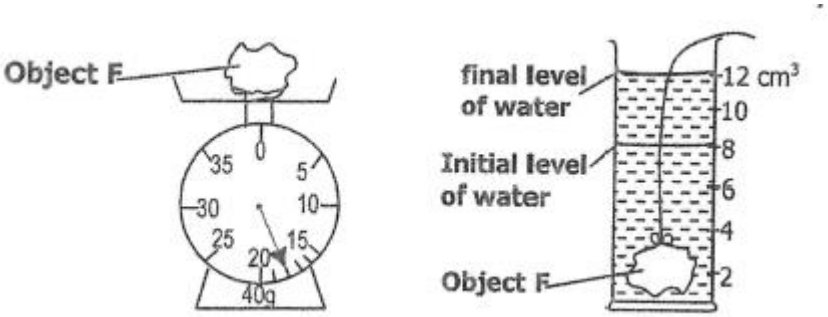
(f) What name is given to the kind of separation technique used in the industries to separate the mixture mentioned in (e) above? [2010.P2.Q9f]

9.1 MASS, WEIGHT, VOLUME & DENSITY – PAPER 1

1 What is the weight of a packet of salt with a mass 500g? (Take acceleration due to gravity as 10N/kg) [2016.P1.Q14]

A 0.5N B 5.0N C 50.0N D 5 000.0N

2 Study the diagram below on determination of density. Calculate the density of object F. The density of F is ... [2016.P1.Q15]



A 1.5g/cm3 B 2.5g/cm3 C 4.0g/cm3 D 4.5g/cm3

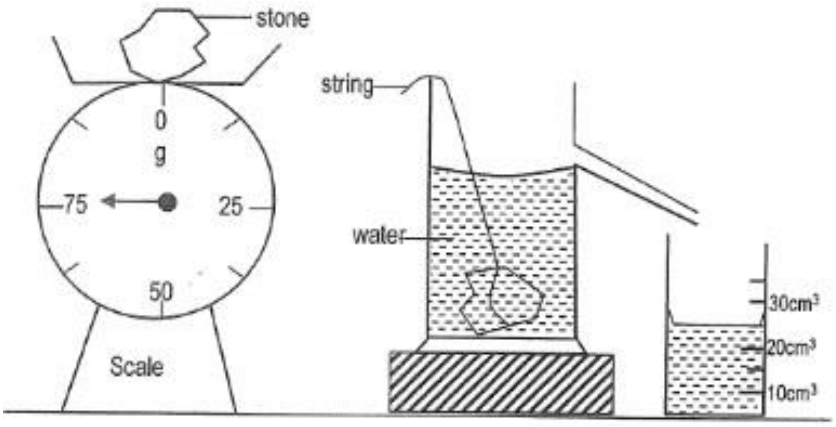
3 Calculate the weight of a 2Kg packet of sugar taking the acceleration due to gravity to be 10m/s2. The weight of the packet of sugar is ..... [2015.P1.Q15]

A 0.2N B 8.0N C 12N D 20N

4 A stone of mass 20g was completely immersed in 10cm3 of water in a measuring cylinder. The level of water rose to 15cm3. Calculate the density of the stone. The density of the stone is ..... [2015.P1.Q16]

A 0.8g/cm3 B 1.38g/cm3 C 2.0g/cm3 D 4.0g/cm3

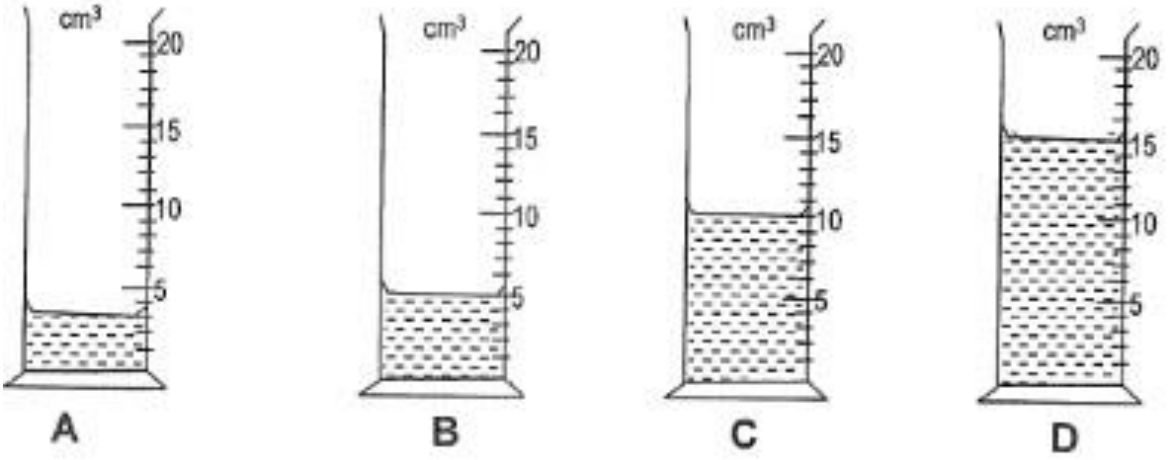
5 The diagrams below show an experiment carried out by the pupils to determine the density of an irregular solid. [2012.P1.Q8]



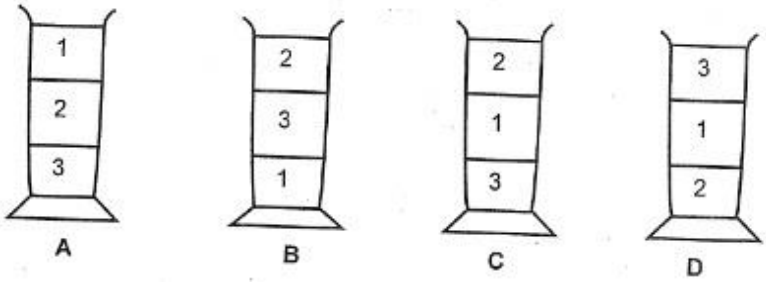
What is the density of the irregular solid shown in the diagrams above?

A 0.33 g/cm^3 B 3.0 g/cm^3 C 50 g/cm^3 D 100 g/cm^3

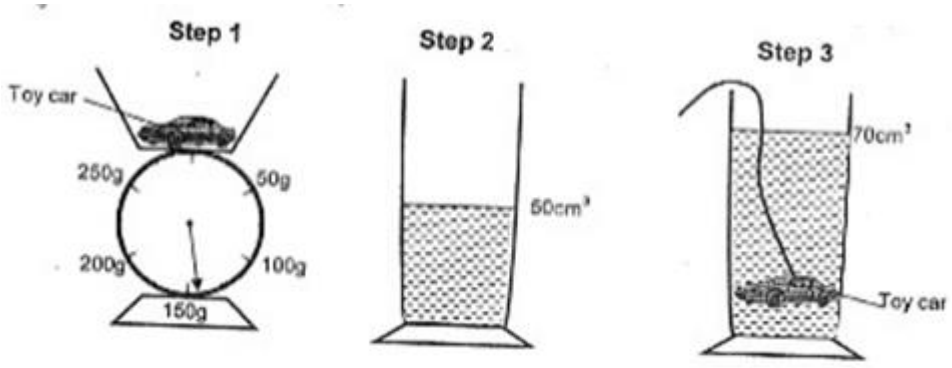
6 The diagrams below show four liquids each having a mass of 5g but different volumes as shown.
Which liquid A, B, C or D is pure water? [2012.P1.Q9]



7 A pupil was given a clean measuring cylinder and equal volumes of the following liquids;
1 Water
2 Mercury
3 Oil
Which diagram below shows how the three liquids settle down when poured in a measuring cylinder? [2011.P1.Q6]



8 The diagrams below show steps used to determine the density of a toy car.

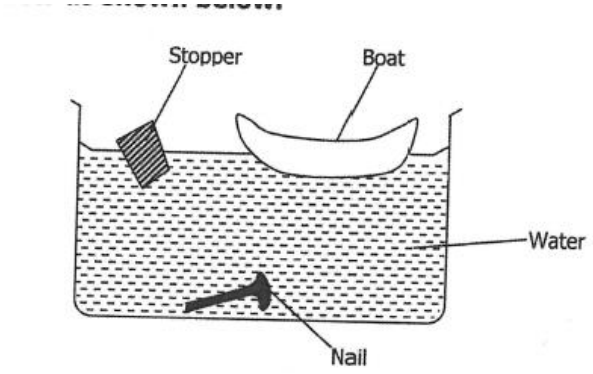


Calculate the density of the toy car. [2011.P1.Q7]
A 7 g/cm^3 B 5 g/cm^3 C 3 g/cm^3 D 2 g/cm^3

9.2 MASS, WEIGHT, VOLUME & DENSITY – PAPER 2

1 DENSITY

A toy boat of mass 500g made of metal, a stopper of mass 80g made of wood and an iron nail of mass 50g were put into a container filled with water as shown below. [2016.P2.Q4]



+

(a) Define density. [2016.P2.Q4a]

(b) Calculate the density of the stopper if its volume was found to be 100cm3. (Show your working) [2016.P2.Q4b]

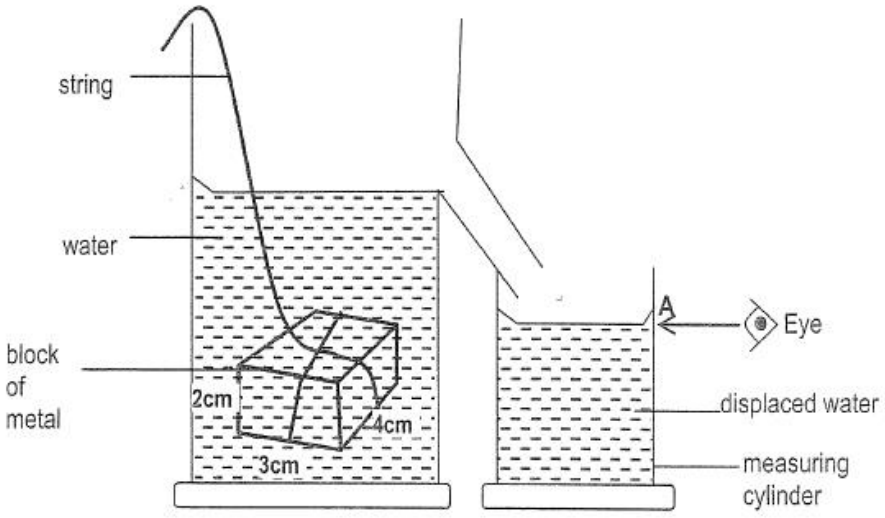
(c) What term is given to objects which stay on top of water as shown by the stopper and boat? [2016.P2.Q4c]

(d) Explain why the boat stayed on top of the water but the nail went down to the bottom of the water. [2016.P2.Q4d]

(e) Predict what would happen to the boat if it was filled with water. [2016.P2.Q4e]

2 QUESTION 2 – MATTER [2013.P2.Q2]

The diagram below shows an arrangement during an experiment on density.



(a) What is the relationship between the volume of the metal and that of the water it displaced? [2013.P2.Q2a]

(b) Find the volume at point marked 'A' on the cylinder. [2013.P2.Q2b]

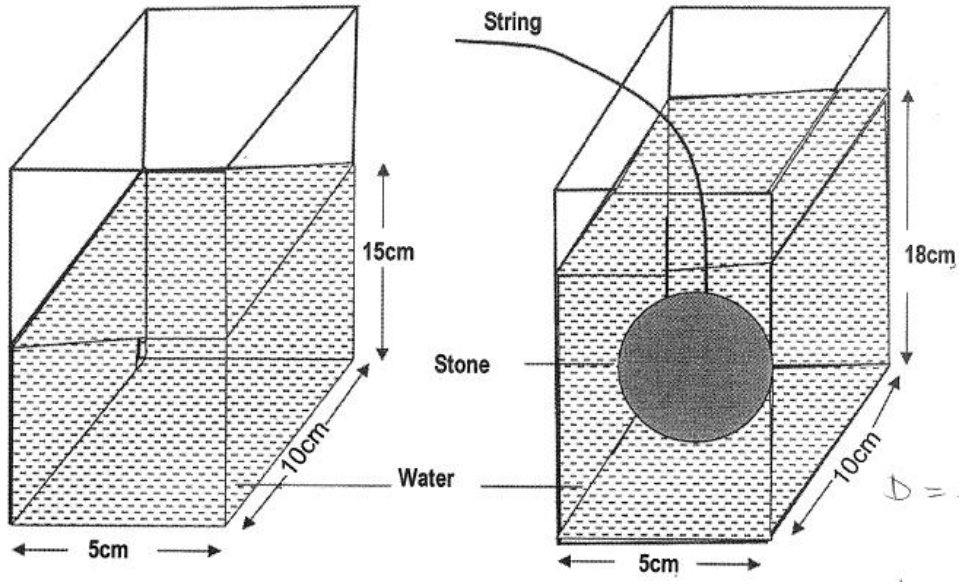
(c) If the mass of the metal block is 48 grams, calculate its density. [2013.P2.Q2c]

(d) Suggest why the metal block could not float on water. [2013.P2.Q2d]

(e) Describe how you can make the metal block float on water. [2013.P2.Q2e]

3 DENSITY [2011.P2.Q2]

Study the diagrams on the experiment on density of a stone.



(a) Calculate the volume of water used in the experiment. [2011.P2.Q2a]

(b) What is the increase in height of water in the vessel when the stone is placed in it? [2011.P2.Q2b]

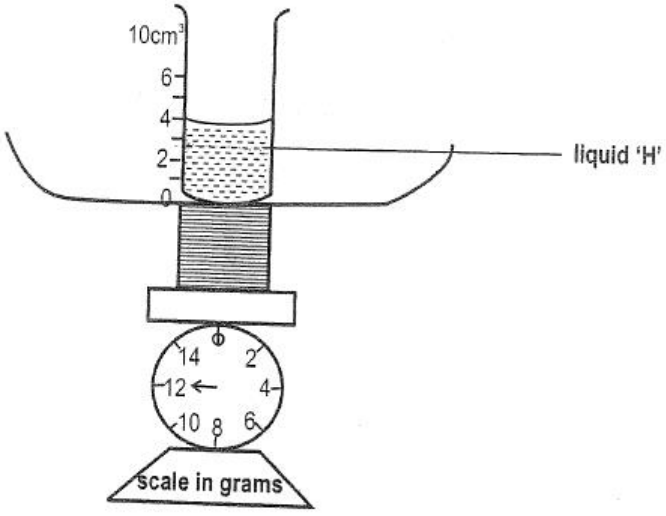
(c) Calculate the volume of the stone in the experiment. [2011.P2.Q2c]

(d) If the mass of the stone used is 300g, calculate the density of the stone. (Show your working) [2011.P2.Q2d]

(e) Why could the experiment not be used to determine the density of a different solid of the same volume but mass of 50g? [2011.P2.Q2e]

4 MATTER [2010.P2.Q3]

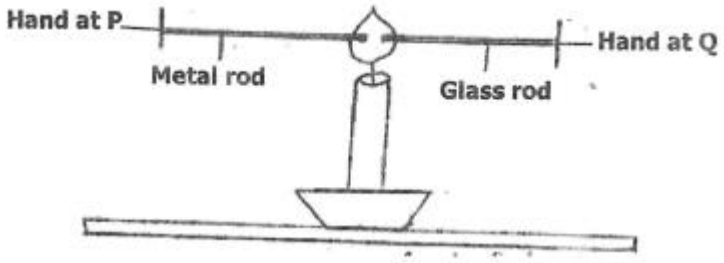
The diagram below shows a measuring cylinder of mass 10g being used to measure the volume of liquid 'H'.



(a) What is the mass of liquid 'H'? [2010.P2.Q3a]
(b) Find the volume of liquid 'H'. [2010.P2.Q3b]
(c) Determine the density of liquid 'H' (show your working) [2010.P2.Q3c]
(d) Suggest what would happen to liquid 'H' if it was poured in a beaker containing water. [2010.P2.Q3d]
(e) Before volume of liquid 'H' was measured, the pupils made sure that the measuring cylinder was clean and dry. State one reason for this precaution. [2010.P2.Q3e]

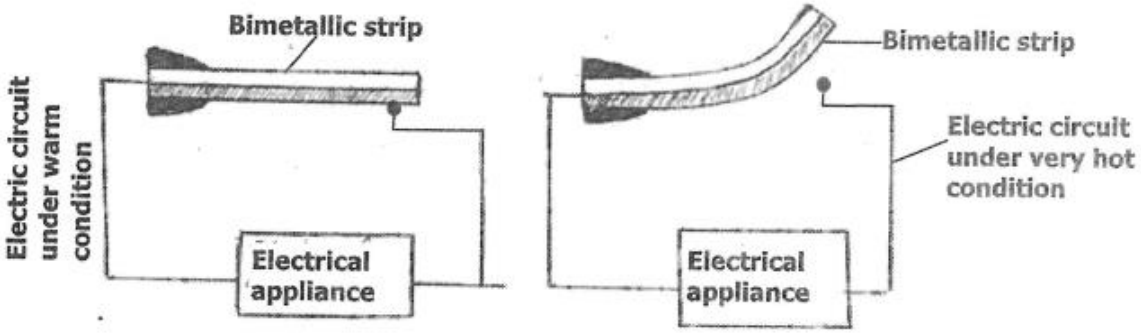
10.1 HEAT TRANSFER & EXPANSION OF SUBSTANCES – PAPER 1

1 The hand at P feels the heat faster than that at Q. This is because ... [2016.P1.Q16]



A metal is a poor conductor of heat. B glass is a good conductor of heat.
C metal is a good conductor of heat. D glass is a better conductor of heat.

2 Study the diagrams below showing the bimetallic strip under warm condition and under very hot condition.



In which electrical appliance would such a type of circuit be used? [2016.P1.Q17]

A Computer B Electric fan
C Electric pressing iron D Thermos flask?

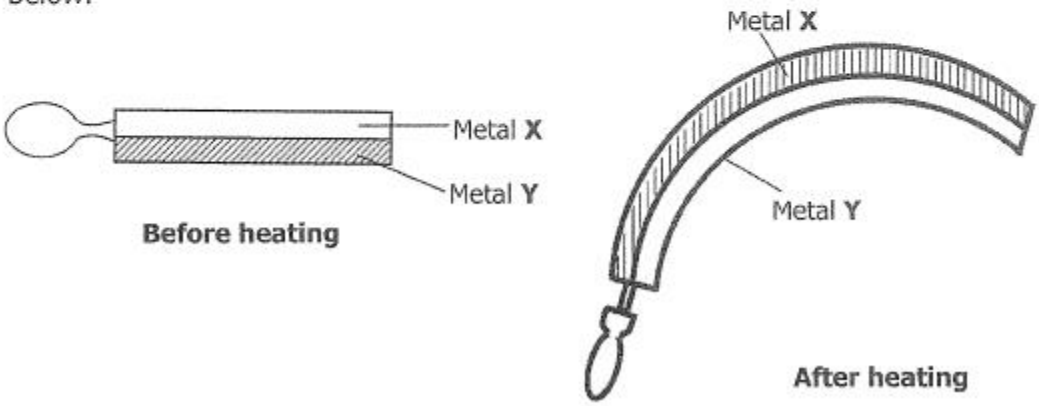
3 Cooking sticks are made of wood because wood is a ..... [2015.P1.Q17]

A bad conductor of heat.
B good conductor of heat.
C bad radiator of heat.
D good radiator of heat.

4 The table below shows the increase in length of four different metals when heated to the same temperature.

Metal	Initial length (cm)	Increase in length (cm)
K	5	0.30
L	5	0.20
M	5	0.09
N	5	0.10

A bimetallic strip made from two metals above when heated bends as shown below.

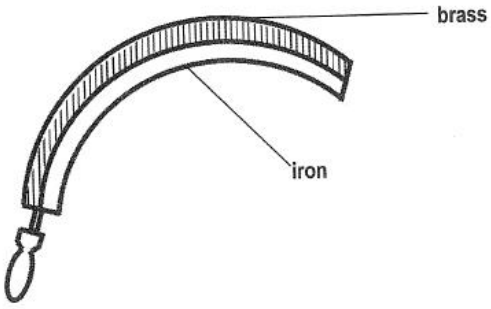


A bimetallic strip made from two metals above when heated bends as shown below.

Which metals from the table produce the above effect? [2015.P1.Q18]

	Metal X	Metal Y
A	K	M
B	L	K
C	N	L
D	M	N

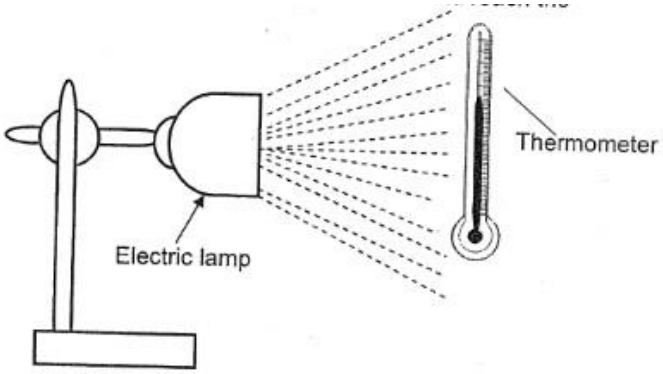
5 A pupil heated a bi-metallic strip made of iron and brass, the result of which is shown in the diagram below.



The statement that best explains the result is that ..... [2013.P1.Q9]

A brass is a good conductor of heat.
B iron is a good conductor of heat.
C brass expands more than iron.
D iron expands more than brass.

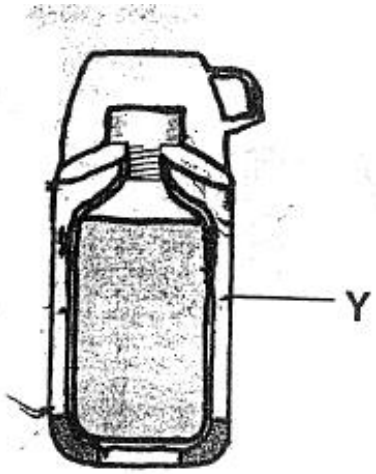
6 An experiment was set up as shown to find out if heat could reach the thermometer.



The thermometer measured an increase in temperature. Which of the following suggests how most of the heat could have reached the thermometer? [2011.P1.Q10]

A Conduction by air
B Electro-magnetic waves in air
C Convection currents by air
D Radiation in air

7 Study the diagram showing a thermos flask.



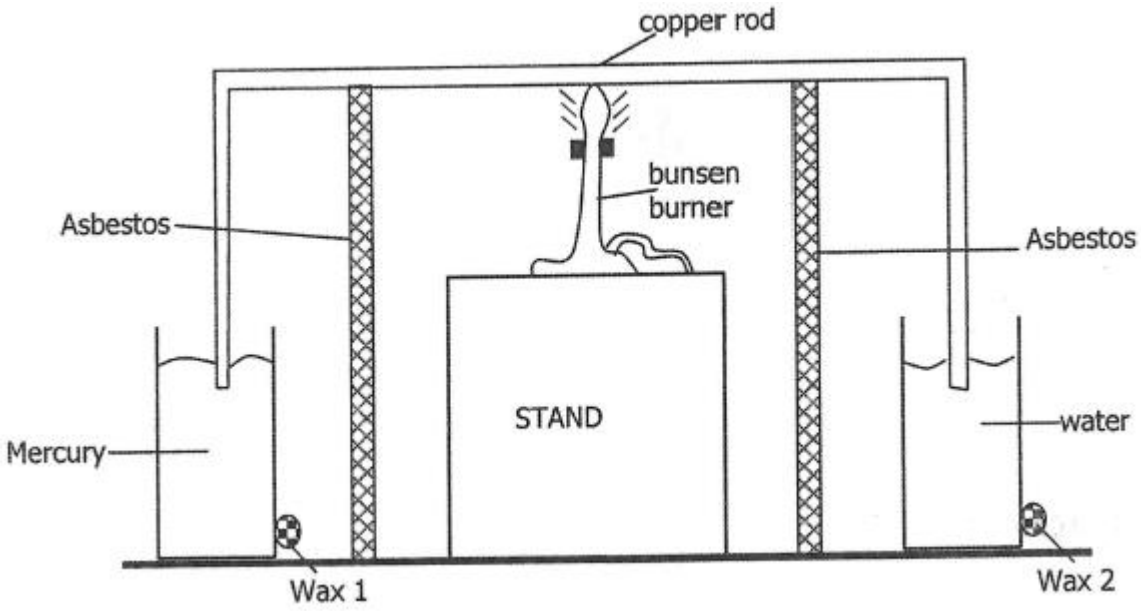
The part marked Y prevents heat transfer by ..... [2011.P1.Q12]

A convection.
B radiation.
C both convection and radiation.
D both convection and conduction

10.2 HEAT TRANSFER & EXPANSION OF SUBSTANCES – PAPER 2

1 HEAT TRANSFER

The diagram below shows a copper rod dipped into a beaker containing mercury on one side and into a beaker containing water on the other side. The rod was then heated in the middle using a Bunsen burner. [2016.P2.Q3]



(a) (i) State the term given to the movement of heat in the copper rod. [2016.P2.Q3a(i)]

(ii) Explain why asbestos boards were placed in the positions shown. [2016.P2.Q3a(ii)]

(b) (i)Which wax would melt first? [2016.P2.Q3b(i)]

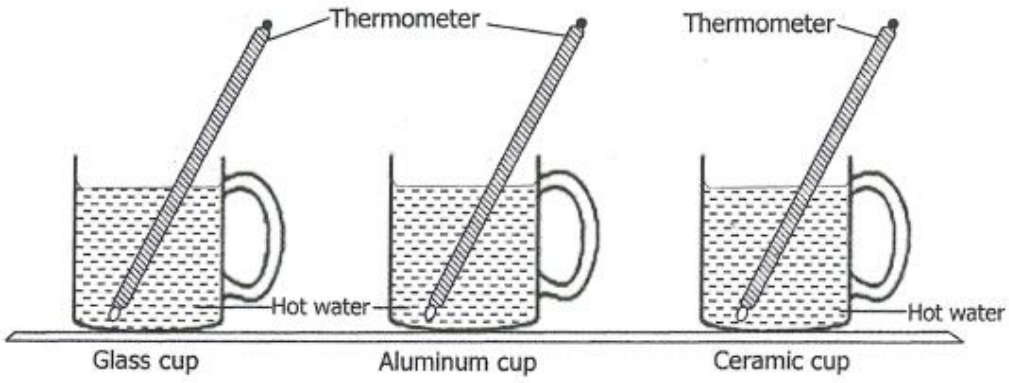
(ii)Give a reason for your answer in question (b) (i) above. [2016.P2.Q3b(ii)]

(iii) Explain why the wax on the other beaker did not melt at the same time as the one mentioned in (b) (i) above. [2016.P2.Q3b(iii)]

(c) Suggest a method of heat transfer not demonstrated in the experiment above. [2016.P2.Q3c]

2 HEAT TRANSFER

Grade 8 learners carried out an experiment by putting 100ml of hot water into three different cups of the same thickness but made of glass, aluminum and ceramic. This was followed by inserting thermometers in the cups as shown below.



One of the learners touched the outside of the cups and compared how hot each cup felt. [2015.P2.Q3]

(a) Which cup felt the hottest? [2015.P2.Q3a]

(b) Give a reason for your answer in question (a) above. [2015.P2.Q3b]

(c) If the cups where allowed to cool for 10 minutes, and then the thermometers read, which cup would the temperature drop;

(i) fastest? [2015.P2.Q3c(i)]

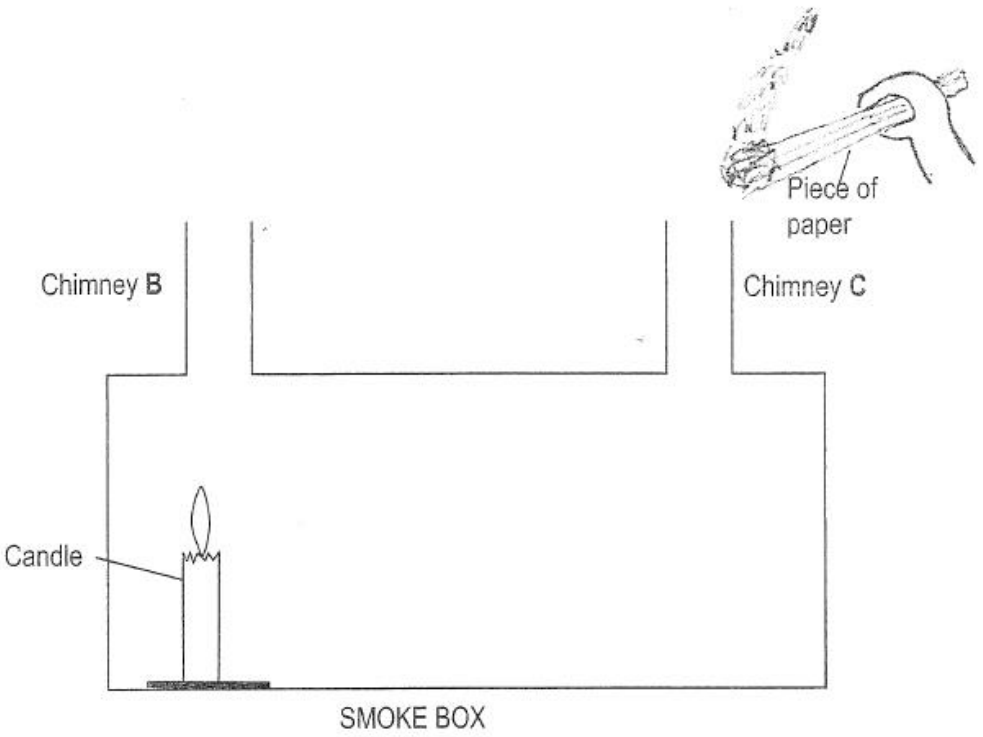
(ii) slowest? [2015.P2.Q3(ii)]

(d) What conclusion can be drawn from the experiment above? [2015.P2.Q3d]

(e) Name one device used to keep hot liquids hot and cold liquids cold. [2015.P2.Q3e]

3 QUESTION 3 – HEAT [2013.P2.Q3]

The diagram below shows an experiment on movement of heat.



(a) Using arrows, show the movement of smoke from the smouldering piece of paper between Chimney B and Chimney C. (Draw in Answer Booklet) [2013.P2.Q3a]

(b) What is the purpose of the burning candle? [2013.P2.Q3b]

(c) What type of heat movement is being demonstrated in the experiment above? [2013.P2.Q3c]

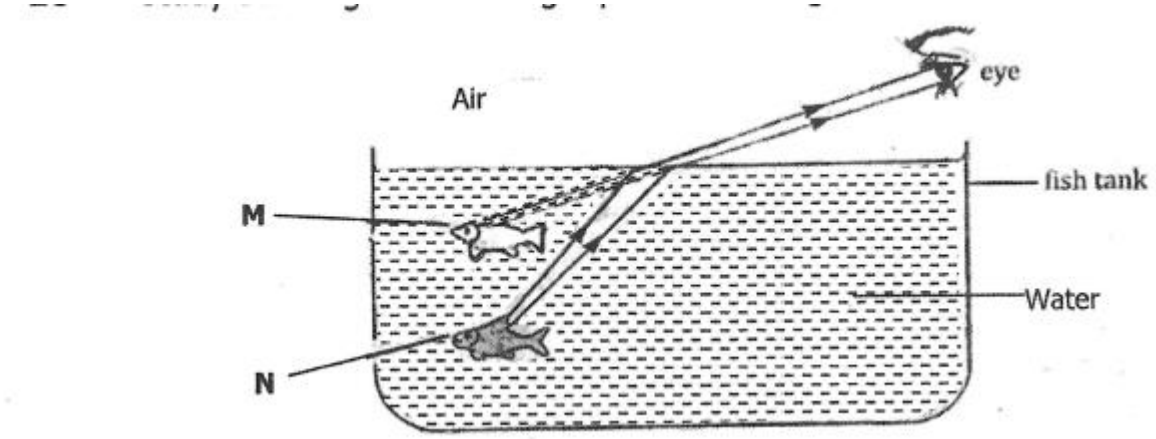
(d) What is the medium of heat transfer in the experiment? [2013.P2.Q3d]

(e) Give one example of an everyday application of this heat movement. [2013.P2.Q3e]

(f) Name one other method of heat transfer. [2013.P2.Q3f]

11.1 REFLECTION AND REFRACTION OF LIGHT – PAPER 1

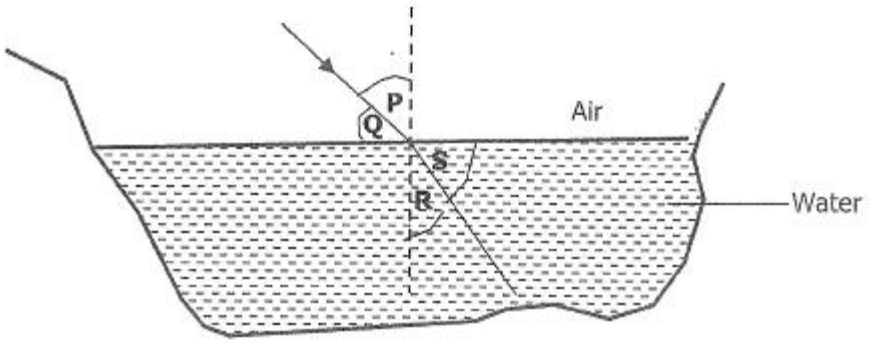
1 Study the diagram showing a person looking at a fish in a fish tank.



Which of the following statements is correct observation of the real position of the fish? The real position of the fish is ... [2016.P1.Q18]

A N because light is refracted as it moves from water to air.
B N because light travels in straight lines.
C M because light travels in straight lines.
D M because light is refracted as it moves from water to air.

2 The diagram below shows a path of a ray of light travelling towards and into a pool of water.

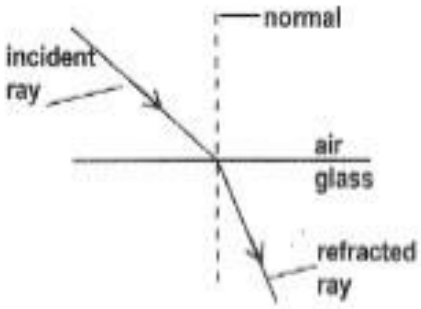


From the four labeled angles, which two show correctly the angles of incidence? [2015.P1.Q19]

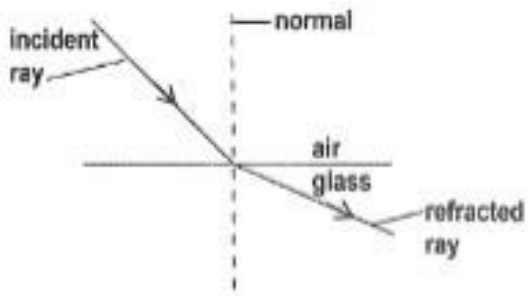
	Incidence	Refraction
A	P	R
B	P	S
C	Q	S
D	Q	R

3 Which diagram below shows the correct refraction of a ray of light as it moves from air to glass? [2013.P1.Q22]

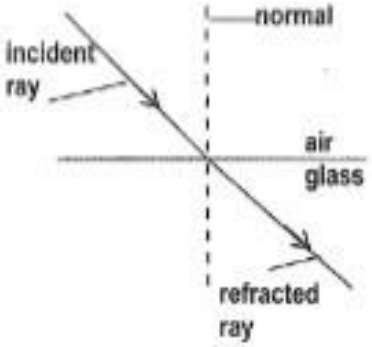
A



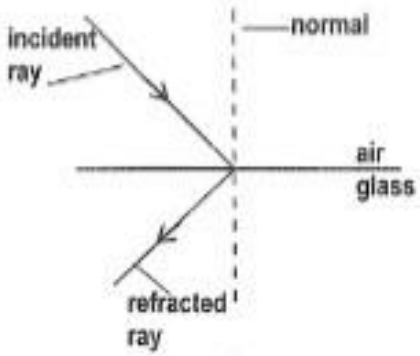
B



C



D



11.2 REFLECTION AND REFRACTION OF LIGHT – PAPER 2

1 QUESTION 4 – LIGHT [2012.P2.Q4]

Figure 1.0 shows light passing from glass into water while Figure 2.0 shows light passing from air to water.

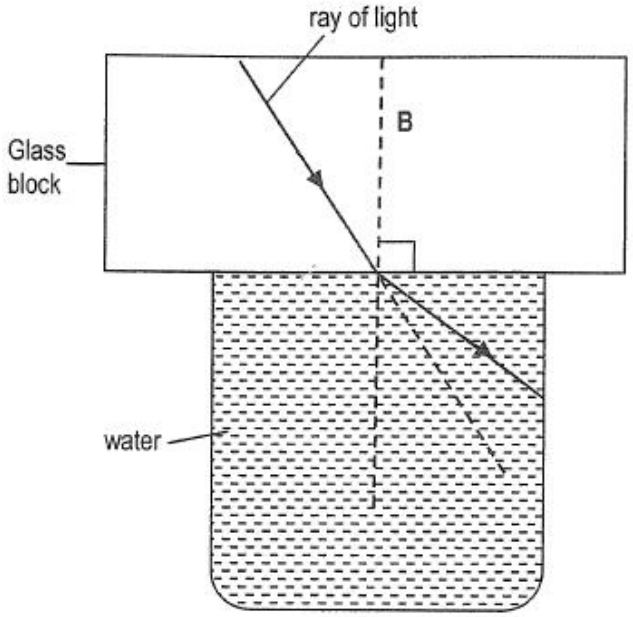


Figure 1.0

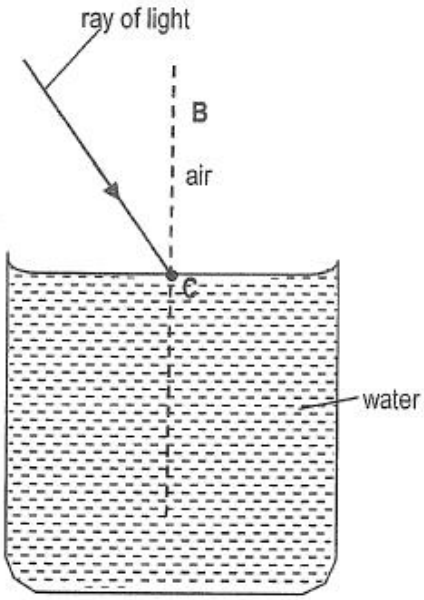


Figure 2.0

(a) What do the broken lines represented by letters labelled B stand for in both Figures 1.0 and 2.0 above? [2012.P2.Q4a]

(b) State the term given to the bending of light shown in Figure 1.0 above. [2012.P2.Q4b]

(c) Explain why the rays of light bend away from line B when it passes into the water in Figure 1.0. [2012.P2.Q4c]

(d) Extend the rays of light from a point labelled C to show how it passes through water in Figure 2.0. (Draw on the diagram in the Answer Booklet) [2012.P2.Q4d]

(e) In which of the three media in the experiments (glass, water or air) is the density the highest? [2012.P2.Q4e]

12.1 COMPOSITION OF AIR – PAPER 1

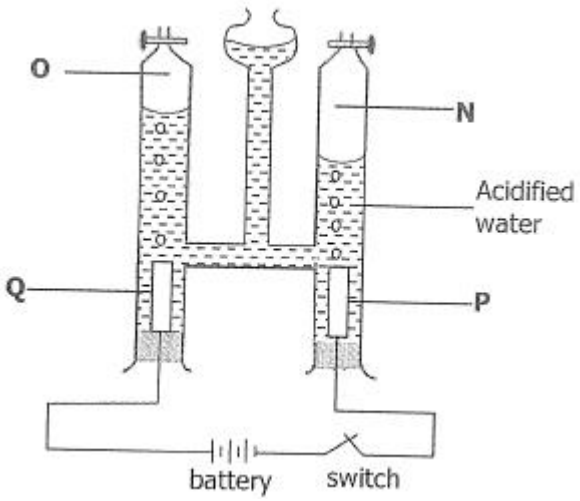
1 Which of the following correctly describes the uses of oxygen, carbon dioxide and hydrogen? [2016.P1.Q19]

	Oxygen	Carbon dioxide	Hydrogen
A	Used in fizzy drinks	Hardening vegetable oils	Used in hospitals
B	Used in steel works	Used in fizzy drinks	Hardening vegetable oils
C	Hardening vegetable oils	Used in steel works	Used in fizzy drinks
D	Used in fire extinguishers	Used in fizzy drinks	Used in hospitals

2 Which of the following gives the correct gas found in air and its properties? [2015.P1.Q20]

	Gas	Solubility in water	Density
A	Oxygen	Not soluble	Less dense than air
B	Nitrogen	Not soluble	Same density
C	Hydrogen	Soluble	Denser than air
D	Carbon dioxide	Soluble	Denser than air

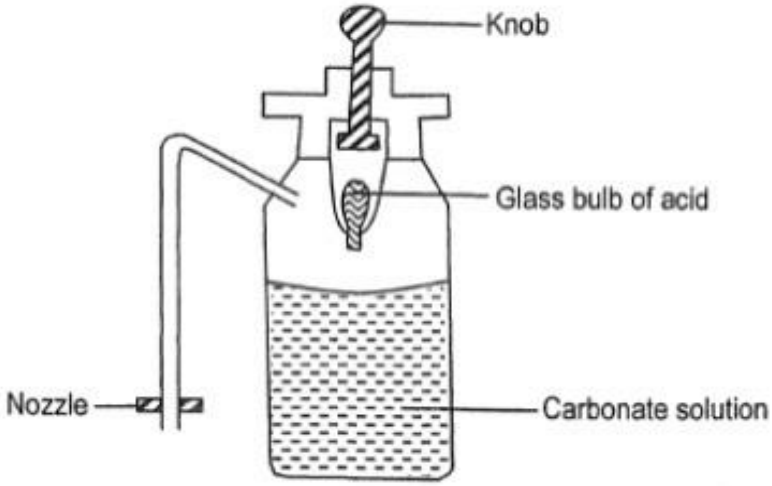
3 Study the diagram below.



In the diagram, which of the following will help you identify gas 0 as hydrogen? [2015.P1.Q30]

A Hydrogen is formed at the anode Q
B Hydrogen is formed twice as much more than oxygen
C Oxygen is formed at the cathode
D Oxygen is formed twice much more than hydrogen

4 Study the diagram below.



The use of the acid in the glass bulb of the fire extinguisher is to ..... . [2012.P1.Q16]

A keep the carbonate solution fresh.
B make carbonate solution to be released through the nozzle.
C react with carbonate solution to produce carbon dioxide to extinguish fire.
D react with carbonate solution to produce water for extinguishing fire.

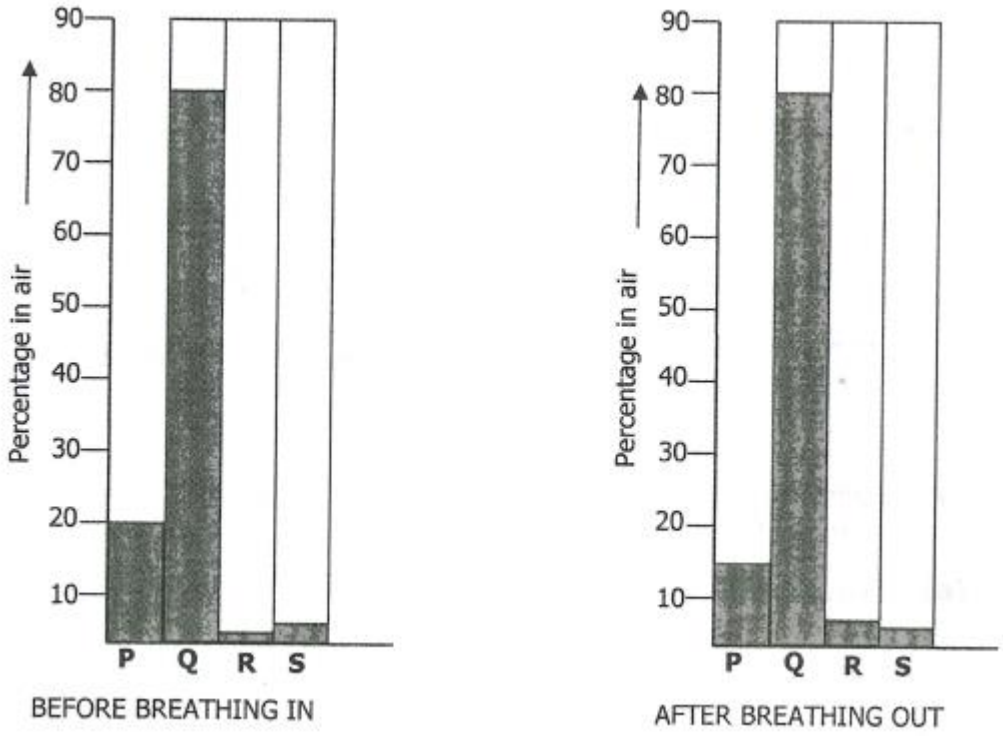
5 What name is given to the flame when acetylene burns in oxygen? [2012.P1.Q17]

A Acetyl-oxygen flame
B Cutting flame
C Welding flame
D Oxy-acetylene flame

12.2 COMPOSITION OF AIR – PAPER 2

1 COMPOSITION OF AIR

The graphs below show the composition of air before it is breathed in and after it is breathed out. [2015.P2.Q5]



(a) (i) Identify the gases marked P and Q. [2015.P2.Q5a(i)]

(ii) Suggest two uses of gas P identified in question (a) (i) above. [2015.P2.Q5a(ii)]

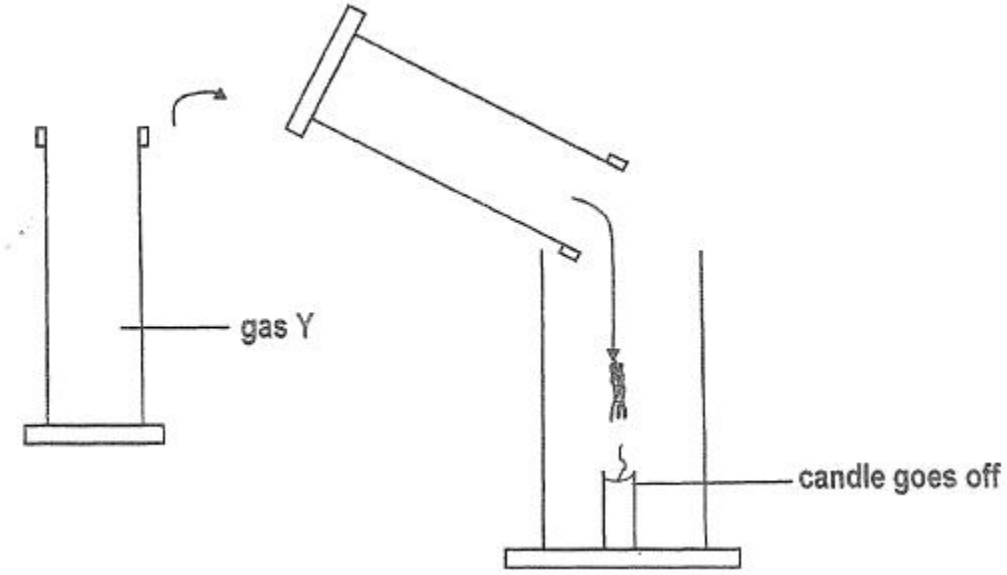
(b) Which letter corresponds to a gas which is used in fire extinguishers? [2015.P2.Q5b]

(c) Explain why water vapour which is a component of air has no fixed percentage. [2015.P2.Q5c]

2 **GASES** [2011.P2.Q4]

A pupil found a gas jar with a label showing that it contained a gas labelled Y.

The pupil decided to pour the gas into a beaker containing a burning candle as shown.



(a) Identify the gas **Y**. [2011.P2.Q4a]

(b) What two properties of the gas Y are demonstrated in the experiment? [2011.P2.Q4b]

(c) Give one reason why the gas jar containing gas Y was not covered. [2011.P2.Q4c]

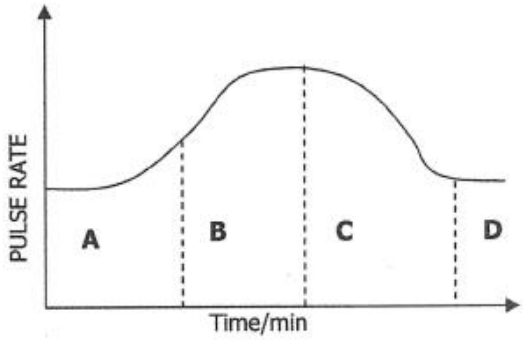
(d) What is the other test which could be done to prove the identity of the gas Y? [2011.P2.Q4d]

(e) Give one use of the gas Y not demonstrated in the experiment above. [2011.P2.Q4e]

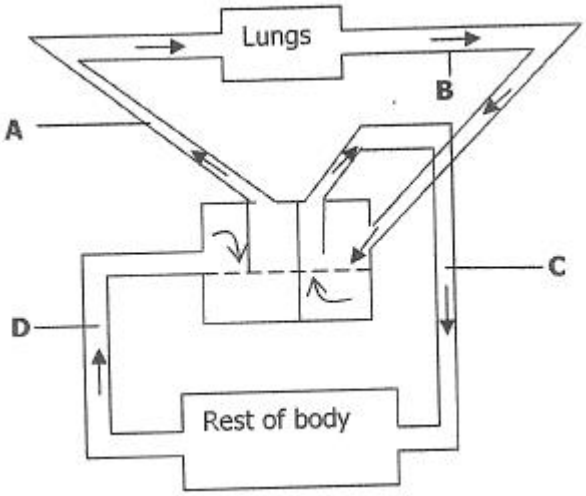
GRADE 9 TOPICS

13.1 CIRCULATORY SYSTEM – PAPER 1

1 The graph below shows the pulse rate of a person before, during and after an exercise.
Which letter shows the period of exercising? [2016.P1.Q20]



2 The diagram below shows the movement of blood in human beings.
Which of the labeled parts show a vein that carries blood which is bright red in colour? [2015.P1.Q21]



3 What is the name of the instrument used to listen to the heart beat? [2013.P1.Q4]
A Hydrometer B Stethoscope C Telescope D Thermometer

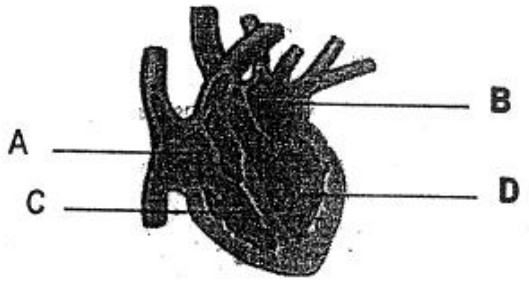
4 A blood vessel has the following characteristics:
(i) has a thin wall
(ii) carry deoxygenated blood
(iii) has valves
Which of the following blood vessels has these characteristics? [2013.P1.Q5]
A Pulmonary artery B Pulmonary vein
C Aorta D Vena cava

5 In which chambers of the heart will you find a high concentration of oxygen and less concentration of carbon dioxide in the blood? [2012.P1.Q3]
A Left Atrium and Left Ventricle B Left Atrium and Right Ventricle
C Right Atrium and Right Ventricle D Right Ventricle and Left Ventricle

6 The main characteristic feature of the red blood that makes it perform its function efficiently is the ..... [2012.P1.Q5]

A presence of haemoglobin. B biconcave shape.
C lack of a nucleus. D round shape.

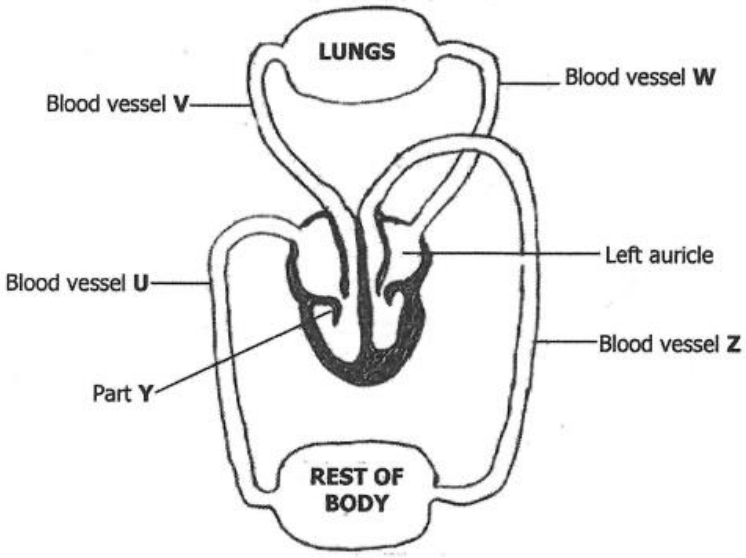
7 The diagram shows the internal structure of the heart. [2011.P1.Q3]



Which labelled part of the heart is responsible for generating the pulse rate felt on the wrist or neck?

13.2 CIRCULATORY SYSTEM – PAPER 2

1 THE CIRCULATORY SYSTEM The diagram below shows parts of a blood circulatory system in a human being. [2016.P2.Q6]



(a) Identify the part labelled Y. [2016.P2.Q6a]

(b) Name one blood vessel which is ...

(i) an artery [2016.P2.Q6b(i)]

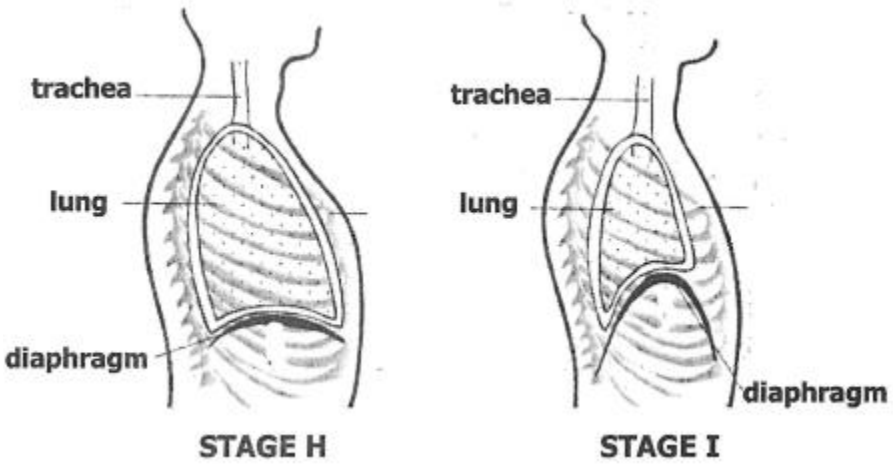
(ii) a vein. [2016.P2.Q6b(ii)]

(c) Suggest the part where oxygen from the air enters the blood. [2016.P2.Q6c]

(d) Using the labels on the diagram, describe how blood which is in the left auricle will move in the circulatory system and arrive back. [2016.P2.Q6d]

14.1 RESPIRATORY SYSTEM – PAPER 1

1 The graph below shows the pulse rate of a person before, during and after an exercise.



Which of the stages correctly shows inspiration? [2016.P1.Q21]

A Stage H because the diaphragm is flattened.

B Stage H because the lung space is reduced.

C Stage I because the diaphragm is dome shaped.

D Stage I because the lung space is increased.

2 The table below shows the composition of oxygen and carbon dioxide in inhaled and exhaled air by a worker at a factory.

	Oxygen	Carbon dioxide
Inhaled (cm^3)	20	0.03
Exhaled (cm^3)	14	0.3

Calculate the percentage of oxygen used up by the body of the worker. The percentage of oxygen used up is ..... [2015.P1.Q22]

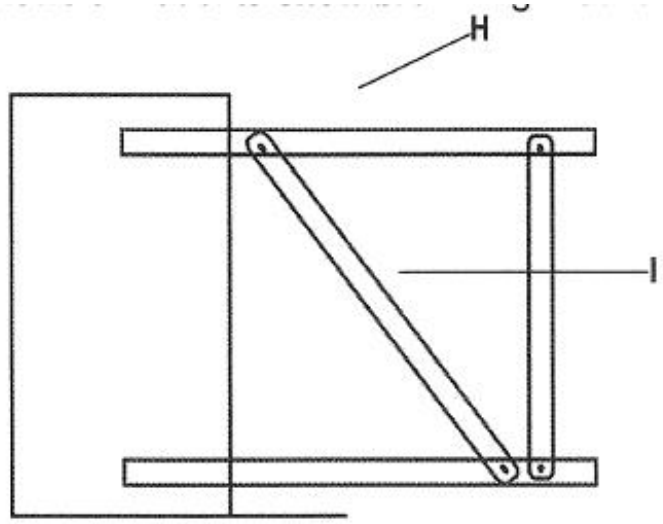
A 14%

B 16%

C 20%

D 30%

3 The diagram below shows a model to show breathing mechanism in human beings.



What do the parts **H** and **I** represent? [2013.P1.Q3]

	H	**I**
A	Diaphragm	Mouth
B	Mouth	Diaphragm
C	Intercostal muscle	Rib
D	Rib	Intercostal muscle

4 Study an incomplete table showing the heart beat and pulse rate of a person at rest and during running.

	ACTIVITY	
	AT REST	**DURING RUNNING**
Heart beat	**72/min**	**X**
Pulse rate	**72/min**	**Y**

What would happen to the heart beat and pulse rate for X and Y respectively? [2011.P1.Q4]

	X	**Y**
A	Increase	Decrease
B	Increase	Increase
C	Decrease	Increase
D	Remain the same	Remain the same

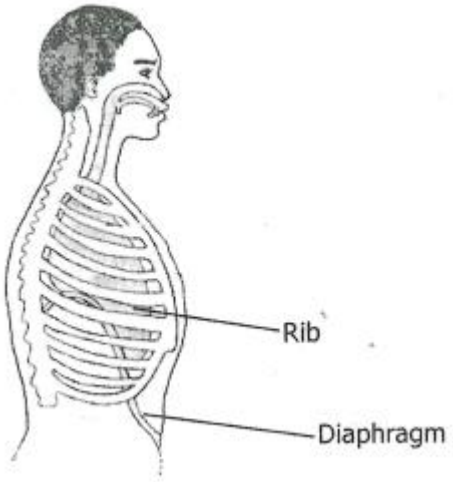
5 The correct sequence that outlines the structures through which expired air passes is .....
[2011.P1.Q5]

A alveolus → bronchus → trachea → nose
B bronchus → nose → alveolus → trachea
C nose → trachea → bronchus → alveolus
D alveolus → trachea → bronchus → nose

14.2 RESPIRATORY SYSTEM – PAPER 2

1 RESPIRATORY SYSTEM

The diagram below shows a stage during the breathing in and out in human beings. [2015.P2.Q6]



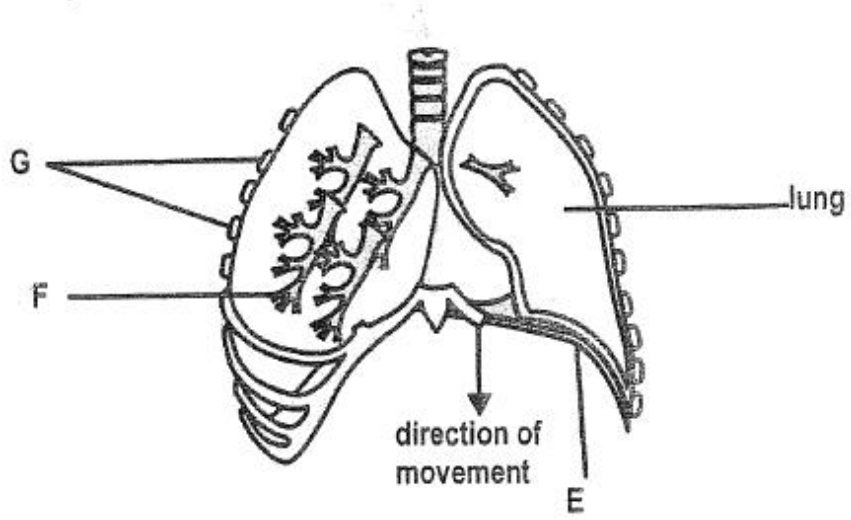
(a) Label on the diagram the following;
(i) nose, [2015.P2.Q6a(i)]
(ii) trachea [2015.P2.Q6a(ii)]

(b) (i) Which stage during breathing is illustrated (shown) in the diagram? [2015.P2.Q6b(i)]
(ii) Give a reason for your answer in question (b) (i) above. [2015.P2.Q6b(ii)]

(c) Describe changes in the diaphragm and ribs which would allow air to go into the lungs. [2015.P2.Q6c]

2 MAN [2010.P2.Q2]

The diagram shows structures related with breathing and gaseous exchange in the human body.



(a) Name the parts labelled E and F. [2010.P2.Q2a]

(b) What is the function of parts labelled G. [2010.P2.Q2b]

(c) If the part marked E moves in the direction of the arrow, explain what would happen to:
(i) the lungs [2010.P2.Q2c(i)]
(ii) parts marked G. [2010.P2.Q2c(ii)]

(d) What do you understand by the term "lung capacity'? [2010.P2.Q2d]

15.1 SEXUALLY TRANSMITTED INFECTIONS (STIS) – PAPER 1

1 Which of the following describes the best way of prevention of Sexually Transmitted Infections (STIs) among adolescents? [2016.P1.Q22]

A Consistence use of condoms. B Correct use of condoms.

C Abstinence from sex until marriage. D Having sex with a faithful partner only.

2 Which of the following diseases is transmitted mainly through sexual intercourse? [2016.P1.Q23]

A Malaria B Syphilis C Typhoid D Scabies

3 Which of the following is not an impact of HIV and AIDS? [2015.P1.Q23]

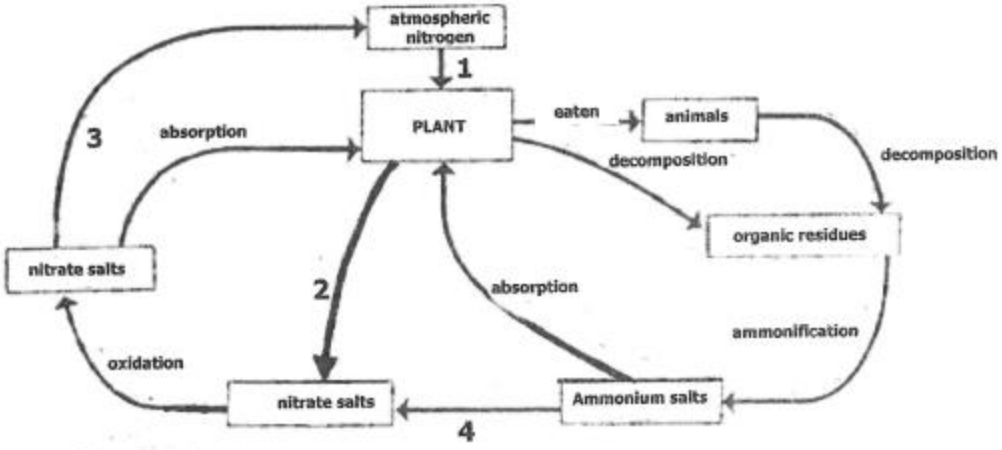
A Increased poverty B Increased number of orphans

C Decreased life expectancy D Decreased pressure on health services

16.1 CYCLES IN THE BIOSPHERE – PAPER 1

1 The diagram below shows a cycle of the biosphere.

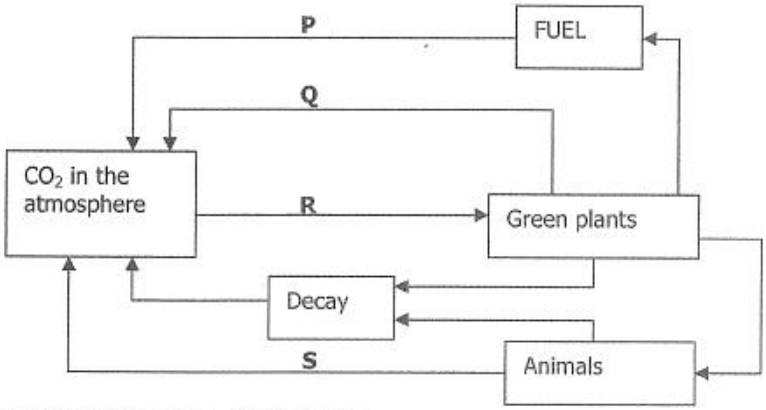
Identify processes 1, 2, 3 and 4. [2016.P1.Q24]



Identify processes **1, 2, 3** and **4.**

	1	2	3	4
A	Lightning	Denitrification	Nitrification	Nitrogen fixation
B	Nitrification	Nitrogen fixation	Denitrification	Lightning
C	Denitrification	Nitrification	Lightning	Nitrogen fixation
D	Lightning	Nitrogen fixation	Denitrification	Nitrification

2 The diagram below shows the carbon cycle.



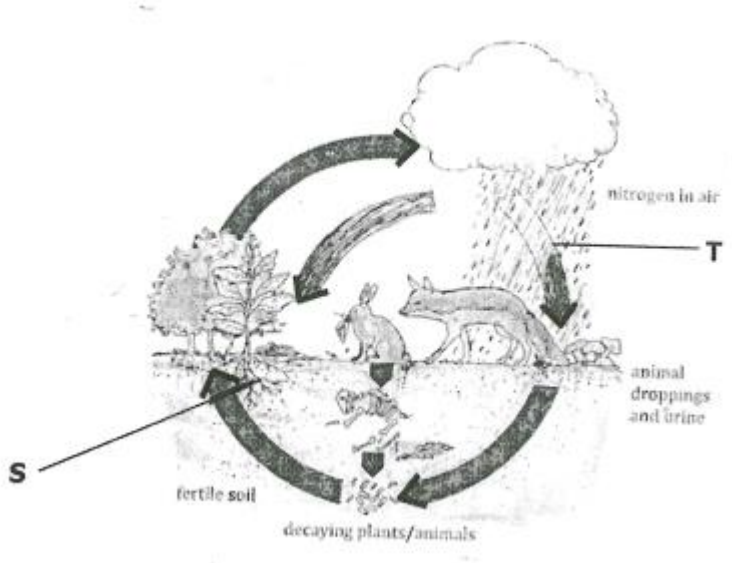
Identify processes **P**, **Q**, **R** and **S**.

	P	Q	R	S
A	Combustion	Transpiration	Respiration	Photosynthesis
B	Transpiration	Combustion	Respiration	Transpiration
C	Photosynthesis	Respiration	Combustion	Transpiration
D	Combustion	Respiration	Photosynthesis	Respiration

Identify processes P, Q, R and S. [2015.P1.Q24]

16.2 CYCLES IN THE BIOSPHERE – PAPER 2

1 CYCLES IN THE BIOSPHERE The diagram below shows the nitrogen cycle. [2015.P2.Q7]



(a) Name the process at T by which nitrogen moves from the air into the soil. [2015.P2.Q7a]
(b) Name the organism involved in changing nitrogen from the air in the soil into the plant roots at S. [2015.P2.Q7b]

(c) Suggest the name given to plants which are able to change nitrogen from the air into the plant. [2015.P2.Q7c]
(d) From the diagram, identify two ways in which animals replace nitrogen back into the soil. [2015.P2.Q7d]
(e) How can human beings add more nitrogen to the soil in order to make the soil more fertile? [2015.P2.Q7e]

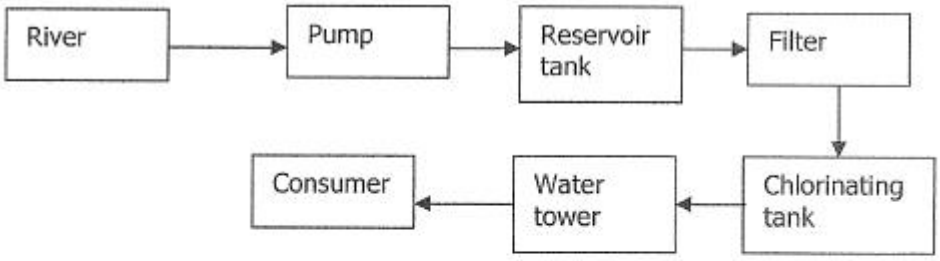
17.1 WATER MANAGEMENT – PAPER 1

1 Which one of the reasons for good water management below does not help to maintain animal and plant life? Water for... [2016.P1.Q25]

A agriculture purposes. B domestic supply.

C fish farming. D recreation.

2 Study the flow chart showing purification of water.



What is the function of the filter and water tower in the process of purification?

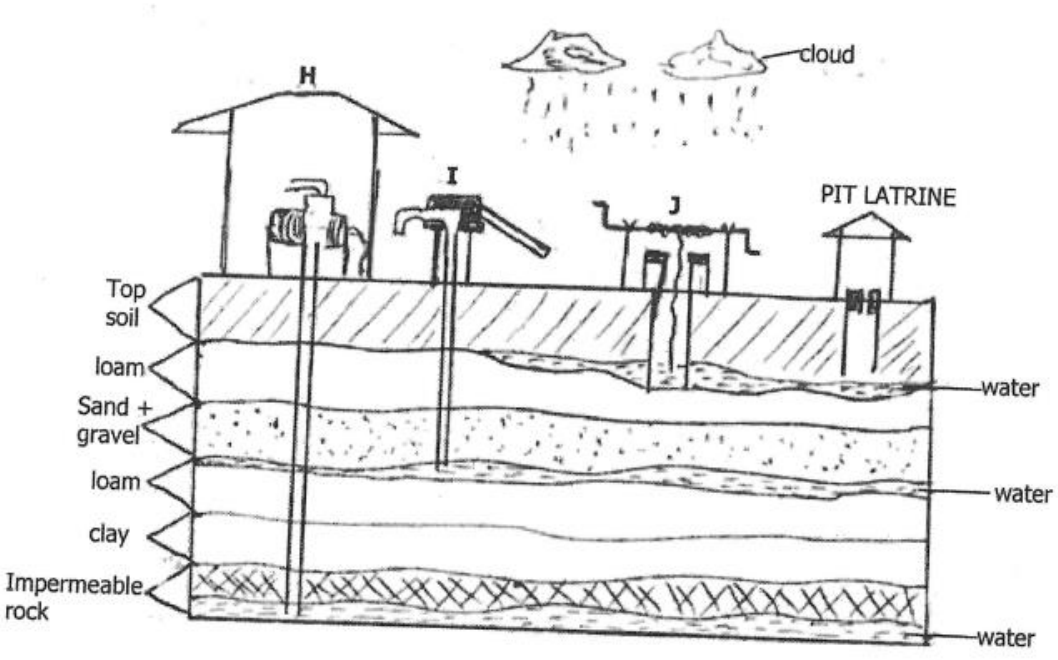
	Filter	Water tower
A	To remove solids	To increase water pressure
B	To get rid of germs	To cool the water
C	To settle solids	To increase pressure
D	To pump the water	To decrease the pressure

What is the function of the filter and water tower in the process of purification? [2015.P1.Q25]

17.2 WATER MANAGEMENT – PAPER 2

1 WATER MANAGEMENT

The diagram below shows some water sources labelled H, I and 3. [2016.P2.Q7]



(a) Which water source is the safest for drinking by humans? [2016.P2.Q7a]

(b) State one disadvantage of obtaining drinking water from water source J. [2016.P2.Q7b]

(c) State one way water is prevented from finishing in the water sources shown in the diagram above. [2016.P2.Q7c]

(d) Explain the role played by sand and gravel in the quality of water from the bore holes. [2016.P2.Q7d]

(e) What term is given to the sources of water shown in the diagram above? [2016.P2.Q7e]

(f) Suggest one other source of fresh water for drinking. [2016.P2.Q7f]

18.1 CONSERVATION OF ANIMALS AND PLANTS – PAPER 1

1 Below is a list of some animals domesticated by man.

1. Cattle 4. Goat

2. Cat 5. Dog

3. Sheep 6. Duck

Which two animals above are domesticated for milk production? [2016.P1.Q26]

A 1 and 3 B 3 and 4

C 4 and 6 D 1 and 4

2 Which plant is not threatened by extinction in Zambia? [2015.P1.Q26]

A Mukwa B Mutondo C Mukula D Musuku

3 Which of the following pairs correctly shows food which grazers and browsers feed on? [2013.P1.Q31]

	GRAZERS	**BROWSERS**
A	Grass	Flesh
B	Flesh	Leaves above ground
C	Grass	Leaves above ground
D	Leaves above ground	Grass

4 Which of the following animals has strong legs for kicking away its predators? [2013.P1.Q32]

A Antelope B Buffalo C Dog D Leopard

5 The correct list of animals domesticated mainly for transport is that which contains..... [2013.P1.Q37]

A sheep, cattle, goats. B dogs, cats, mice.
C donkeys, sheep, dogs. D donkeys, camels, horses.

6 Before cross-breeding the two animals, the farmer should make sure the animals are pure. This is done through ..... [2013.P1.Q38]

A in-breeding. B artificial insemination.
C taming. D detasselling.

7 Detasselling of one species of maize plants prevents the maize plants from carrying out ..... [2012.P1.Q35]

A self-breeding. B self-pollination.
C cross-pollination. D cross-breeding.

8 Which one of the following animals is in danger of extinction as a result of its named characteristic feature? [2012.P1.Q36]

	Animal	**Feature**
A	Zebra	Skin
B	Baboon	Skull
C	Black lechwe	Meat
D	Crocodile	Teeth

9 The table below shows some characteristics associated with some cattle.

TYPE OF CATTLE	**CHARACTERISTICS**
1	Big, high yield of milk
2	Small, good quality of milk
3	Small, high yield of milk
4	Big, good quality of meat

Which two types of cattle would you cross-breed to produce a hybrid of big cattle with high yield of milk and good quality meat? [2011.P1.Q35]

A Type 1 and 4
B Type 1 and 2
C Type 2 and 3
D Type 3 and 4

10 How can the diminishing of the following resources best be reduced? [2011.P1.Q37]

	TREES	COPPER	IRON
A	reduced dependence	recycling	recycling
B	Recycling	afforestation	reduced dependence
C	Recycling	afforestation	recycling
D	reduced dependence	reduced dependence	afforestation

18.2 CONSERVATION OF ANIMALS AND PLANTS – PAPER 2

1 QUESTION 8 – MAN [2012.P2.Q8]

In modern farming, farmers practice cross and in breeding.

(a) Why do farmers practice;
  (i) cross-breeding? [2012.P2.Q8a(i)]
  (ii) in-breeding? [2012.P2.Q8a(ii)

(b) Write down one quality a farmer would like to have in each of the following.
  (i) chickens that produce eggs (layers) [2012.P2.Q8b(i)]
  (ii) maize [2012.P2.Q8b(ii)]

(c) In each case state one advantage and one disadvantage of artificial insemination. [2012.P2.Q8c]

2 MAN'S INTERVENTION IN NATURE [2011.P2.Q9]

Zambia has game parks nearly in all its nine provinces.

(a) Name one of the major living attractions found on land in game parks. [2011.P2.Q9a]

(b) State one important feature of a game park which will enable the living attraction mentioned in (a) above to live there. [2011.P2.Q9b]

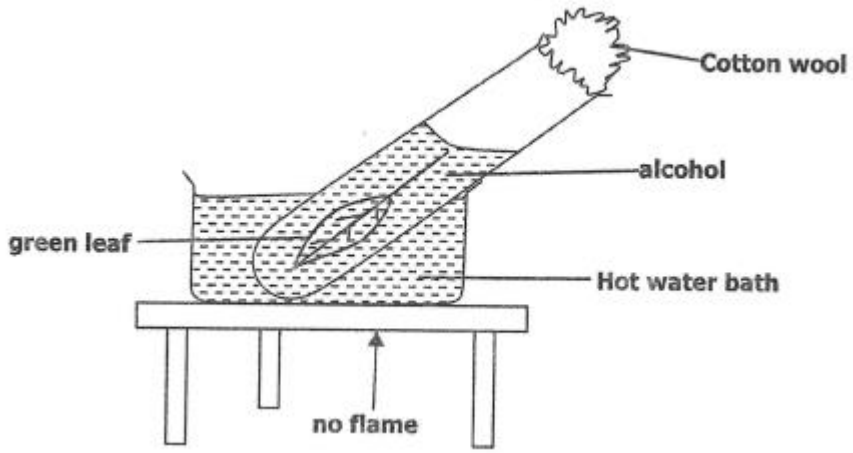
(c) State one regulation that visitors to the national park have to observe. [2011.P2.Q9c]

(d) Name two structures that must be built in order to attract more tourists to a game park. [2011.P2.Q9d]

(e) State one advantage of a game park to a country like Zambia. [2011.P2.Q9e]

19.1 PHOTOSYNTHESIS – PAPER 1

1 The diagram below shows a stage in testing a leaf for starch.

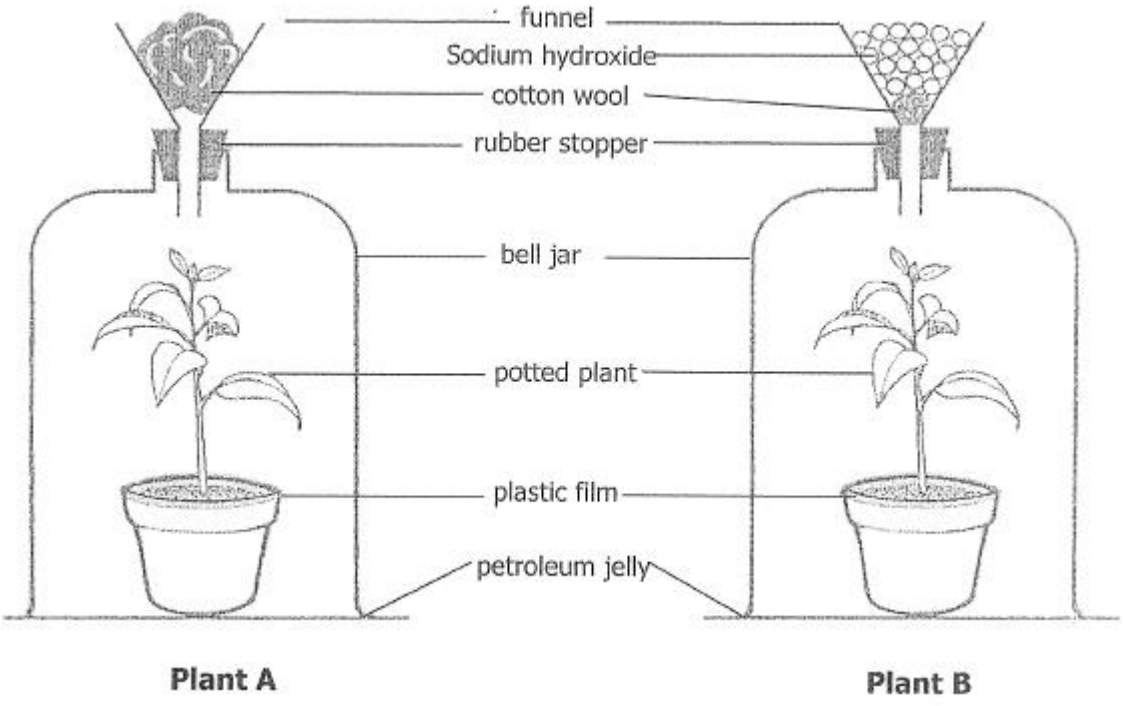


What is the purpose of the stage in the diagram? [2016.P1.Q27]

A Killing protoplasm B Removing chlorophyll

C Making a leaf soft D Hardening the leaf

2 The diagrams below shows an experiment using two destarched plants that were exposed to sunlight for 8 hours.



Which statement is true about the experiment? The leaf from plant..... [2015.P1.Q27]

A stained blue-black. B stained blue-black.

C contained starch. D stained brown.

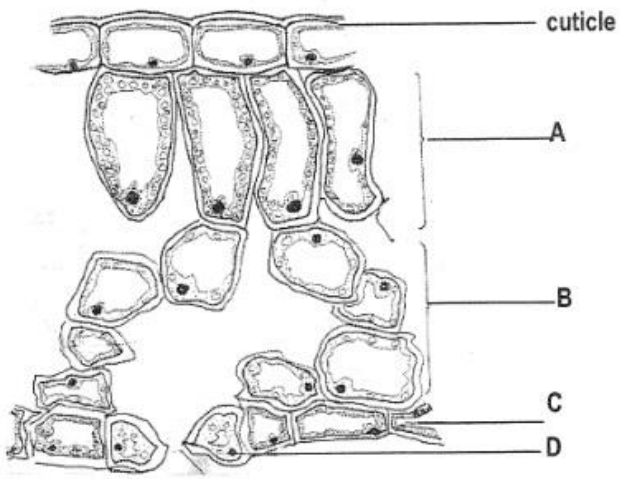
3 What is the correct identity test of starch? [2013.P1.Q25]

A Iodine changes starch to blue/black.
B Alcohol changes starch to milky white.
C Lime water changes starch to blue/black.
D Lime water changes starch to milky white.

4 Which of the following is the correct equation for respiration? [2013.P1.Q26]

A Carbon dioxide + water ---> glucose + oxygen
B Carbohydrates + water + energy ---> oxygen + carbon dioxide
C Glucose + water ---> carbon dioxide + water +energy
D Oxygen + glucose ----> carbon dioxide + water + energy

5 Study the diagram of the internal features of the leaf. [2013.P1.Q27]

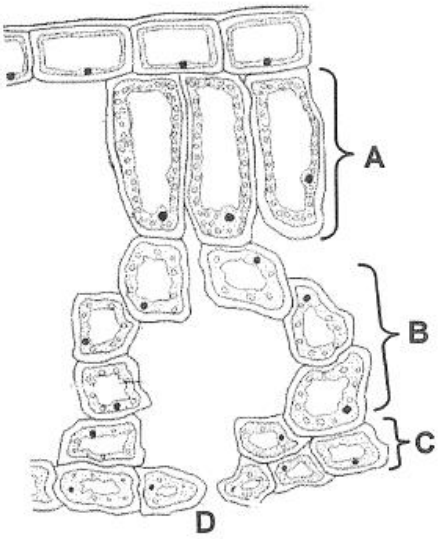


Which letter A, B, C or D shows where you find cells that help to regulate the rate of transpiration?

6 The correct identity test for the gas released during photosynthesis is that it..... [2013.P1.Q28]

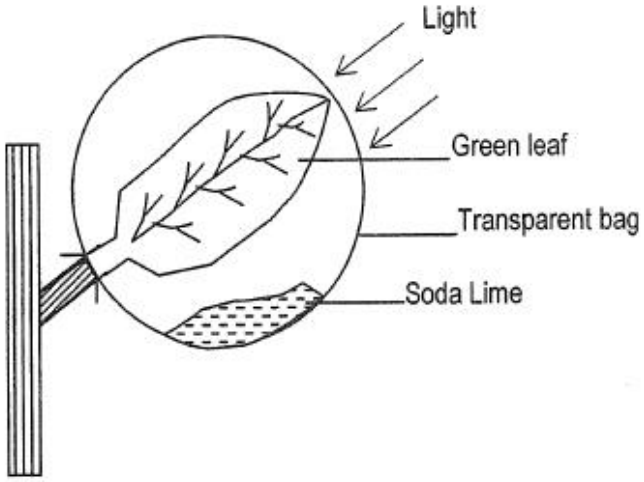
A turns lime water milky.
B burns with a 'pop' sound.
C turns cobalt chloride paper pink.
D relights a glowing splint.

7 Below is an internal structure of a leaf.



Which letter represents a region where the rate of photosynthesis is highest? [2012.P1.Q24]

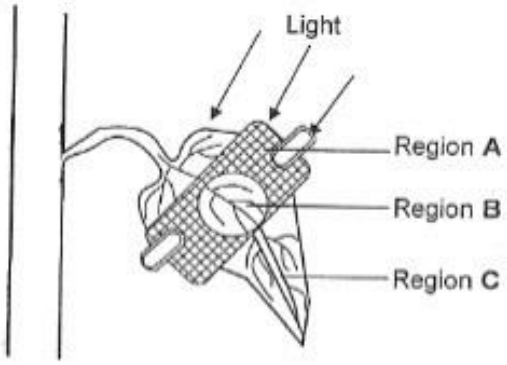
8 The diagram below shows a leaf enclosed in a transparent plastic bag for 24 hours.



What would be the result when the above leaf is tested for starch with iodine solution? [2012.P1.Q25]

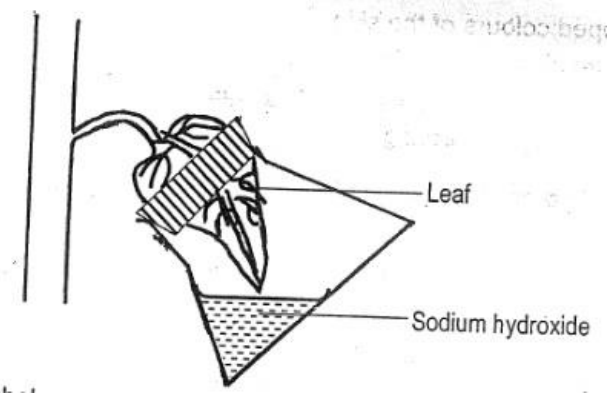
A Leaf turns blue/black because of chlorophyll.
B Leaf turns blue/black because of sunlight.
C Leaf is stained brown due to absence of oxygen.
D Leaf is stained brown due to absence of carbon dioxide.

9 The diagram below shows a leaf with different parts exposed to different conditions.
After six hours, the leaf was tested for starch using iodine solution. Which of the following best fits the results obtained if the leaf was put in darkness before the experiment was carried out? [2011.P1.Q25]



	REGION A	REGION B	REGION C
A	Brown	Blue /black	Blue/black
B	Brown	Brown	Brown
C	Blue/black	Blue /black	Blue /black
D	Blue/black	Brown	Brown

10 The diagram below shows an experiment carried out by a Grade 9 pupil to show substances necessary for photosynthesis. [2011.P1.Q26]

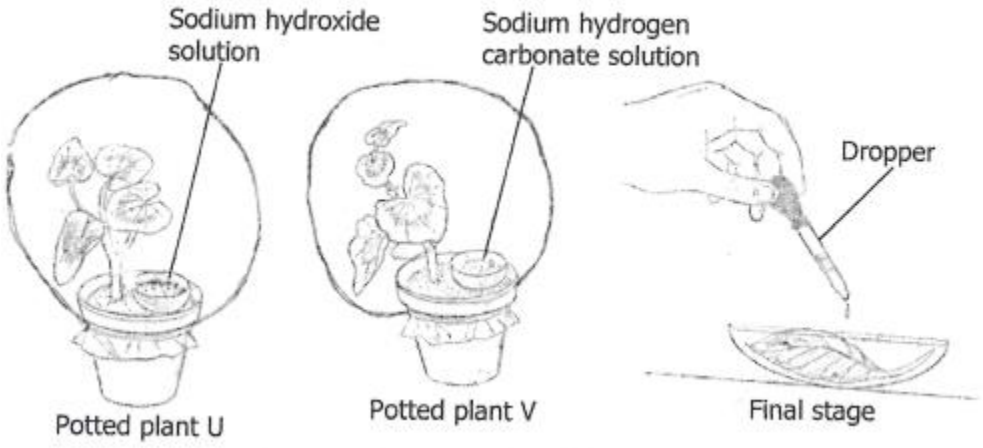


Suggest the substance necessary for photosynthesis that the pupil was trying to remove by using sodium hydroxide in the experiment above.

A Carbon dioxide
B Chlorophyll
C Light
D Water

19.2 PHOTOSYNTHESIS – PAPER 2

1 PLANTS AND ANIMALS The diagram below shows two potted plants (u) and (V) undergoing an experiment on photosynthesis. [2015.P2.Q8]



(a) Explain the purpose of enclosing the plant with the following;
(i) Petri-dish with sodium hydroxide solution, [2015.P2.Q8a(i)]
(ii) Petri-dish with sodium hydrogen carbonate solution. [2015.P2.Q8a(ii)]

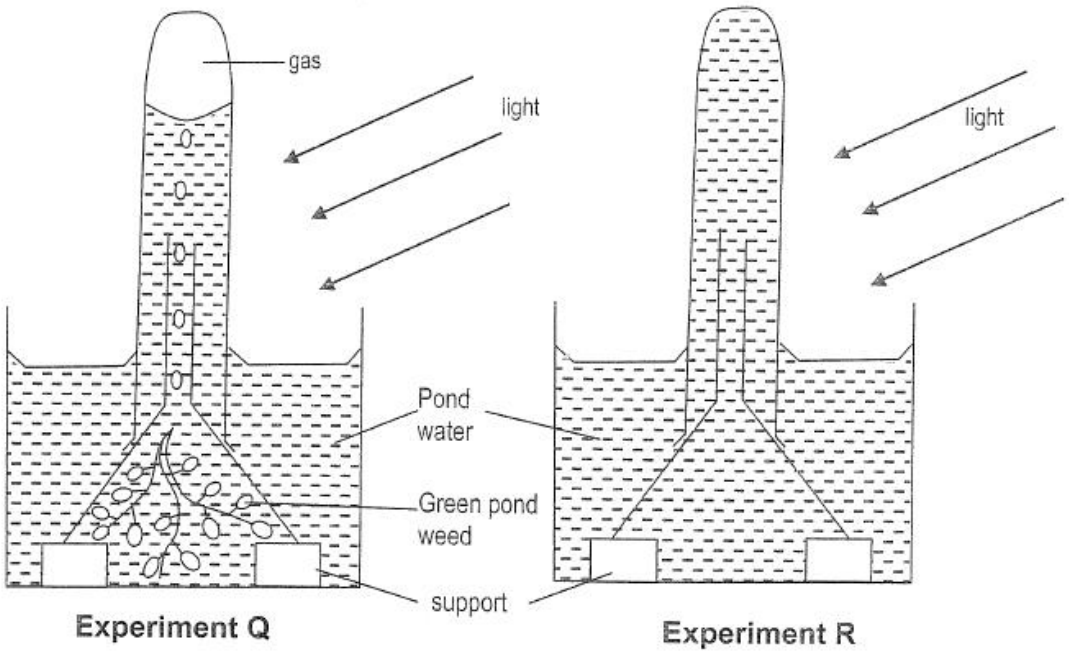
(b) In which potted plant would leaves turn blue-black when tested for starch after 24 hours? [2015.P2.Q8b]

(c) What is the name of the solution in the dropper in the final stage which is being used to test the leaf for starch? [2015.P2.Q8c]

(d) State two steps which will have to be carried out in order to test the leaf for starch, before final stage is done in the diagram above. [2015.P2.Q8d]

2 QUESTION 7 – PLANTS [2013.P2.Q7]

The diagrams show and experiment on a pond weed.



(a) What are the experiments trying to demonstrate? [2013.P2.Q7a]

(b) Why is it recommended to use pond water instead of water from the tap? [2013.P2.Q7b]

(c) What is the purpose of the experiment R? [2013.P2.Q7c]

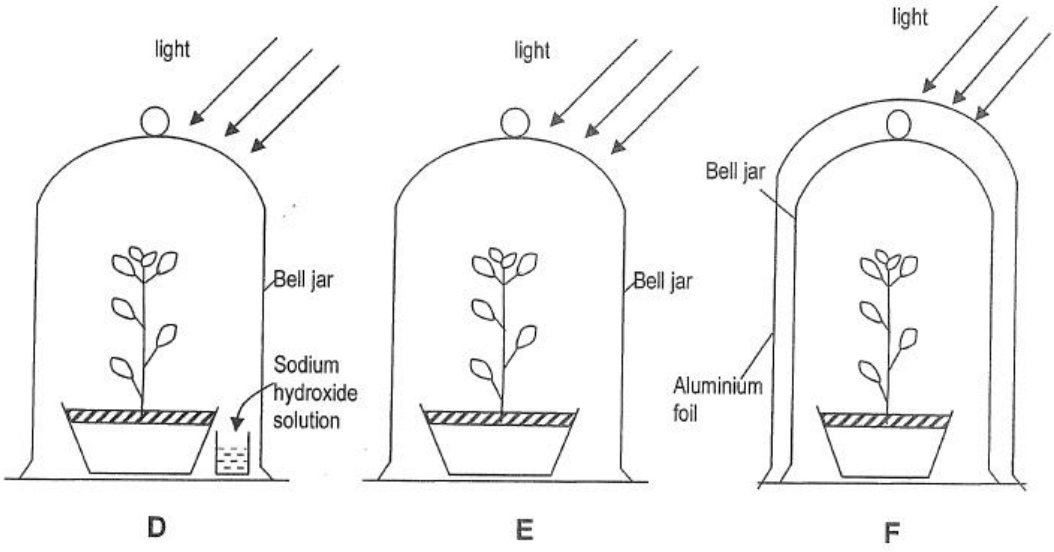
(d) Name the gas collected in experiment Q. [2013.P2.Q7d]

(e) What is the identity test for the gas collected in Q? [2013.P2.Q7e]

(f) Write a word equation on the process producing the gas in experiment Q. [2013.P2.Q7f]

3 QUESTION 5 – PLANTS [2012.P2.Q5]

Three potted plants were kept in a dark cupboard for two days before being used for an experiment. Each plant was then put under a bell jar under different conditions as shown for six hours.



(a) Explain why there was no starch in the leaves at the end of the two days in the cupboard. [2012.P2.Q5a]

(b) What was the purpose of sodium hydroxide solution before the plants were put under bell jar in D? [2012.P2.Q5b]

(c) Suggest why the bell jar in F was covered with aluminium foil. [2012.P2.Q5c]

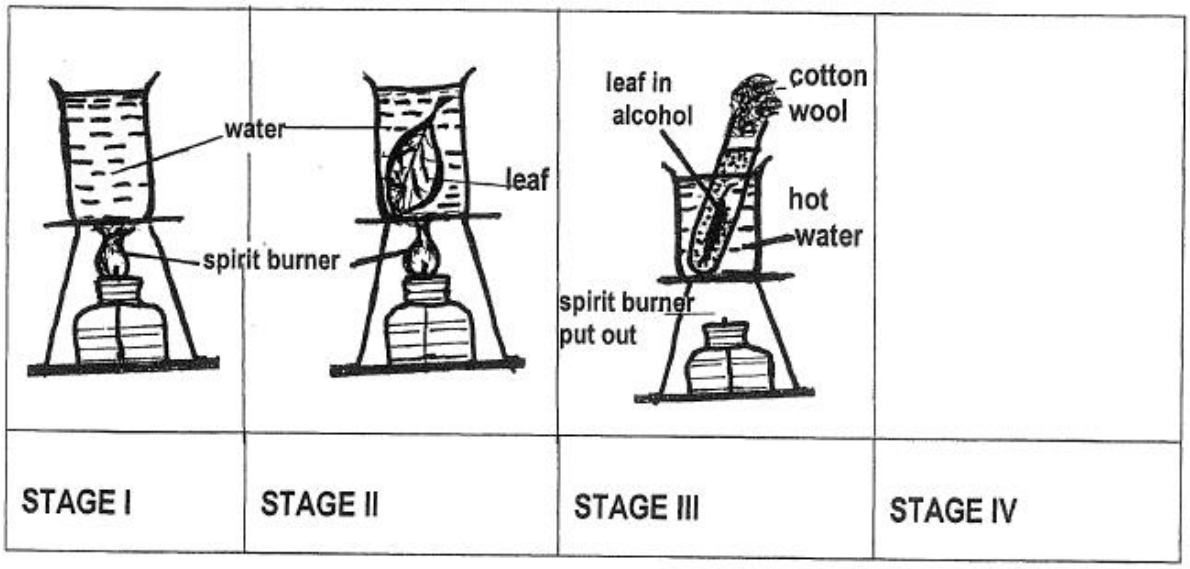
(d) After five hours, a leaf from each plant was taken and tested for starch.

(i) State whether starch will be present or absent in experiment D. [2012.P2.Q5d(i)]

(ii) What would be the colour changes in leaves in experiments E and F? [2012.P2.Q5d(ii)]

4 QUESTION 6 – PLANTS

Study the diagrams showing three stages in an experiment on a leaf. [2011.P2.6]



(a) State the substance in the leaf that is being tested in the experiment. [2011.P2.6a]

(b) Name the chemical solution that is used to identify the substance you have mentioned in (a) above. [2011.P2.6b]

(c) Explain why the leaf was treated as shown in;

(i) Stage II [2011.P2.6c(i)]

(ii) Stage III [2011.P2.6c(ii)]

(d) Give a reason why the flame is put out in Stage III. [2011.P2.6d]

(e) Suggest what would happen in stage IV not shown in the diagrams. [2011.P2.6e]

20.1 TRANSPIRATION – PAPER 1

1 A student treated four identical leaves as follows:

Leaf 1: Petroleum jelly was applied to both lower and upper surfaces.

Leaf 2: Petroleum jelly was applied only to lower surface.

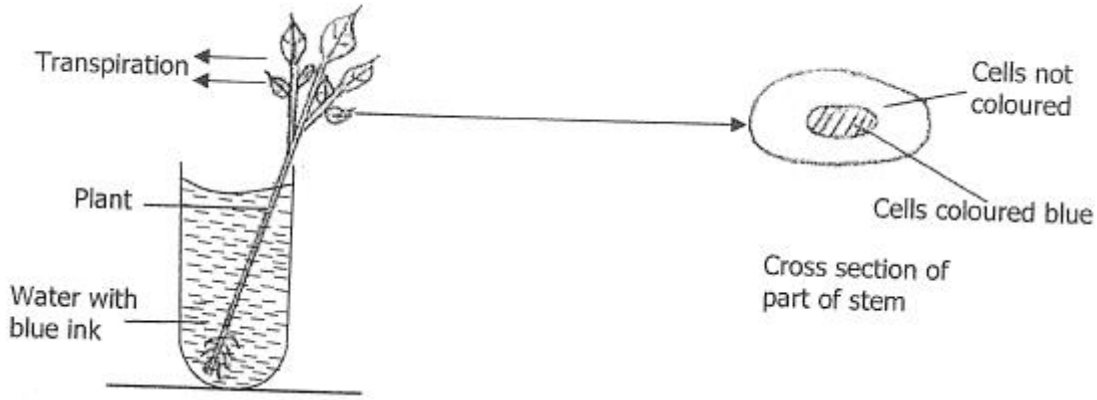
Leaf 3: Petroleum jelly was applied only to upper surface.

Leaf 4: was left untreated.

Which leaf stays fresh for a long time? [2016.P1.Q28]

A 1 B 2 C 3 D 4

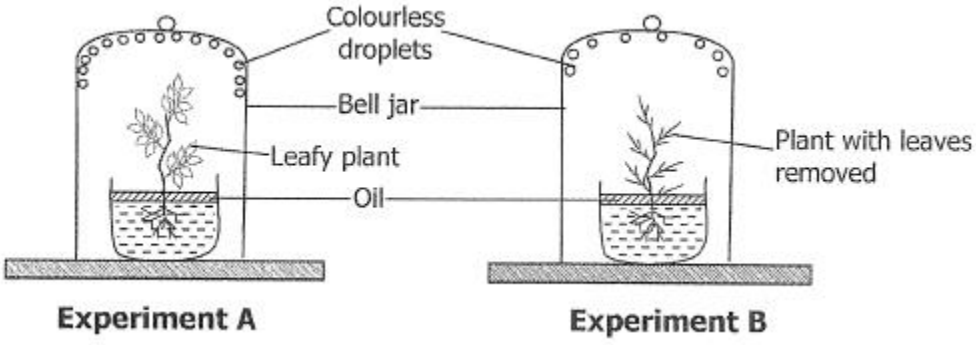
2 Study the diagram below.



The cells which were coloured blue are called ..... [2015.P1.Q9]

A xylem B phloem C palisade cells D guard cells

3 Two experiments were set up to demonstrate transpiration.



What conclusion can be made from the observation in the experiment? Transpiration occurs ..... [2015.P1.Q28]

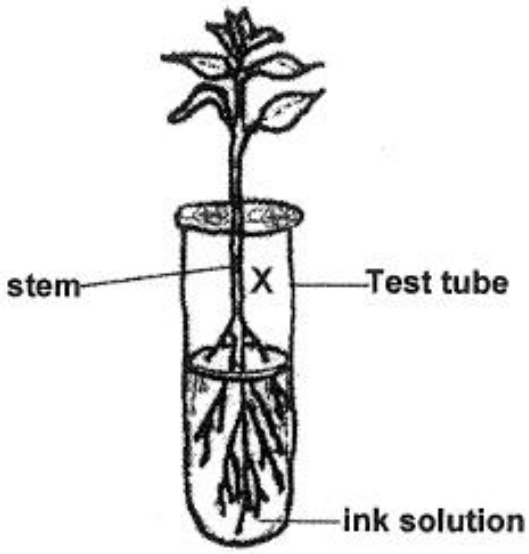
A only in the leaves.

B only in stems.

C mainly in the leaves.

D mainly in stems.

4 Study the diagram below.

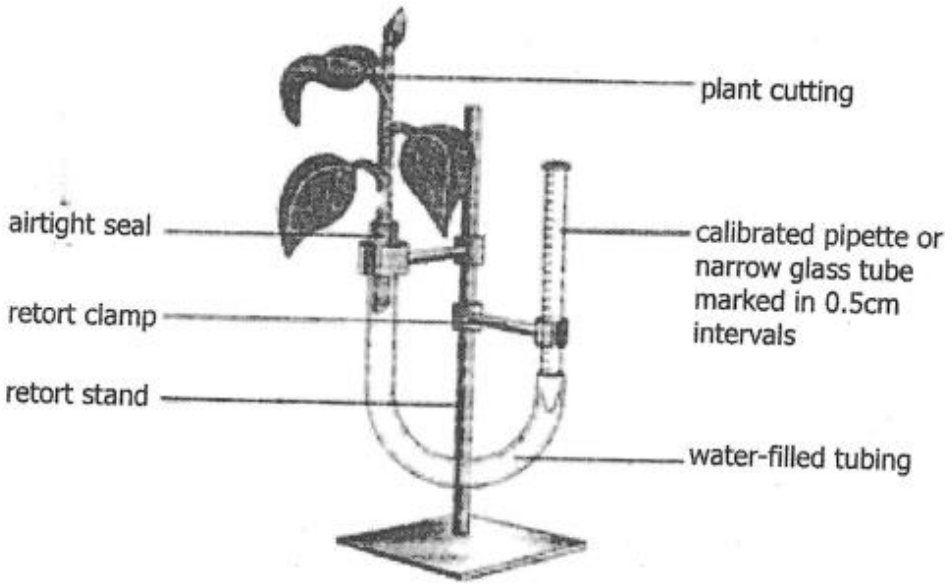


After some days, the plant stem was cut across at point X. Which of the following cells will be coloured with ink? [2012.P1.Q7]

A	Palisade	B	Phloem
C	Xylem	D	Spongy

20.2 TRANSPIRATION – PAPER 2

1 PLANTS AND ANIMALS The diagram below shows a simple potometer. [2016.P2.Q8]



(a) Explain the importance of the air tight seal in the potometer. [2016.P2.Q8a]

(b) State the function of the potometer. [2016.P2.Q8b]

(c) What substance would you use to prove that water is being lost by the plant in the potometer? [2016.P2.Q8c]

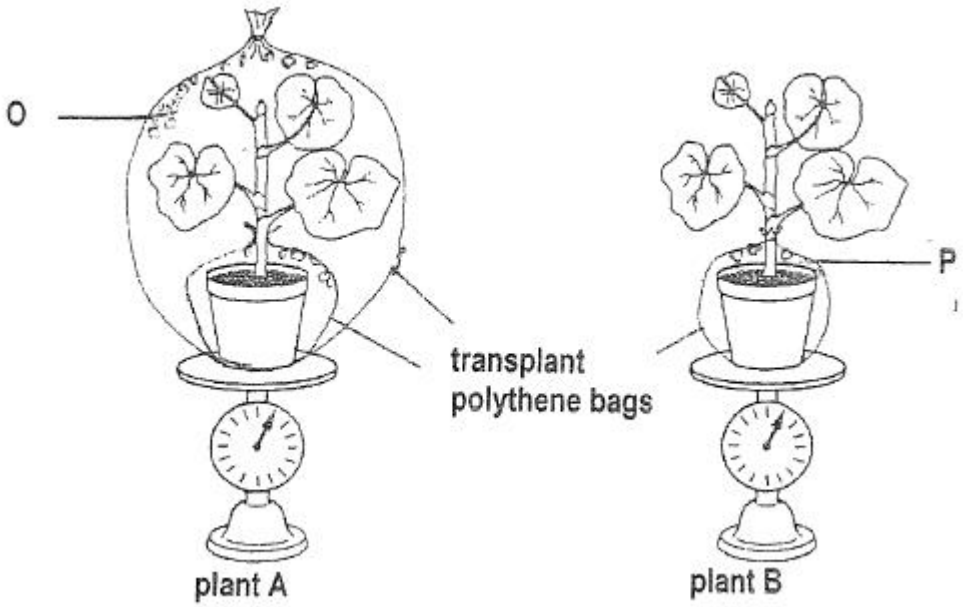
(d) Explain the colour changes the substance mentioned in question (c) above would undergo when testing the water from the potometer. [2016.P2.Q8d]

(e) Why is transpiration necessary in plants? [2016.P2.Q8e]

(f) Suggest one way by which plants reduce water loss through transpiration [2016.P2.Q8f]

2 PLANTS [2010.P2.Q6]

Study the diagram showing experiments set up to find out about the changes in the mass of two plants growing in pots filled with black soil over a long period of time.



(a) Which plant will have loss of mass. [2010.P2.Q6a]

(b) Name the process in the plant which causes the loss of mass. [2010.P2.Q6b]

(c) What is the identity test for the liquid droplets labelled **0**. [2010.P2.Q6c]

(d) Suggest a reason for covering the pots with transparent polythene bag. [2010.P2.Q6d]

(e) State the sources of liquid droplets **0** and **P**. [2010.P2.Q6e]

21.1 CHEMICAL REACTION – PAPER 1

1 The law that states that mass of a substance before a chemical reaction is equal to the total mass of the substances that are produced is called the law of... [2016.P1.Q29]

A conservation of matter. B conservation of energy.
C conversion of matter D conversion of energy?

2 Sodium and hydrogen chloride react as shown in the chemical equation below.

Sodium + hydrogen chloride → sodium chloride + hydrogen

The type of chemical reaction shown in the equation above is called ... [2016.P1.Q30]

A Synthesis B Single replacement

C Double decomposition D Double replacement

3 Study the word equation below.

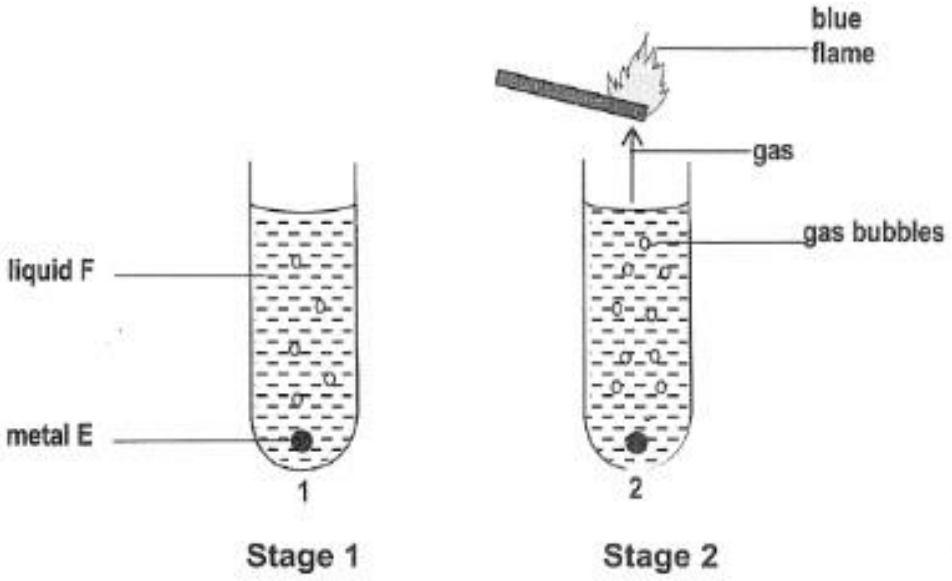
Copper + oxygen --> copper oxide

This is a good example of a chemical reaction called ..... [2015.P1.Q29]

A decomposition. B synthesis.

C double replacement. D single.

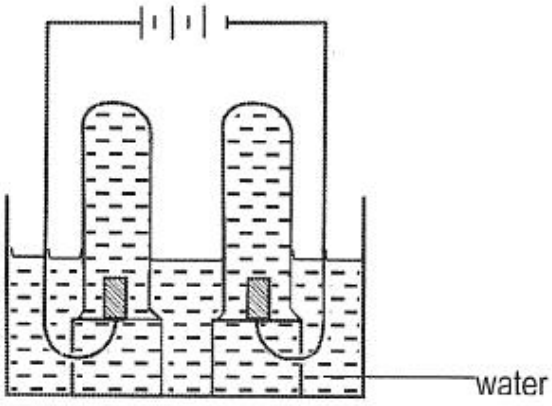
4 The diagrams below show stages in an experiment on reaction of a metal E with a liquid F.



What could be liquid F and the gas produced? [2013.P1.Q1]

	Liquid F	**Gas**
A	hydrochloric acid	carbon dioxide
B	hydrochloric acid	oxygen
C	carbon dioxide	hydrogen
D	water	hydrogen

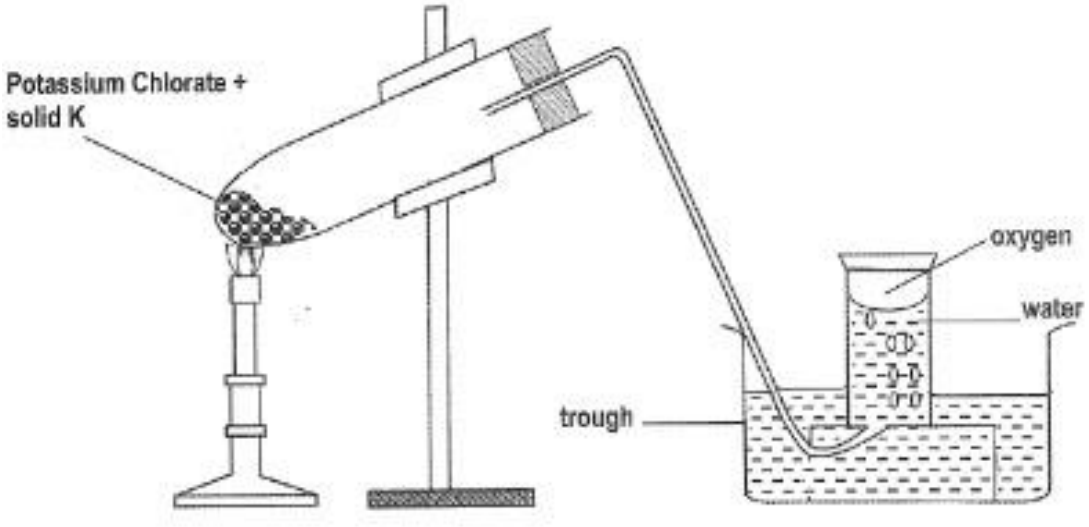
5 The diagram shows the set up for electrolysis of water.



Which substance must be added to the water to promote electrolysis? [2013.P1.Q6]

A Sulphuric acid
B Sodium hydroxide
C Calcium carbonate
D Hydrogen peroxide

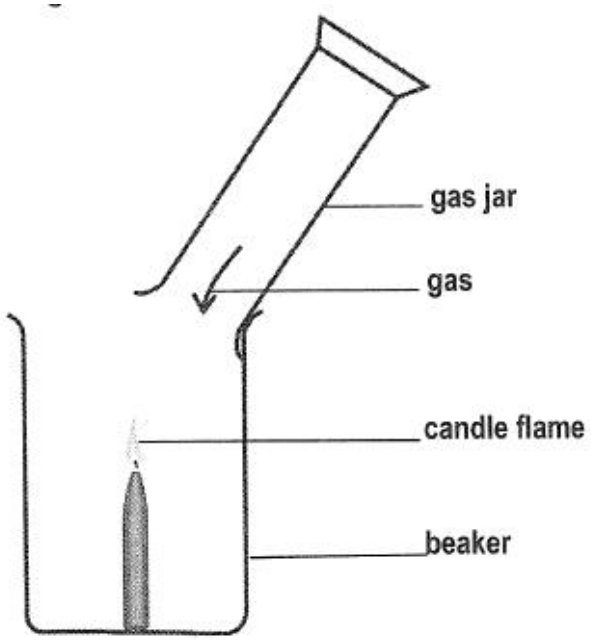
6 The diagram shows preparation of oxygen in the laboratory.



Name solid K. [2013.P1.Q15]

A Magnesium dioxide.
B Manganese dioxide.
C Potassium dioxide.
D Potassium chloride.

7 The diagram below shows an experiment carried out by a pupil to investigate some properties of a certain gas. It was observed that the candle was put off. [2013.P1.Q16]



The property of the gas being demonstrated in the experiment is that the gas ..... [2013.P1.Q16]

A supports burning. B burns in air.
C burns with a pop sound. D is denser than air.

8 Which of the following gases changes the cobalt chloride paper pink? [2013.P1.Q17]

A Carbon dioxide B Hydrogen C Water vapour D Nitrogen

9 The following is an equation of a reaction.

Magnesium + water → Magnesium hydroxide + Hydrogen

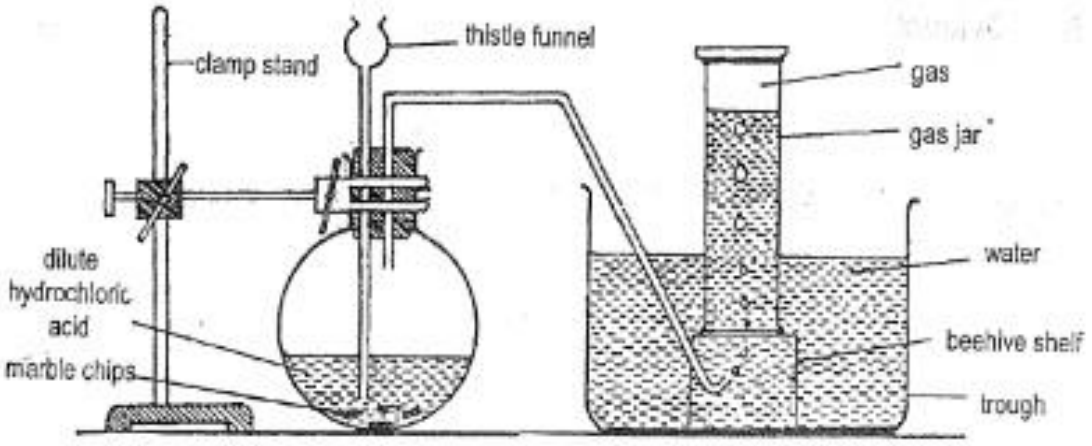
Which of the following will increase the rate of the above reaction? [2012.P1.Q2]

A Using cold magnesium instead of hot magnesium.
B Using steam instead of cold water.
C Using magnesium ribbon instead of powder.
D Using distilled water instead of tap water.

10 Which metal would react with cold water very slowly to produce a hydroxide of that metal and hydrogen gas? [2011.P1.Q2]

A Potassium B Magnesium C Calcium D Iron

11 Study the diagram below showing preparation and collection of a gas.



Suggest three substances that will be produced when the reaction in the flask is completed. [2011.P1.Q17]

A Potassium chlorate + oxygen + water
B Water + calcium chloride + oxygen
C Dilute hydrochloric acid + water + carbon dioxide
D Calcium chloride + water + carbon dioxide

21.2 CHEMICAL REACTION – PAPER 2

1 MATERIALS AND ENERGY

The table below shows the chemical reactions and their products. [2016.P2.Q10]

S/N	CHEMICAL REACTION	PRODUCTS
1	Copper heated with oxygen	Copper (II) oxide
2	Heating calcium carbonate	Calcium oxide and carbon dioxide
3	Calcium reacts with hydrochloric acid	Calcium chloride and hydrogen gas
4	Electrolysis of acidified water	Hydrogen and oxygen gases

(a) Define a chemical reaction. [2016.P2.Q10a]

(b) Identify chemical reactions which can be classified as;

(i) Single replacement. [2016.P2.Q10b(i)]

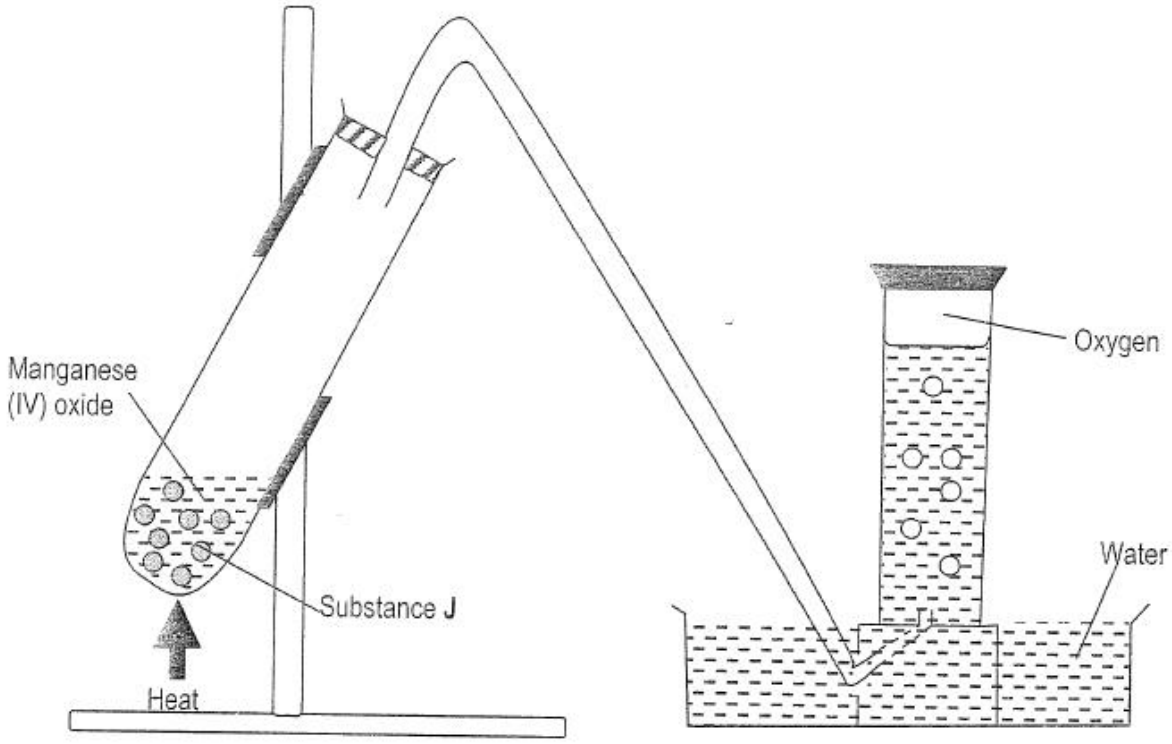
(ii) Decompostion. [2016.P2.Q10b(ii)]

(iii) Synthesis. [2016.P2.Q10b(iii)]

(c) Suggest two endothermic reactions from the table. [2016.P2.Q10c]

2 GASES [2013.P2.Q5]

Study the diagram below that shows preparation and collection of oxygen gas.



(a) Name substance J. [2013.P2.Q5a]

(b) Write a word equation for the reaction in the experiment. [2013.P2.Q5b]

(c) What is the method of collecting the gas used in the experiment? [2013.P2.Q5c]

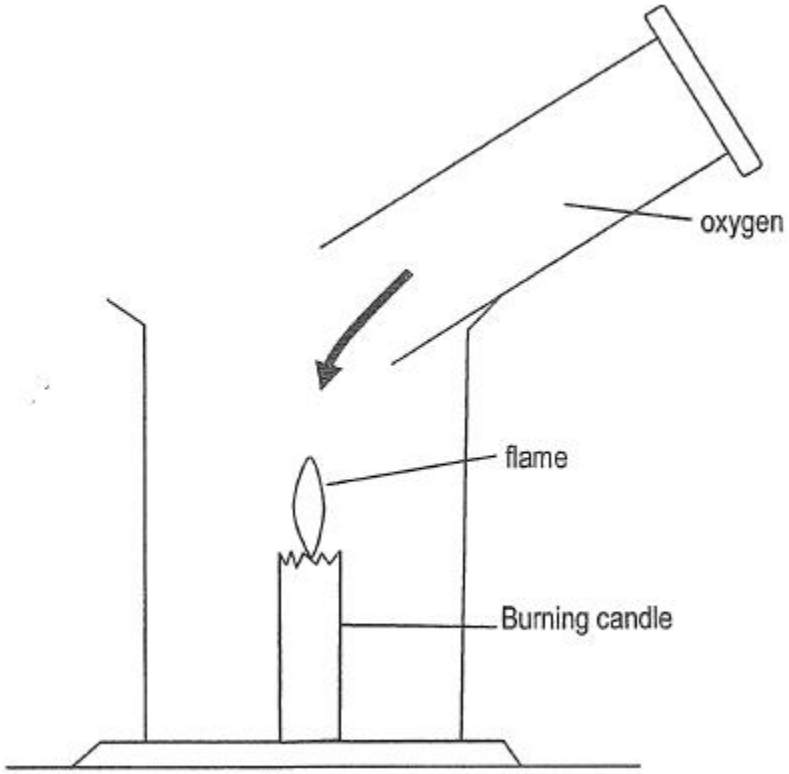
(d) What is the function of the manganese (IV) oxide in the experiment? [2013.P2.Q5d]

(e) Explain why the first gas collected in the gas jar was not pure oxygen. [2013.P2.Q5e]

(f) State one industrial use of oxygen gas. [2013.P2.Q5f]

3 **GASES** [2012.P2.Q3]

The diagram below shows an experiment with oxygen gas.



(a) What will be observed when the oxygen gas is poured on to the burning candle? [2012.P2.Q3a]

(b) When the experiment above is repeated with a different gas and the flame is put off, what would be the gas? [2012.P2.Q3b]

(c) What property of oxygen makes it possible to be poured downwards? [2012.P2.Q3c]

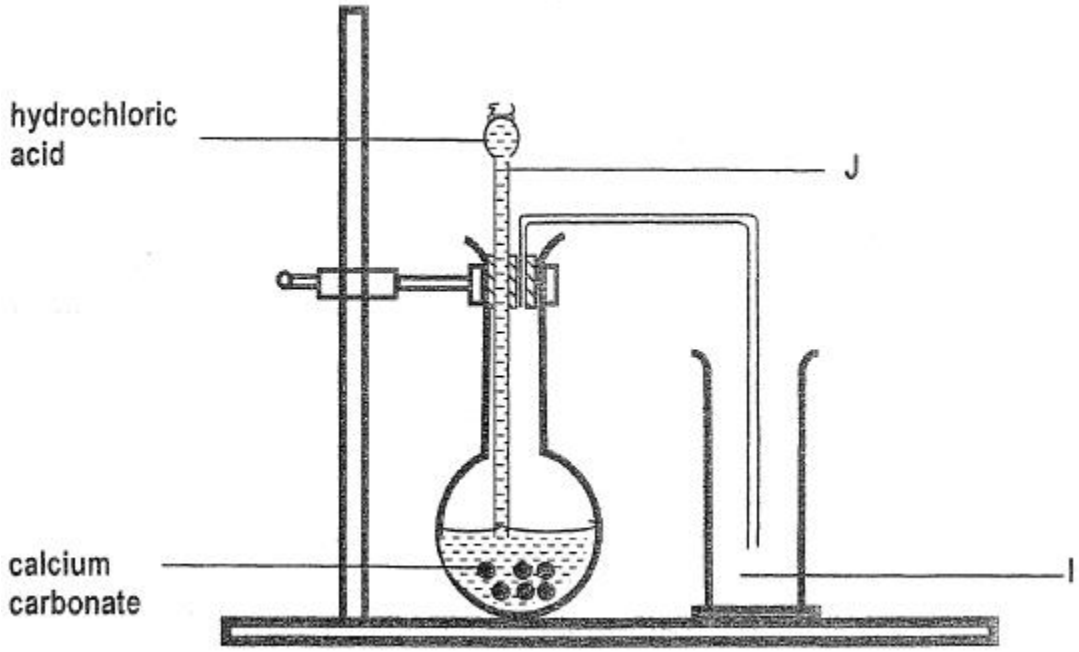
(d) Name one natural process that releases oxygen into the atmosphere. [2012.P2.Q3d]

(e) What catalyst is used when oxygen is prepared from hydrogen peroxide? [2012.P2.Q3e]

(f) Give one use of oxygen in manufacturing industries. [2012.P2.Q3f]

4 **GASES** [2010.P2.Q4]

Study the diagram showing an experiment on the laboratory preparation and collection of gas I.



(a) What is the name of the apparatus labelled **J**. [2010.P2.Q4a]

(b) Name gas I. [2010.P2.Q4b]

(c) What is the identity test for gas **I**. [2010.P2.Q4c]

(d) Write a complete word equation on the reaction in the flask. [2010.P2.Q4d]

(e) Suggest the property of the gas **I** which makes it possible for it to be collected as shown in the diagram. [2010.P2.Q4e]

(f) State one use of the gas **I**. [2010.P2.Q4f]

22.1 LIGHT AND ITS NATURE – PAPER 1

1 An object is placed 6cm from a bi-convex lens of focal length 2cm. Which statement below shows the nature of the image formed? [2016.P1.Q31]

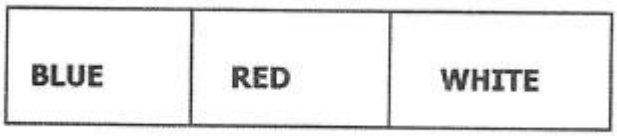
A Magnified, Real and Upright.

B Virtual, Upright and Diminished.

C Real, Upside down and Diminished.

D Virtual, Upside down and Same size as object.

2 The diagram below shows a flag with three different colours.

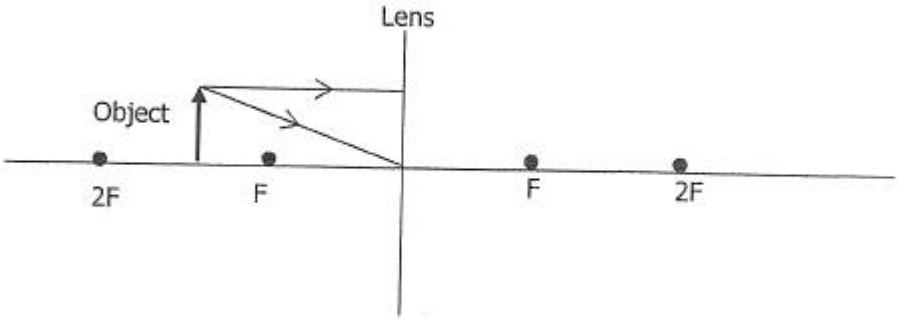


If you looked at the flag through a piece of red glass, which of these colours would you see? [2016.P1.Q32]

A Blue Red White B Red Red Black

C Black Red Red D Red Red Red

3 Study the diagram below carefully.

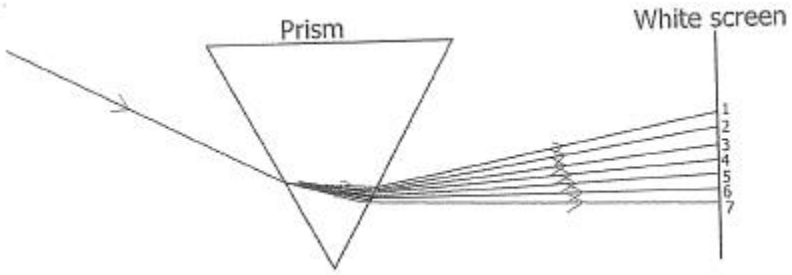


When the diagram is completed, the image formed will be ..... [2015.P1.Q31]

A diminished, real and upside down. B upright, real and diminished.

C magnified, real and upside down. D magnified, virtual and upright.

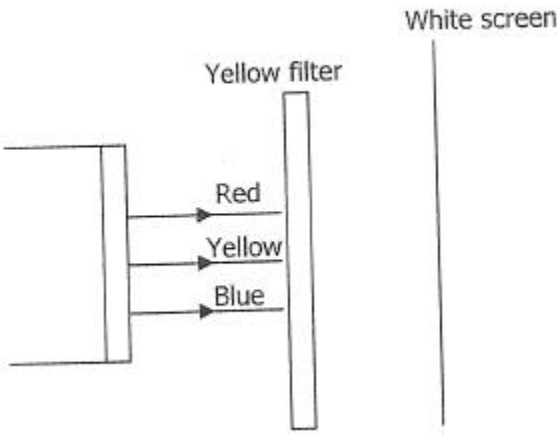
4 The diagram below shows the dispersion of white light by a glass prism.



Which colours will appear on numbered parts 4 and 5? [2015.P1.Q32]

	4	5
A	Green	yellow
B	Orange	green
C	Blue	yellow
D	yellow	green

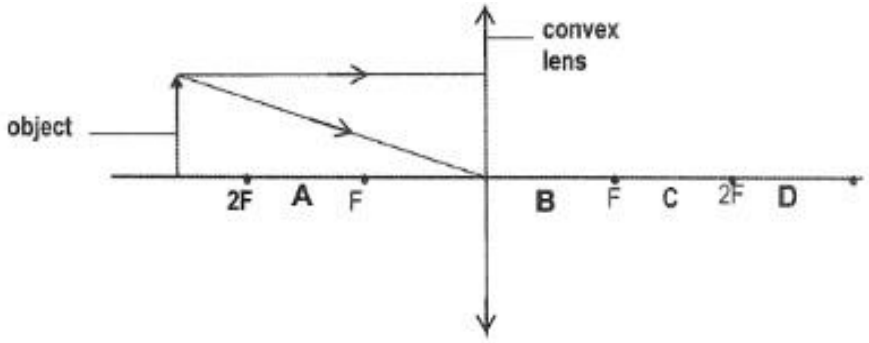
5 The diagram below shows red, yellow and green light shone on a yellow filter.



What will be seen on the screen? [2015.P1.Q33]

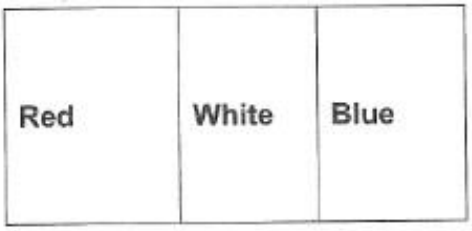
A Red only B Yellow only C Red and green D Red and yellow

6 Study the ray diagram below very carefully.



When the construction of the ray diagram is completed, where will the image be formed? [2013.P1.Q23]

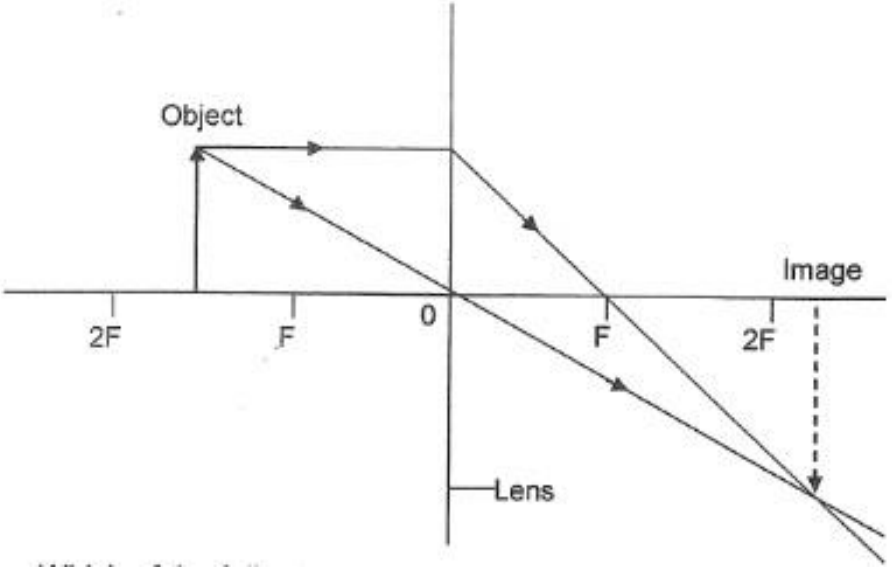
7 Below is a flag of a country. [2013.P1.Q24]



How will the colour of the flag be when viewed in red light?

A	Red	Red	Black
B	Black	White	Blue
C	Red	Red	Blue
D	Red	White	Blue

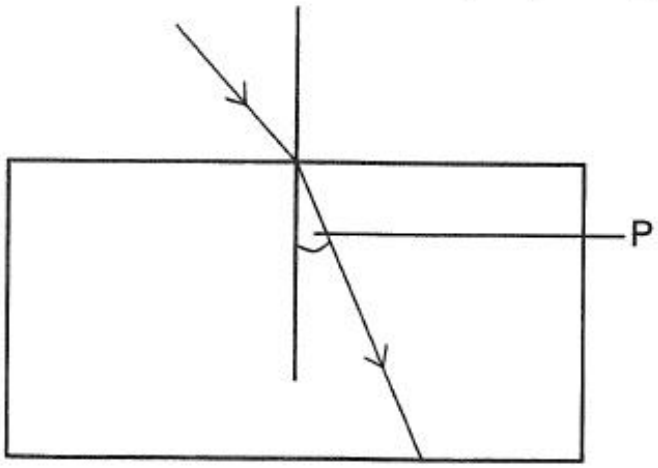
8 Below is a ray diagram.



Which of the following instruments produces the image like the ray diagram above? [2012.P1.Q22]

A Camera B Microscope C Film projector D Telescope

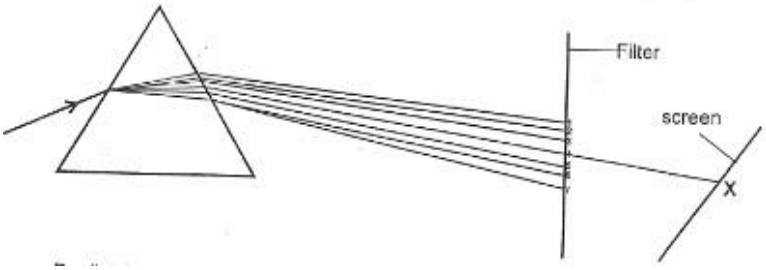
9 The diagram below shows light passing through a rectangular prism.



The part labelled P is called ..... [2012.P1.Q23]

A normal. B angle of refraction.
C angle of incidence. D incident ray.

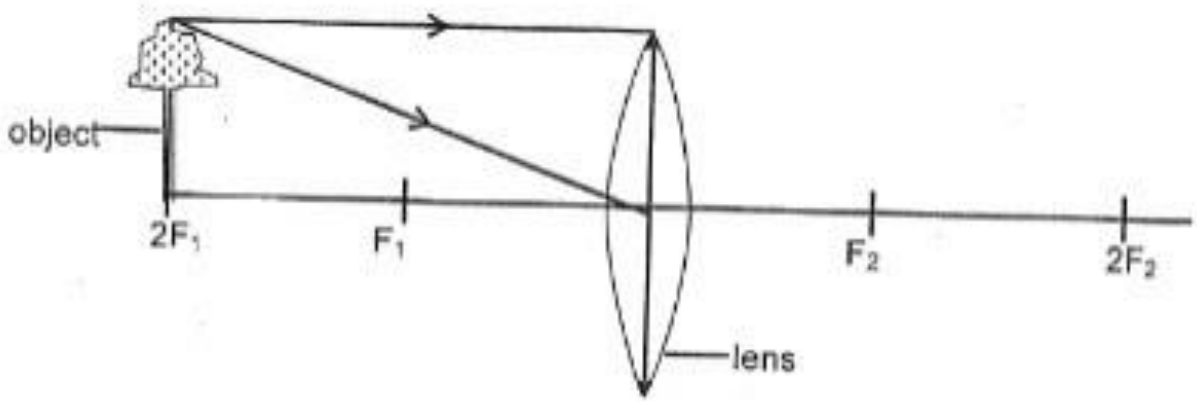
10 Study the diagram below on an experiment on light.



Predict the colour of the filter, if the colour appearing at X is that of the refracted ray numbered 4. [2011.P1.Q23]

A Blue B Green C Indigo D Orange

11 The diagram below shows an incomplete construction of a ray diagram. The lens used in the experiment is a convex (converging) lens. [2011.P1.Q24]



If the object was placed as shown in the diagram, suggest the characteristics of the image which would be formed.

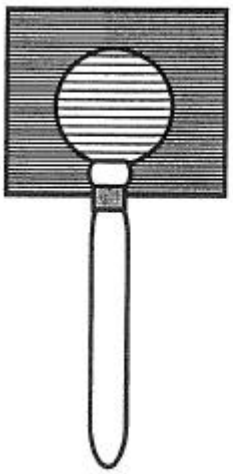
A Real, magnified and upright
B Real, same size and upside down
C Virtual, diminished and upright
D Virtual, same size and upright

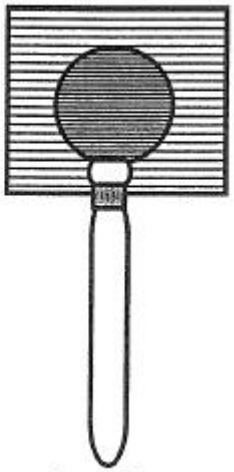
22.2 LIGHT AND ITS NATURE – PAPER 2

1 **LIGHT** [2011.P2.Q5]

A pupil was carrying out experiments to show effects of different types of lenses.

The diagrams below show the results.





(a) State the name of lens;

(i) G, [2011.P2.Q5a(i)]

(ii) H. [2011.P2.Q5a(ii)]

(b) State one visible difference between the lines in lens;

(i) G and the ones on the paper it is on. [2011.P2.Q5b(i)]

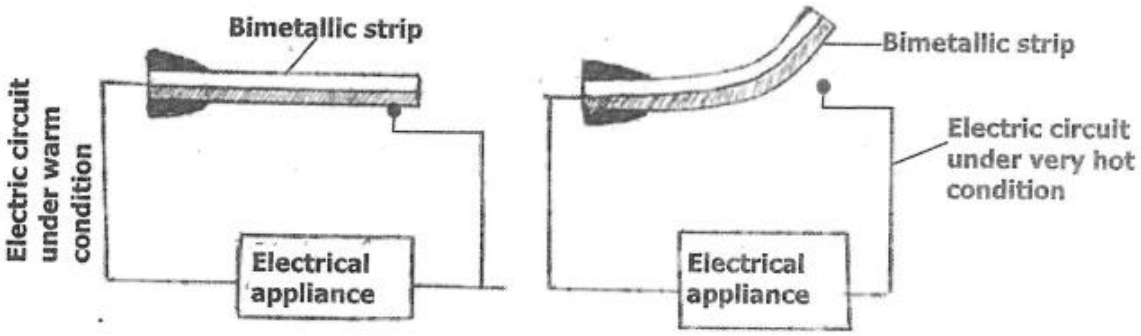
(ii) H and the ones on the paper it is on. [2011.P2.Q5b(ii)]

(c) State one property of the image formed by lens H shown in the diagram. [2011.P2.Q5c]

(d) What effect would you observe if sunlight is focused on the paper by lens G? [2011.P2.Q5d]

23.1 ELECTRIC CURRENT AND VOLTAGE IN CIRCUIT – PAPER 1

1 Study the diagrams below showing the bimetallic strip under warm condition and under very hot condition.

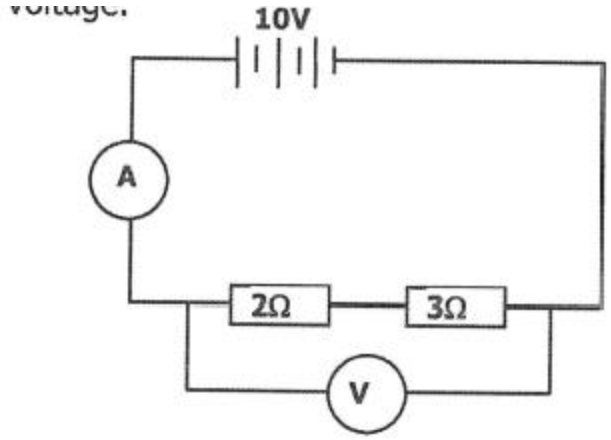


In which electrical appliance would such a type of circuit be used? [2016.P1.Q17]

A Computer B Electric fan

C Electric pressing iron D Thermos flask?

2 Study the diagram showing an experiment on the relationship between current and voltage.



Calculate the current reading on the ammeter. The reading will be ... [2016.P1.Q33]

A 2A

B 5A

C 10A

D 50A

3 Which of the following appliances used in the local environment converts electric current into both light and sound energy? [2016.P1.Q34]

A Heater

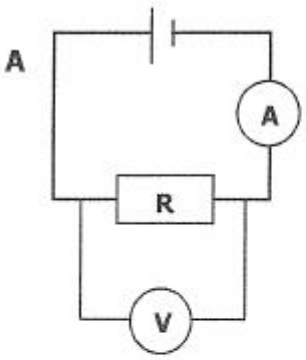
B Cellphones

C Electric fans

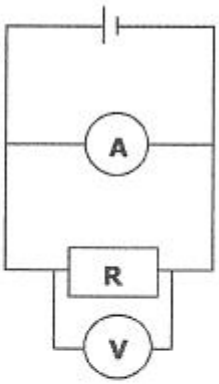
D Energy saving bulbs

4 Which of the following diagrams show the correct use of a voltmeter and an ammeter to measure voltage and current respectively through resistor R? [2015.P1.Q34]

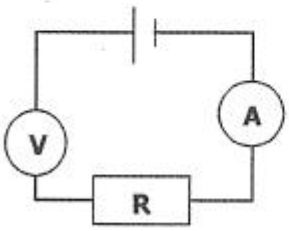
A



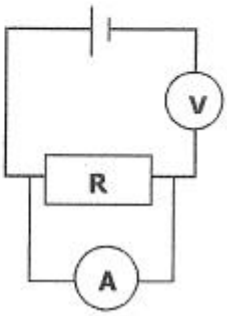
B



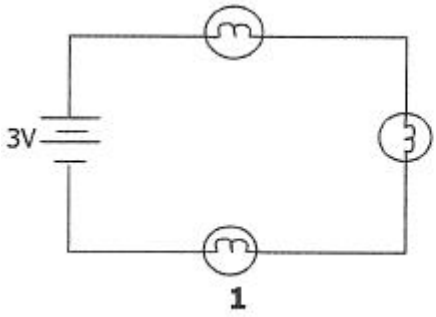
C



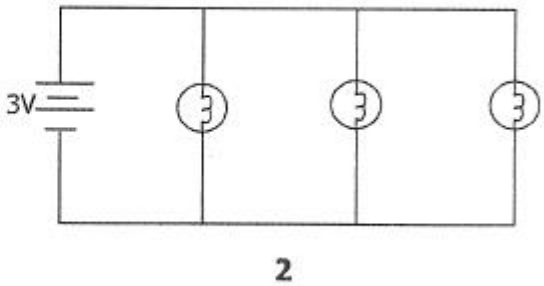
D



5 The diagrams below show circuits with similar cells and bulbs of the same resistance.



1



2

Which statement is true about the bulbs? The bulbs in ... [2015.P1.Q35]

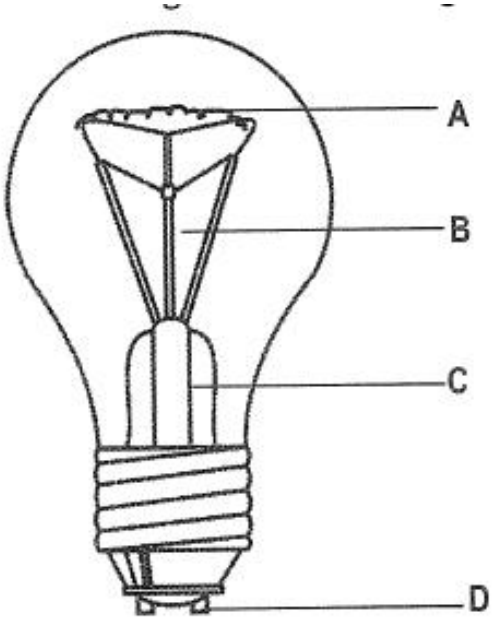
A both circuits will light with the same brightness.

B circuit 1 will light brighter than in 2.

C 2 will light brighter than in 1.

D both circuits will not light up.

6 Which of the labelled parts in the diagram has the highest resistance? [2013.P1.Q12]



7 Which of the following is true about an electromagnet and a permanent magnet? [2013.P1.Q13]

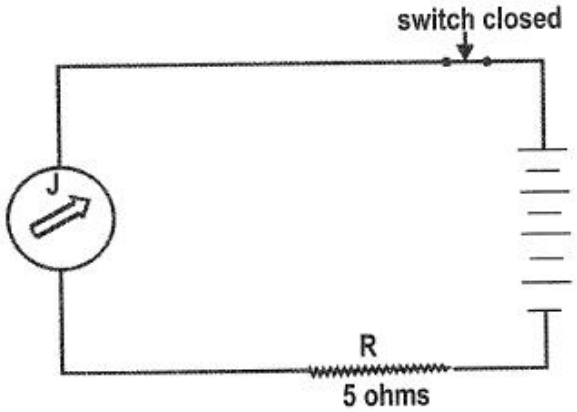
A Electromagnets are weaker than permanent magnets.

B Electromagnets can be made using other magnets while permanent magnets can be made using electricity.

C Permanent magnets can be made using other magnets while electromagnets can be made using electricity.

D Electromagnets cannot be switched off but permanent magnets can be switched off.

8 A magnetic campus was placed along an electric circuit as shown below.

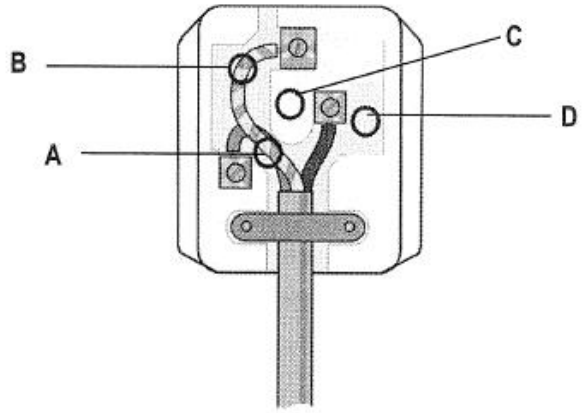


The pointer was seen to deflect to the right from the resting position point '**J**'. [2013.P1.Q14]
What could have been responsible for this deflection?

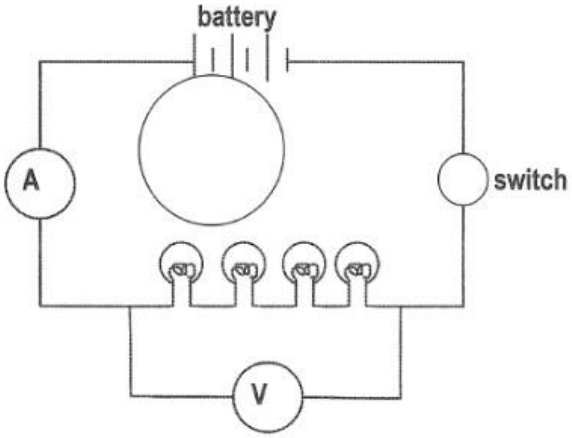
A Electromotive force B Magnetic effect

C Resistance of R D Potential difference

9 Which letter shows the correct position for a fuse in the three pin plug? [2013.P1.Q34]



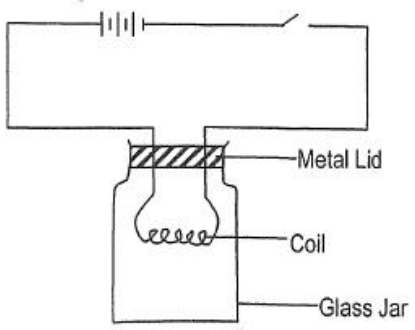
10 Study the circuit diagram below showing cells each of 2.0 V connected to bulbs each of resistance 3 ohms in series. [2013.P1.Q35]



What will be the total voltage and resistance in the circuit?

	Voltage (Volts)	**Resistance (ohms)**
A	12	0.5
B	6	12
C	0.5	12
D	6	0.5

11 Study the diagram below



What is wrong with the simple electric bulb above? [2012.P1.Q14]

A Type of lid B Number of cells C Position of coil D Position of switch

12 Which of the following correctly shows the appliances in which electric motors are used in everyday life? [2012.P1.Q15]

A Water pump, record player, telephone receiver.
B Record player, water pump, electric sewing machine.
C Refrigerator, telephone receiver, electric kettle.
D Electric sewing machine, electric kettle, refrigerator.

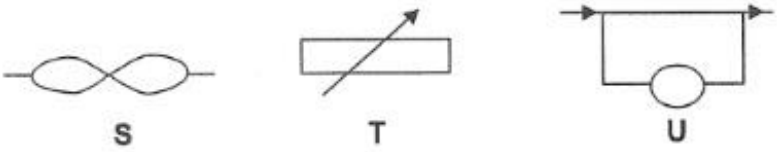
13 The table below shows incomplete results of two electric circuit settings on an experiment on electricity.

CURRENT	RESISTANCE	VOLTAGE	
	3 ohms	12 volts	1st setting
5 amps	2 ohms		2nd setting

What are the correct readings for the missing values of current and voltage? [2012.P1.Q30]

	Current	**Voltage**
A	36 amps	2.5 volts
B	15 amps	4 volts
C	4 amps	15 volts
D	4 amps	10 volts

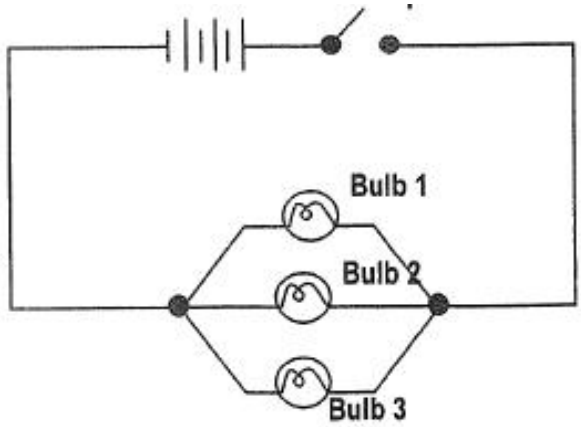
14 The following are symbols used in electricity. [2012.P1.Q31]



Which of the following correctly identifies **S**, **T**, **U**?

	S	T	U
A	Switch	Battery	Ammeter
B	Fuse	Rheostat	Voltmeter
C	Ammeter	Switch	Fuse
D	Battery	Voltmeter	Rheostat

15 The diagram below represents a circuit diagram showing bulbs of equal resistance connected in parallel.



What conclusion can you make about the current passing through the three bulbs? A
More in bulb 2 than 1 and 3. [2012.P1.Q32]

B More in bulb 3 and 1 than 2.

C Equal in bulb 1 and 3 but not in 2.

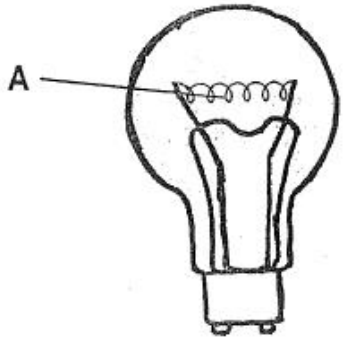
D Equal in all the three bulbs.

16 The name of the electric effect which makes a plotting compass needle to deflect is the ..... [2011.P1.Q14]

A heating effect. B magnetic effect.

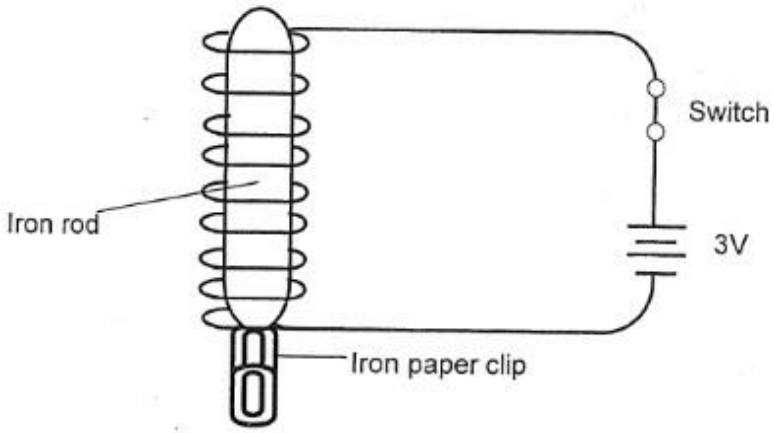
C motor effect. D electromagnetic effect.

17 Which of the following properties makes **A** to be used in the electric bulb? [2011.P1.Q15]



	RESISTANCE TO ELECTRICITY	**MELTING POINT OF A**
A	Very high	Very low
B	Very low	Very high
C	Very high	Very high
D	Very low	Very low

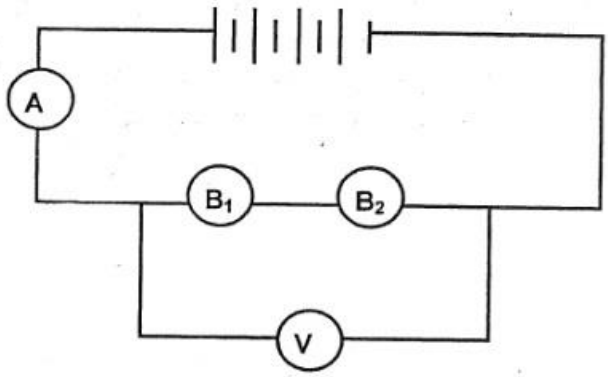
18 Study the diagram showing the effect of electricity on an iron rod.



A pupil decided to add two more paper clips, all the paper clips dropped. Describe the adjustment that could be done in order to hold four or even more paperclips. [2011.P1.Q16]

A Increase both the voltage and the number of turns in the coil.
B Decrease the voltage and increase the number of turns in the coil.
C Increase the voltage and decrease the number of turns in the coil.
D Decrease the voltage but maintain the number of turns in the coil.

19 Study the circuit diagram below showing two bulbs B1 and B2 of equal resistance connected to cells, each of 1.5V.



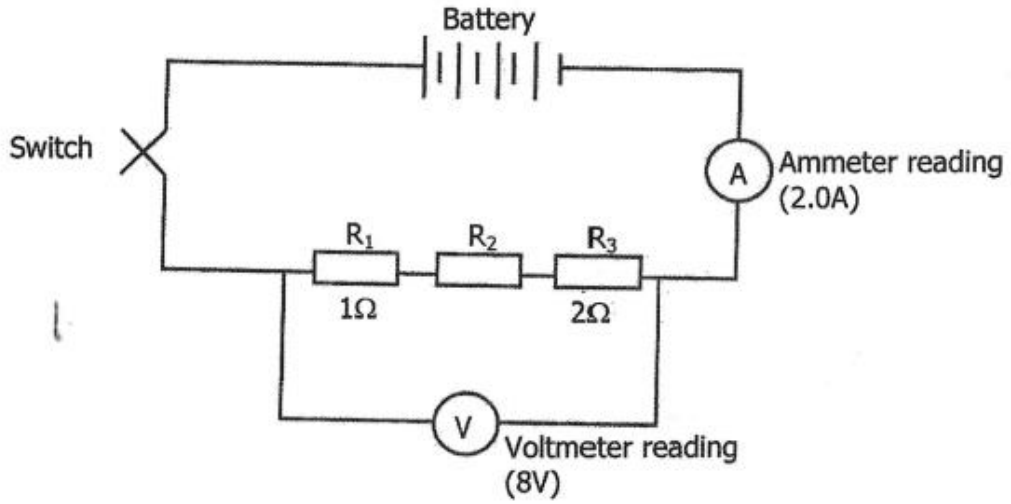
If the current through the ammeter is 2 amps, calculate the resistance of bulb labelled B1. The resistance in ohms will be ..... [2011.P1.Q31]

A 6 B 3 C 2 D 1.5

23.2 ELECTRIC CURRENT AND VOLTAGE IN CIRCUIT – PAPER 2

1 ELECTRICITY

Carefully study the electric circuit diagram below. [2016.P2.Q9]



(a) How many cells are in the battery? [2016.P2.Q9a]

(b) Calculate the resistance of resistor R2. (Show your working) [2016.P2.Q9b]

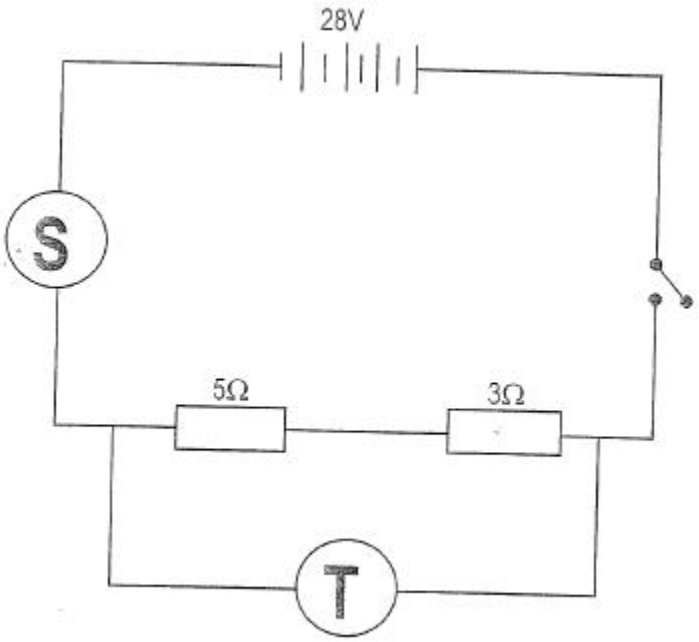
(c) If the cells in the battery are of equal voltage, calculate the voltage of one cell. [2016.P2.Q9c]

(d) If the resistor R3 is removed and the circuit with the remaining resistors (Ri and R2) is completed by closing the switch, what will be the reading of the ammeter? [2016.P2.Q9d]

(e) Calculate the total power with the remaining resistors R1 and R2 in the circuit when R3 is not used. [2016.P2.Q9e]

2 **ELECTRICITY** [2013.P2.Q8]

Study the circuit diagram shown below.



(a) What would be the readings on instruments **S** and **T**? [2013.P2.Q8a]

(b) Explain why the instrument labelled T is connected parallel to resistors. [2013.P2.Q8b]

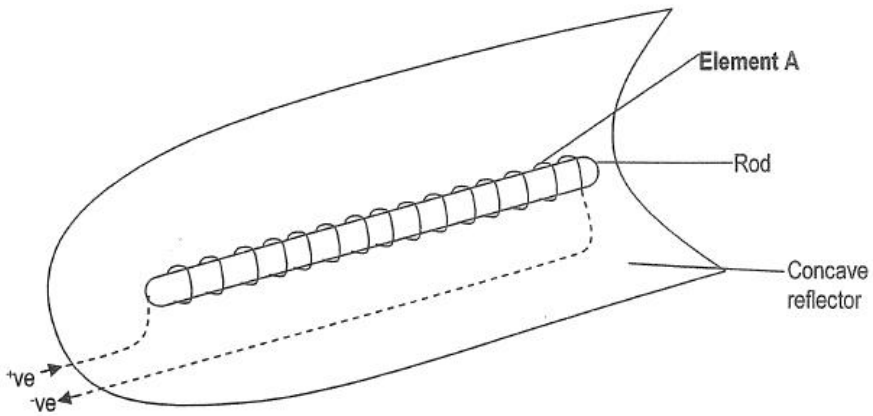
(c) Calculate the power in the circuit. [2013.P2.Q8c]

(d) Calculate the voltage across the 3 ohm resistor. [2013.P2.Q8d]

(e) Explain how a fuse protects an electrical appliance. [2013.P2.Q8e]

3 **ELECTRICITY** [2012.P2.Q2]

The diagram below shows an electric appliance.



(a) What effect of an electric current is being demonstrated in the diagram? [2012.P2.Q2a]

(b) (i) Which metal element could be used as a coil for element A? [2012.P2.Q2b(i)]

(ii) Give a reason for your answer. [2012.P2.Q2b(ii)]

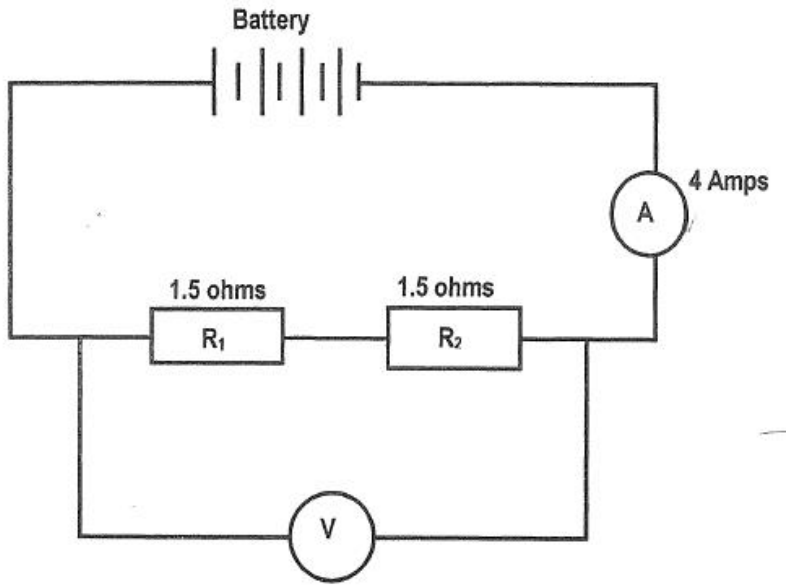
(c) Name one other appliance which uses the electric current effect demonstrated above. [2012.P2.Q2c]

(d) State one difference between element A and a similar element in an electric bulb? [2012.P2.Q2d]

(e) Which effect of electricity is experienced in both the heater and the bulb? [2012.P2.Q2e]

4 ELECTRICITY [2011.P2.Q8]

Study the electric circuit diagram below.



(a) How are the resistors R1 and R2 arranged in the circuit? [2011.P2.Q8a]

(b) Calculate the total resistance offered by the two resistors. [2011.P2.Q8b]

(c) Calculate the;

(i) total voltage across the two resistors. [2011.P2.Q8c(i)]

(ii) voltage for a single cell used in the experiment. [2011.P2.Q8c(ii)]

(d) Calculate the total power of R2 in the circuit. (Show your working). [2011.P2.Q8d]

5 ELECTRICITY [2010.P2.Q8]

The table below shows the results obtained from an experiment on Ohm's Law.

Voltage (V)	3	6	9	12	15
Current (A)	1	2	3	4	5

(a) State the Ohm's Law. [2010.P2.Q8a]

(b) Using the values in the table, plot a graph of voltage against current. [2010.P2.Q8b]

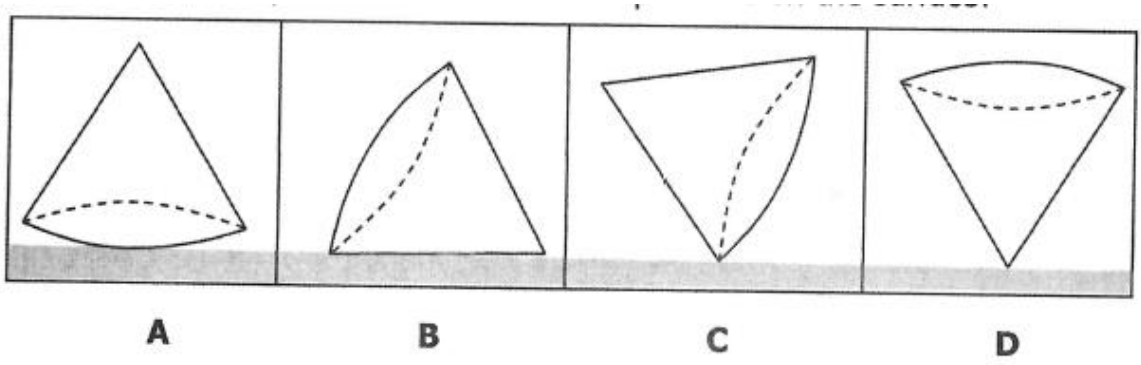
(c) Using the graph you have plotted, determine the current when the voltage is 8V. [2010.P2.Q8c]

(d) Calculate the resistance of the appliance used in the experiment to obtain the results in the table. [2010.P2.Q8d]

(e) Calculate the power when the voltage is 15V during the same experiment. [2010.P2.Q8e]

24.1 PRESSURE – PAPER 1

1 The diagrams below show the same cone placed in different positions on a surface. Which position exerts the most pressure on the surface? [2016.P1.Q35]



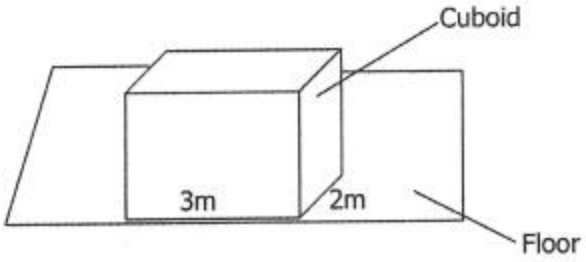
2 What pressure is produced when a force of 600N acts on an object of area 6m squared. [2015.P1.Q36]

A 100 N/m2 B 594 N/m2 C 606 N/m2 D 3600 N/m2

24.2 PRESSURE – PAPER 2

1 PRESSURE

The diagram below shows a cuboid of weight 90N. [2015.P2.Q9]



(a) Define pressure. [2015.P2.Q9a]

(b) If the base of the cuboid measures 3m x 2m as shown above, calculate the;

(i) area of the base of the cuboid, [2015.P2.Q9b(i)]

(ii) pressure exerted by the cuboid on the floor (show your working). [2015.P2.Q9b(ii)]

(c) Explain what would happen to the pressure exerted by the cuboid shown above if the area of its base was increased. [2015.P2.Q9c]

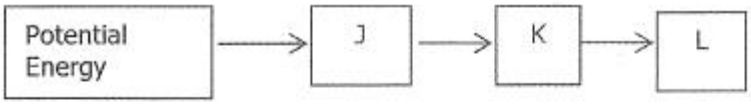
25.1 ENERGY AND ITS CONSERVATION – PAPER 1

1 Which one of the following is not the best way of conserving energy? [2016.P1.Q36]

A Re-use waste to generate energy. B Avoid using energy to avoid wasting.

C Reduce the amount of energy wasted. D Recycle waste into new products.

2 The diagram below shows a flow chart of power production from water. [2015.P1.Q37]



What is represented by **J**, **K** and **L**?

	J	K	L
A	Mechanical energy	Electrical energy	Kinetic energy
B	Electrical Energy	Mechanical energy	Kinetic energy
C	Kinetic Energy	Electrical energy	Mechanical energy
D	Kinetic Energy	Mechanical Energy	Electrical energy

3 Which of the following correctly shows types of electrical energy generation mainly practiced in Zambia? [2013.P1.Q39]

A Hydro-electric and nuclear generation.

B Diesel and hydro electric generation.

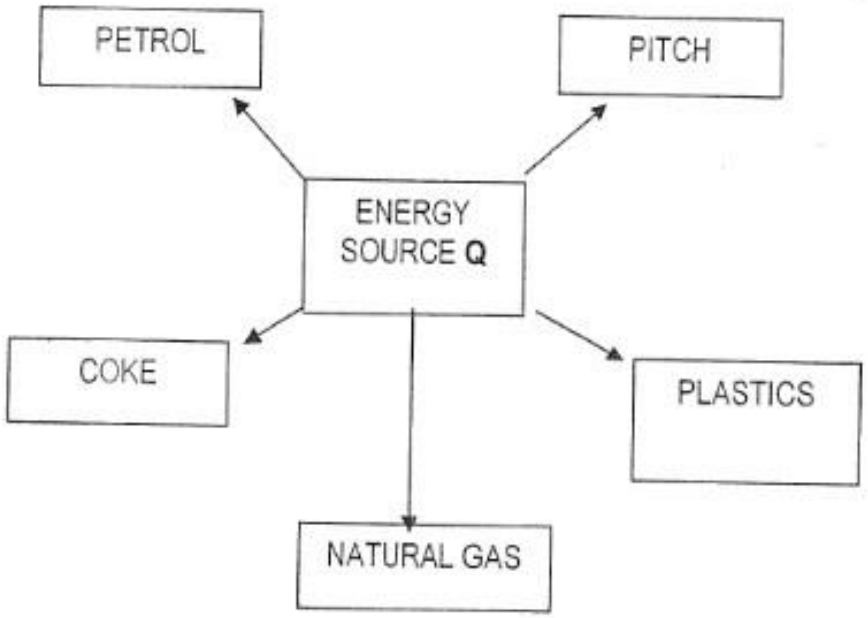
C Nuclear and thermal generation.

D Diesel and wind power generation.

4 Which of the following is an alternative source of energy to a diminishing resource? [2012.P1.Q38]

A Hydro-electricity B Natural gas C Coal D Oil

5 Study the chart below showing one energy source and its useful products.



Which energy source is represented by Q? [2011.P1.Q38]

A Uranium B Coal C Crude oil D Bio fuel

6 The list below consists of different forms of energy involved during energy generation at a hydroelectric power station. [2011.P1.Q40]

1 electric energy
2 kinetic energy
3 potential energy
4 mechanical energy
5 light energy

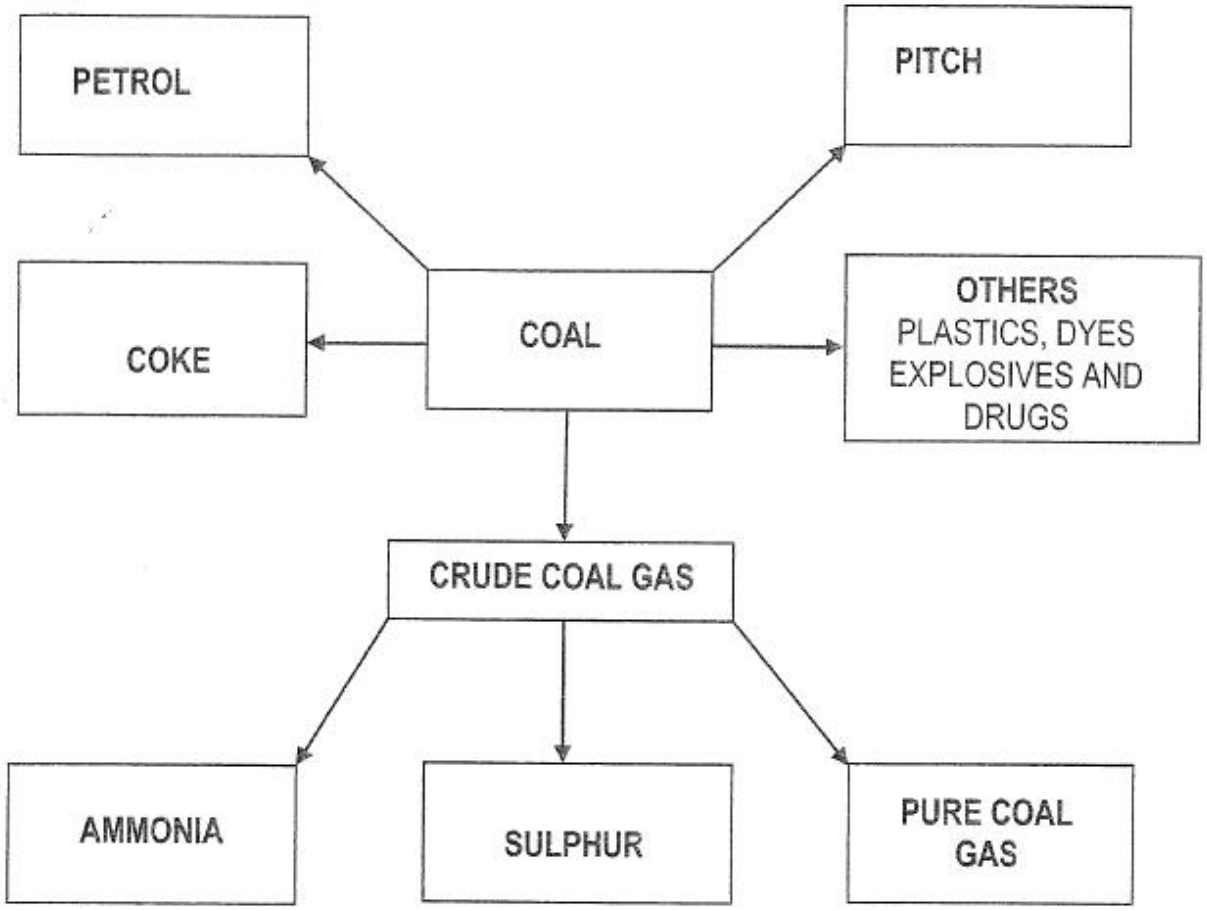
The sequence that best describes energy generation at the hydroelectric power station is .....

A 1, 5, 3, 4, 2. B 3, 2, 4, 1, 5.
C 4, 1, 2, 5, 3. D 2, 3, 5, 1, 4.

25.2 ENERGY AND ITS CONSERVATION – PAPER 2

1 ECOLOGY [2013.P2.Q10]

Study the chart below that shows different products obtained from coal as a resource.



(a) What kind of resource is coal? [2013.P2.Q10a]

(b) Name the process of obtaining products from coal shown in the chart. [2013.P2.Q10b]

(c) In the chart identify the product that contribute to:

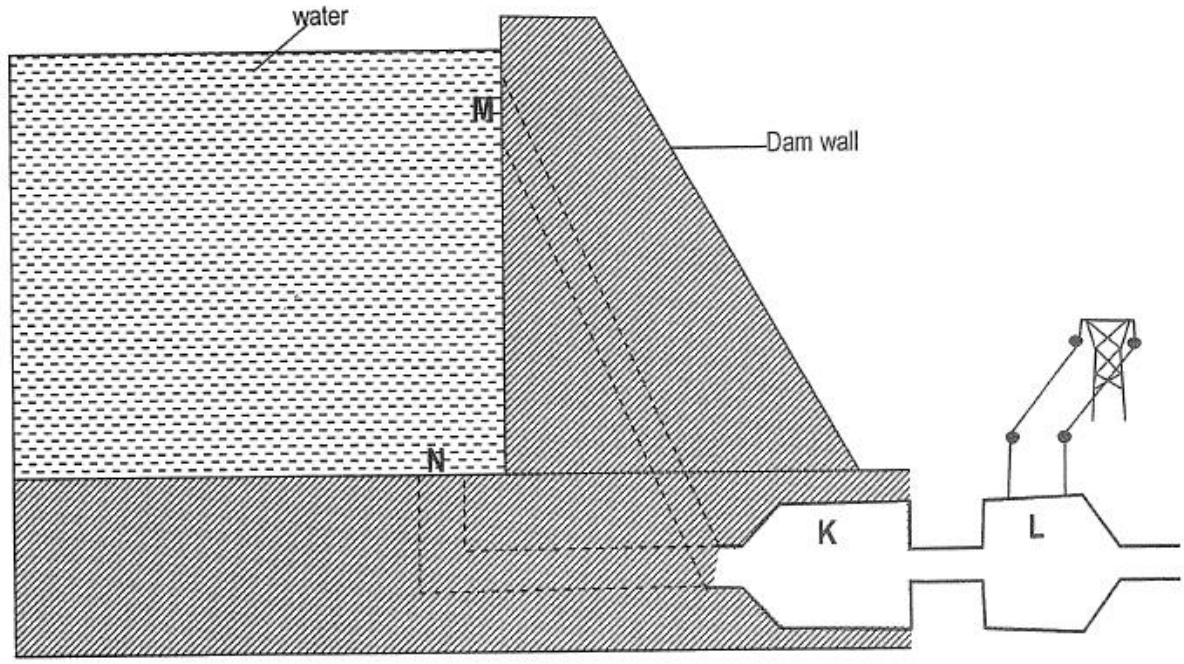
(i) increase of food production. [2013.P2.Q10c(i)]

(ii) land pollution. [2013.P2.Q10c(ii)]

(d) Suggest how pitch obtained from the coal can be of importance to the standard of living in Zambia. [2013.P2.Q10d]

(e) Explain why the petrol used in engines is unleaded. [2013.P2.Q10e]

2 ENERGY [2012.P2.Q9] The diagram below shows one method of producing electricity.



(a) What type of power station is shown in the diagram above? [2012.P2.Q9a]
(b) What does the letter K represent? [2012.P2.Q9b]
(c) Explain why the dam wall is thicker at the bottom than at the top. [2012.P2.Q9c]
(d) Which place M or N will be most suitable for constructing a tunnel to deliver water to K? [2012.P2.Q9d]
(e) (i) State one advantage of such a type of a power station. [2012.P2.Q9e(i)]
(ii) State one disadvantage of such a type of a power station. [2012.P2.Q9e(ii)]

26.1 COMMUNICATION – PAPER 1

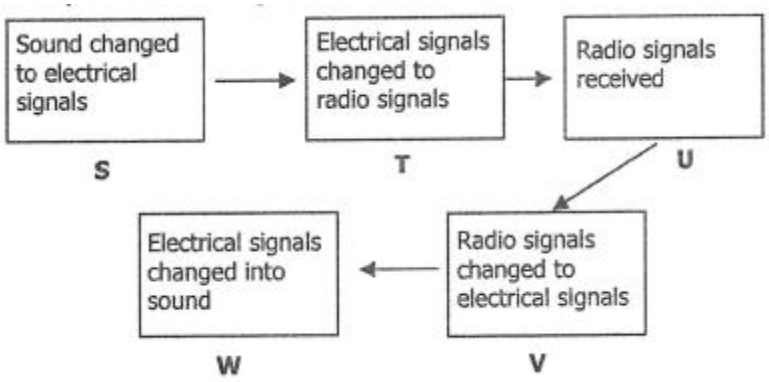
1 Which pair correctly shows communication using visual signals? [2016.P1.Q37]

A Smoke and telegraph B Traffic lights and letters
C E-mail and messenger D Radio and heliograph

2 Which one is not an advantage of a method of sending information through text messaging using a cellphone? [2016.P1.Q38]

A It is cheaper than a phone call.
B It is good for recording messages and their contents.
C Its message can be read at any time.
D Its messages are short.

3 Study the block diagram on communication shown below.



What communication devices are represented at T and W? [2016.P1.Q39]

	T	W
A	Microphone	Loud speaker
B	Microphone	Antenna
C	Transmitter	Loud speaker
D	Transmitter	Antenna

4 A geostationary satellite is one which ... [2016.P1.Q40]

A orbits in the atmosphere as it receives information.

B is stationed on earth as it transmits information.

C moves beyond the earth exploring the space.

D moves so as to remain in the same area above the earth.

5 Which of the following is not a way of sending of messages electronically? By... [2015.P1.Q38]

A facsimile

B radio message.

C mail through post.

D short message system (SMS)

6 Which of the following is correct for both digital and analogue transmission? [2015.P1.Q39]

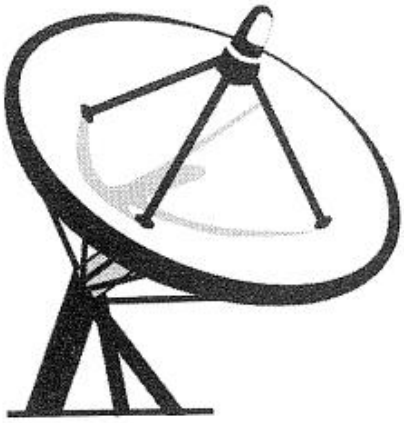
	Analogue Transmission	Digital Transmission
A	Generated by frequency modulation (IM)	Generated by amplitude modulation (AM)
B	Signal is conveyed as continuous	Signal is conveyed as discontinuous
C	Transfers data discretely	Transfers data varyingly
D	Conveys information using electromagnetic waves	Conveys information using air

7 Which of the following is not necessary during transmission of a live COSAFA match relayed from Harare, Zimbabwe to towns in Zambia? [2015.P1.Q40]

A Satellite B Microphone

C Tape recorder D Video camera

8 The structure drawn is found at an earth station. What is it called? [2013.P1.Q29]



A Tracking disc B Space satellite

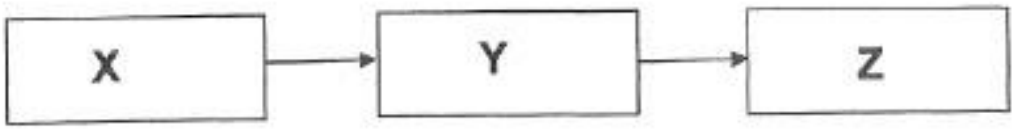
C Dish D T.V. antenna

9 Which device changes sound into electrical signals during recording in a television studio? [2012.P1.Q26]

A Loud speaker B Tuner

C Microphone D Aerial

10 The block diagram below represents the three stages of an educational broadcast. [2012.P1.Q27]



Which of the following do the letters X, Y and Z represent?

	X	Y	Z
A	Recording	Transmission	Reception
B	Reception	Transmission	Recording
C	Recording	Reception	Transmission
D	Transmission	Recording	Reception

11 Identify the correct sequence that shows a reception stage in an educational radio broadcast. [2011.P1.Q27]

A Radio teacher → microphone → tape recorder

B Amplifier → transmitter → repeater

C Receiving aerial → amplifier → loud speaker

D Lesson → microphone → amplifier

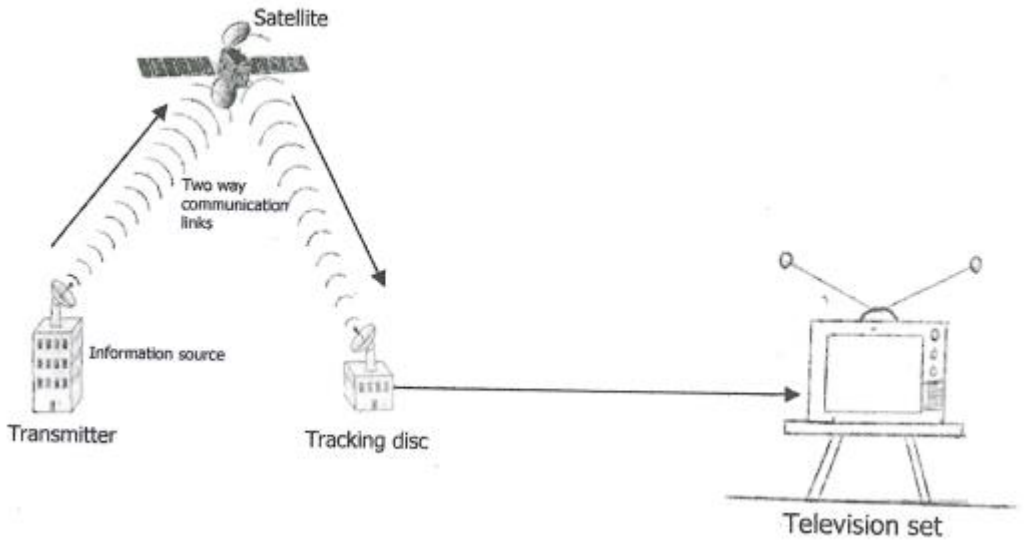
12 State the instrument that changes sound energy into electrical energy during a live television news broadcast. [2011.P1.Q28]

A Loud speaker
B Repeater
C Satellite
D Microphone

26.2 COMMUNICATION – PAPER 2

1 COMMUNICATION

The diagram shows parts of the television communication system (not drawn to scale). [2015.P2.Q10]



(a) In the diagram, which device will only receive but not send signals? [2015.P2.Q10a]

(b) By using a block diagram, show the two stages of television [2015.P2.Q10b]

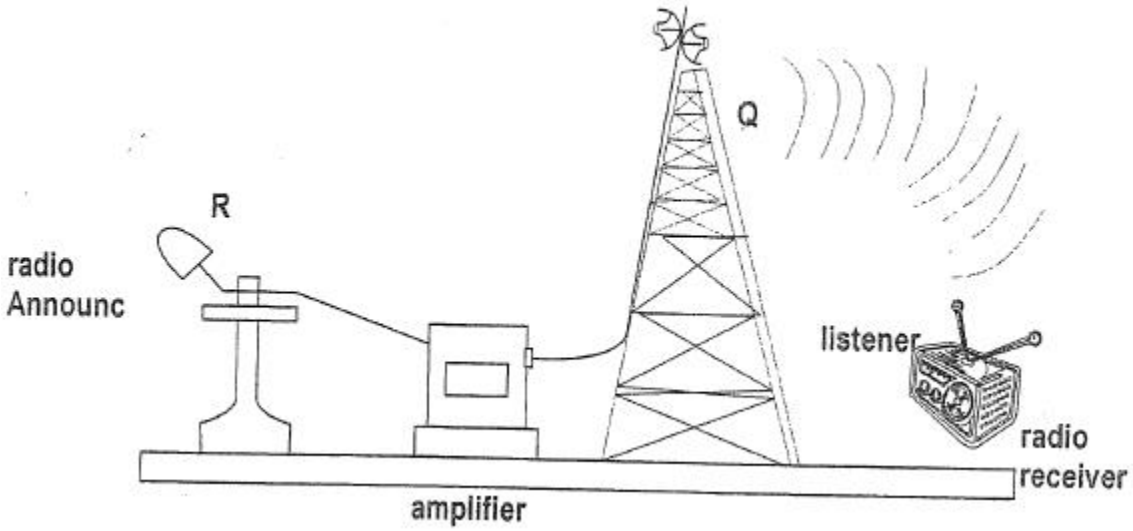
(c) What would be the effect of transmission if the tracking disc was removed? [2015.P2.Q10c]

(d) Suggest a communication device that can be placed between the tracking disc and television set. [2015.P2.Q10d]

(e) What effect will the inclusion of the named device in question (d) above have on transmission? [2015.P2.Q10e]

2 COMMUNICATION [2010.P2.Q7]

Study the drawings illustrating radio transmission and reception within a town in Zambia.



(a) Identify the devices R and Q. [2010.P2.Q7a]

(b) State the functions of:

(i) Part marked R. [2010.P2.Q7b(i)]

(ii) amplifier. [2010.P2.Q7b(ii)]

(c) Suggest a device that can be included in the transmission of Radio signals from this town to another town:

(i) within Zambia [2010.P2.Q7c(i)]

(ii) in Europe [2010.P2.Q7c(ii)]

27.3 PRACTICAL EXAMS – PAPER 3

1 2016 QUESTION 1

You are provided with a regularly shaped object P.

(a) Measure the mass (m) of object P.

(b) Measure and record in centimetres, the length, breadth (width) and height of object P

(c) Calculate the volume (V) of object P. Show your working here.

(d) Calculate the density of object P using the formula $D = \frac{m}{V}$.

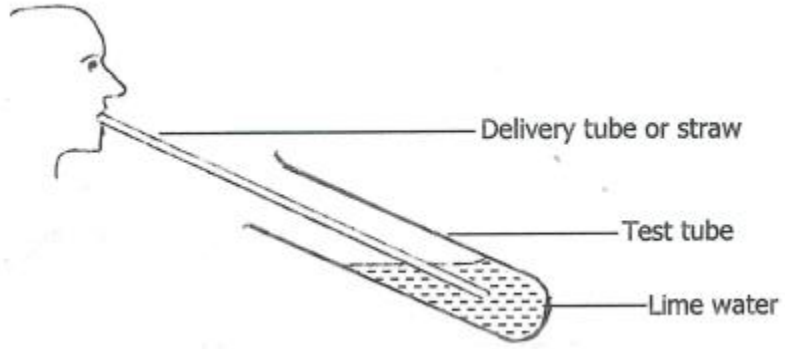
(e) In your experiment, in which area would you have made a mistake?

(f) State the precaution you took during this experiment.

2 2016 QUESTION 1

You are provided with limewater and liquid X in two separate test tubes labelled 'limewater' and 'liquid X' respectively.

(a) (i) Take a deep breath in with the delivery tube NOT dipped into the lime water and then breathe out through a delivery tube or straw into limewater as shown in the diagram below. Caution: Do not suck the limewater into your mouth:



Take another deep breath in and then breathe out through another delivery tube or straw into liquid X. Record your observations in the table below:

Observations with limewater	
Observations with liquid **X**	

(ii) Explain your observations in the test with limewater.

(b) (i) State the name of the gas which is responsible for the change observed in the test with limewater.

(ii) State the name of the gas which is responsible for the change observed in the test with limewater.

(c) What is the role of liquid X in the experiment?

3 2015 QUESTION 1

In this experiment you will test two food samples for the presence of starch and protein.
You are provided with two test tubes labelled S1 and S2 containing food samples suspended in water and two empty test tubes also labelled S1 and S2 respectively.

(a) Put half of the contents of S1 in the test tube labelled S1.
Put half of the contents of S2 in the test tube labelled S2.
Test the contents of each of the two test tubes SI and S2 for the presence of starch.

(i) Record your results in the table below:

	Observation after testing for starch	Conclusion
S1		
S2		

(ii) Describe how you carried out the starch test on each sample.

(b) Another nutrient found in food is protein. To test for the presence of protein, biuret reagent is used.
When biuret reagent is added to a food sample containing protein, the buiret reagent turns purple from blue.
Test the remaining food samples in SI and S2 for the presence of protein by adding three drops of biuret solution to each.

(i) Record your results in the table below.

	Observation after testing for protein	Conclusion
S1		
S2		

(ii) Describe how you carried out the test for protein on each sample.

4 2015 QUESTION 2

In this experiment, you will measure the mass of different volumes of a liquid P. You are provided with a 50cm^3 measuring cylinder, a sample of liquid P in a beaker labelled A, an electronic balance and an empty beaker labelled B.

(a) Measure the mass of the empty beaker labelled B and record the mass.

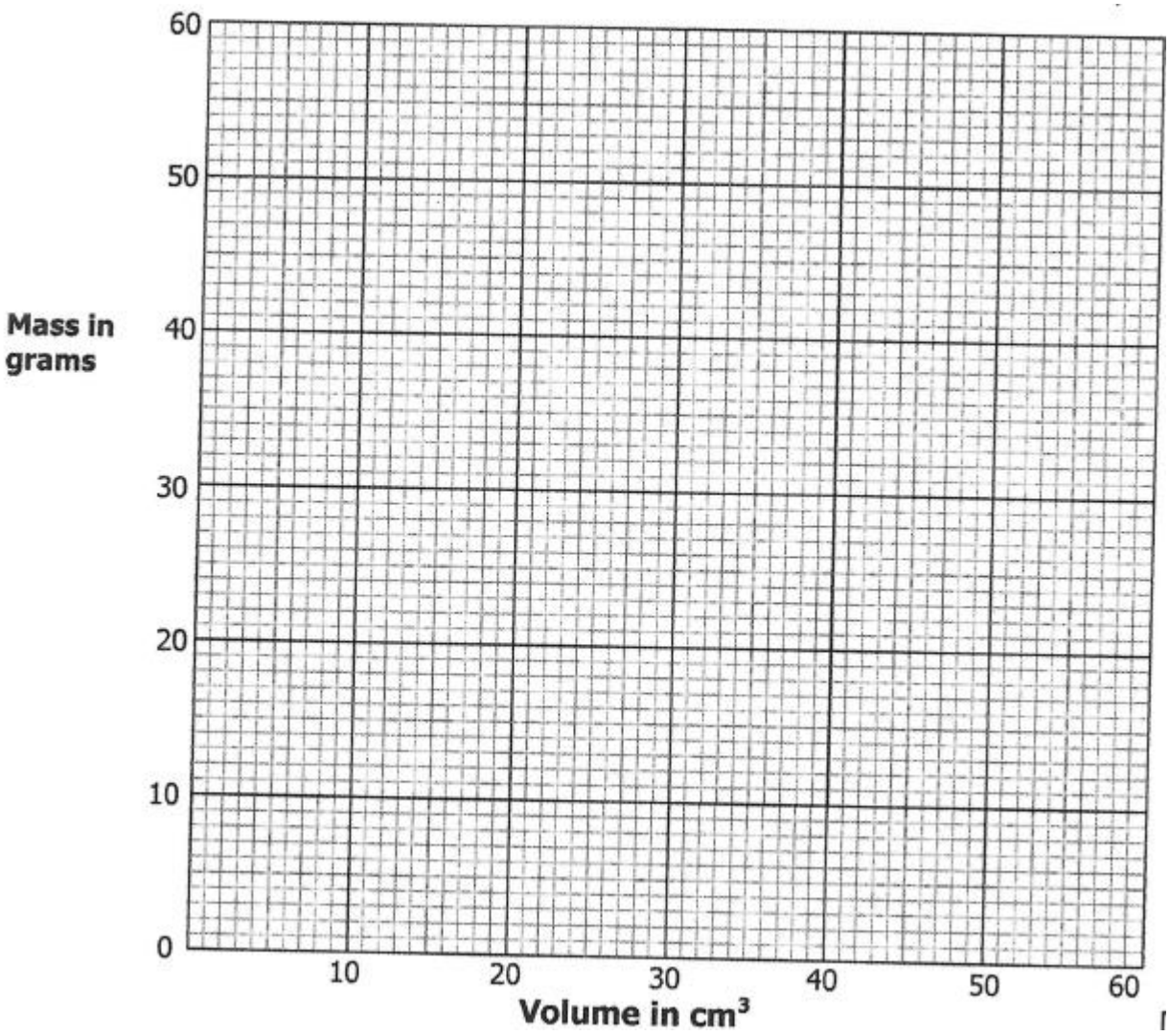
(b) Measure 10cm^3 of liquid P using the measuring cylinder. Pour the 10cm^3 of iiquid P in the beaker labelled B and measure the mass of the beaker plus liquid P. Record the mass obtained.

(c) Calculate the mass of 10cm^3 of liquid P.

(d) Repeat step (b) above for different volumes of liquid P as shown in the table below. For each, calculate the mass of the liquid P and record your results in the table below.

Volume of **P** in cm^3	Mass of beaker plus liquid **P** in grams	Mass of liquid **P** in grams
20		
30		
40		
50		

(e) On the grid below, plot a graph of mass of liquid P against volume of liquid P.



(e) What physical quantity is represented by the gradient of the graph of mass against volume?

SUGGESTED ANSWERS 2011 - 2016

GRADE 8 TOPICS

1.1 HUMAN REPRODUCTIVE SYSTEM AND PUBERTY – PAPER 1

1 B **Q** – Cervix. **S** – Uterus, P – Ovary, R – Oviduct or Fallopian tube

2 B Avoid the organs getting infections.

3 A. 2. sexual intercourse, 1. fusion of sperm and ovum, 3. formation of zygote, 4. attachment of zygote to uterus

4 C Uterus - Organ where embryo develops

5 B Deepening of the voice

6 C Protecting the testes

7 B Gestation period is about 270 days – 280 days (9 months)

8 A

9 B 3. Vagina -> 4. Cervix -> 1. Uterus -> 2. Oviduct

10 **A** – Normal full term pregnancy is 9 months. January 28 – October 28.

11 **B** – Epididymis is a coiled tube attached to the back and upper side of the testicle that stores sperm and is connected to the vas deferens.

12 **C** – Ovulation releases the egg from the ovary, the egg in the fallopian tube is **fertilised** by a sperm to form a zygote which is **implanted** in the uterus.

1.2 HUMAN REPRODUCTIVE SYSTEM AND PUBERTY – PAPER 2

1 (a) **G** – Urethra and **I** - Scrotum.

(b) **J** – the epididymis stores sperms [2016.P2.Q1b]

(c) If part **K** (sperm duct) was to be tied, the sperms would not be released by the penis.

(d) Pubic hair grows. Deepening of the voice. Male sex organs grow bigger. Muscles in the arms, legs and shoulders grow larger and stronger.

2 (a) **M** – Cervix, **P** – Epididymis.

(b) **L** (ovary) – produces and releases the egg.
N (urethra) – carries sperms from the sperm ducts.

(c) **K** (Oviduct or fallopian tube) – (Ovulation releases the egg from the ovary, the egg in the fallopian tube is **fertilised** by a sperm to form a zygote which is **implanted** in the uterus.)

(d) 9 months (270 – 280 days)

3 (a) **K** – umbilical cord and **L** – uterus

(b) **N** – amniotic fluid. It protects the embryo/foetus from damage and external shocks. It provides a suitable environment for the embryo's development.

(c) Food from the mother moves to the embryo through the umbilical cord.

(d) Gaseous exchange between mother and foetus occurs in M, the placenta. Food and oxygen from the mother's blood moves to the foetus. Carbon dioxide and waste products formed in the foetus move to the mother's blood. The foetus does not drown because it does not breathe through its nose or mouth.

(e) A pregnant woman requires a lot of proteins for the normal growth of the foetus. She also requires a lot of carbohydrates for her extra energy needs of carrying the baby.

2.1 NUTRITION – PAPER 1

1 D carbohydrates.

2 C Age of the child's mother.

3 B Note: **Kwashiorkor** (lack of protein), **marasmus** (lack of carbohydrates), **anaemia** ((lack of iron, Vitamin B), **scurvy** and poor healing of wounds (lack of vitamin C), **goitre** (lack of iodine), **rickets** (lack of vitamin D and calcium), **pellagra** and **beri beri** (lack of vitamin B), **dehydration** (lack of water), **night blindness** (lack of vitamin A), **constipation** (lack of dietary fibre) .

4 A (1 – carbohydrates, 2 – fats, 3 – proteins, 4 – mineral salts)

5 A

6 **C** – In a minute 4 kilojoules of energy is used. One gram releases 16 kilojoules of energy. So one gram is used in $\frac{16}{4} = 4$ minutes. Therefore, 4 g will last $4 \times 4 = 16$ minutes.

7 **A** – **Kwashiorkor** (lack of protein), **marasmus** (lack of carbohydrates), **anaemia** ((lack of iron, Vitamin B), **scurvy** and poor healing of wounds (lack of vitamin C), **goitre** (lack of iodine), **rickets** (lack of vitamin D and calcium), **pellagra** and **beri beri** (lack of vitamin B), **dehydration** (lack of water), **night blindness** (lack of vitamin A), **constipation** (lack of dietary fibre) .

8 **B** – Normal weight should be between the two lines. In June, it was above the upper line.

2.2 NUTRITION – PAPER 2

1 (a) (i) Vitamin C

(ii) Carbohydrates

(b) (i) **Proteins** – needed for the foetus to makes its own proteins for growth

(ii) **Iron** – ensures that the foetus can make its own red blood cells and to prevent the mother from becoming anaemic.

(c) - monitoring of baby growth and development; nutritional advice; vaccination (immunisation); supplementary feeding for under-nourished children; deworming;

3.1 WATER, AIR AND LAND POLLUTION – PAPER 1

1 A

2 D harmful substances to the environment.

3 C

4 B

5 **C** – Acid rain is caused by emissions of sulfur dioxide and nitrogen oxide, which react with the water molecules in the atmosphere to produce acids.

6 **B** – Carbon dioxide causes increased air temperature or global warming

3.2 WATER, AIR AND LAND POLLUTION – PAPER 2

1 (a) (i) Trees and water are a diminishing but renewable resource.
(ii) Copper, coal and iron are diminishing and not renewable.
(b) Coal causes air pollution when it burns.
(c) Trees absorb some of the carbon dioxide produced by burning coal.
(d) Copper
(e) Recycling. Re-using. Reducing use of pollutants

4.1 PLANT & ANIMAL CELLS – PAPER 1

1 A – The cell membrane, the nucleus and the cytoplasm together form the protoplasm, which is the living part of the cell.

2 B – cell wall and chlorophyll are found in plant cells.

3 **D** – The mirror can be set at an angle so that light can be reflected through the hole in the stage and through the specimen to the lens. This helps us to make the image bright and clear.

4 C – cell wall is present in plant cells but absent in animal cells.

5 **B** – The **cell wall** is not selectively permeable

6 **C** – Light moves to the eye from V (**Mirror**) → U (**Stage**) → T (**Tube**)→ R (**Eyepiece**)

4.2 PLANT & ANIMAL CELLS – PAPER 2

1 (a) **M** – nucleus and **N** – cytoplasm.

(b) Cell wall and chloroplasts

(d) Microscope

2 (a) **K** – magnifying lens (eyepiece) and **M** – objective lenses.

(b) **K** – magnifying lens (eyepiece) and **M** – objective lenses.

(c) **K** – Eyepiece.

(d) (i) **J** (Adjustment knob) – moves the body tube up and down to bring the image of the specimen into focus

(ii) **O** (Mirror) – reflects rays from a light source through the hole in the stage and through the specimen to the lens. This helps to make the image bright and clear.

5.1 PLANT GROWTH AND NUTRIENTS – PAPER 1

1 C

2 **C** – When there is no Nitrogen, leaves turn yellow. Lack of phosphorous leads to poor root growth.

3 **C** – Nitrogen is needed for formation of strong green leaves. **Phosphorus** for growth of strong roots, **potassium** for healthy leaves.

6.1 COMPOSITION OF MATTER – PAPER 1

1 C

2 A

7.1 PHYSICAL CHANGE OF STATE – PAPER 1

1 A – −130°C is less than the melting point (−100°C).

2 C

3 **C** – **Carbon dioxide** and **iodine** change state from a gas to a solid or solid to a gas. This is called sublimation. Other substances that sublime include **ammonium chloride** and **naphthalene**.

4 **D** – Solid to Liquid → **Melting**; Solid to Gas → **Sublimation**; Liquid to Gas → **Evaporation**
Liquid to Solid → **Freezing**; Gas to Solid → **Sublimation**; Gas to Liquid → **Condensation**

5 **C** – **Carbon dioxide** and **iodine** change state from a gas to a solid or solid to a gas. This is called sublimation. Other substances that sublime include **ammonium chloride** and **naphthalene**.

7.2 PHYSICAL CHANGE OF STATE – PAPER 2

1 (a) (i) Q

(ii) S

(b) When a substance is changing from solid to liquid, **Q**, the temperature remains constant.

When a substance is changing from liquid to gas, **S**, the temperature remains constant.

There are 3 states of matter.

(c) R

(d) Q

2 (a) 0^oC

(b) It must be heated

(c) Melting (change from solid state to liquid state)

(d) Mercury expands at a very constant rate over a large range of temperatures.

(e) Melting point

(f) Sublimation. (Change from solid state into gaseous state or from gaseous state into solid state without becoming a liquid.) Substances that sublime include **Carbon dioxide, iodine, ammonium chloride** and **naphthalene**.

3 (a) (i) A liquid takes up shape of container
(ii) A gas spreads and fills up any space available to it.

(b) When water freezes into ice, the ice can be changed back to water by melting. A physical change is one which can easily be reversed and in which no new substances are formed.

8.1 MIXTURES – PAPER 1

1 B

2 A

3 **C** – Product of distillation is a distillate. Product of filtration is filtrate.

4 A

5 C

6 D

8.2 MIXTURES – PAPER 2

1 (a) Mixture

(b) (i) dissolved

(ii) did not dissolve

(c) (i) Filtration – pour the mixture of sand, water and salt into a filter paper placed inside a funnel. The dissolved salt and water will pass through the filter, but the sand will remain trapped on the filter as residue.

(ii) Evaporation – the filtrate that will be collected after filtration will be boiled, the water will evaporate and the salt will remain.

(d) Fractional distillation.

2 (a) **Fractional distillation**. It is used to separate 2 or more miscible liquids (liquids that mix completely without reacting and without forming layers of separation). The distillates are collected as fractions at different temperatures.

(b) Liquids in the mixture do not react.
The liquids have different boiling points

(c) **U** – Liebig condenser. The cold water circulating through it causes the evaporated gas to condense back to liquid.

(d) Ethanol will be collected first in the conical flask because it has lower boiling point(78^oC) than water (100^oC).

(e) Fractional distillation is used in the separation of various components of crude oil or petroleum. The components are petro, paraffin, diesel, lubricants etc

Note – as the mixtures is heated, ethanol (alcohol) begins to boil at 78°C. Some water evaporates too, so a mixture of alcohol vapour and water vapour rises up the column. The vapours condense on the glass beads in the column, making them hot. When the temperature of the beads reaches 78°C, ethanol vapour no longer condenses on them, but the water vapour still does and drips back into the flask, while the ethano vapour is forced into the condenser where it turns back to liquid.

3 (a) (i) Ink. Dye.
(ii) Water
(b) It is porous. It selectively absorbs pigments
(c) Chromatogram
(d) Analysis of food, drugs, blood and petroleum products.
(e) Distillation

4 (a) **T** – Liebig condenser
(b) Distillation
(c) The cold water circulating through it causes steam to condense to water.
(d) The cold water enters through **U** and leaves through **S**.
(e) Fractionating column packed with glass beads.
(f) Fractional distillation. It is used to separate 2 or more miscible liquids (liquids that mix completely without reacting and without forming layers of separation). The distillates are collected as fractions at different temperatures.

9.1 MASS, WEIGHT, VOLUME & DENSITY – PAPER 1

1 B $Weight = mass \times acceleration = 0.5kg \times 10 = 5\text{N}$ (mass must be in kg, not *g*)

2 D $\text{Density} = \frac{Mass}{Volume} = \frac{18g}{Final - Initial\ level} = \frac{18g}{12 - 8\ cm^3} = \frac{18}{4} = \frac{9}{2} = 4.5g/cm^3$

3 D $Weight = mass \times acceleration = 2kg \times 10m/s^2 = 20\text{N}$

4 D $\text{Density} = \frac{Mass}{Volume} = \frac{20g}{Final - Initial\ level} = \frac{20g}{15 - 10\ cm^3} = \frac{20g}{5\ cm^3} = 4g/cm^3$

5 **B** – Mass = 75 g (as shown on the scale). Volume = 25 cm^3 (volume of displaced water)
Density = $\frac{\text{Mass}}{\text{Volume}} = \frac{75\text{ g}}{25\text{ cm}^3} = 3\text{ g/cm}^3$

6 **B** $\frac{5\text{ g}}{5\text{ cm}^3} = 1\text{g/cm}^3$. Water has the maximum density of 1g/cm^3 only when it is pure water.

7 **D** – Mercury is denser than water. Water is denser than oil.

8 **D** – Density = $\frac{\text{M}}{\text{V}}$, Mass =140 g, Volume = 70cm^3 – 50cm^3 = 20 cm^3. Density=$\frac{140\text{ g}}{20\text{ cm}^3}$ =7 g/cm^3

9.2 MASS, WEIGHT, VOLUME & DENSITY – PAPER 2

1 (a) Density is mass per unit volume of a substance. It is the ratio of mass to the volume of a substance. Density is the amount of matter (mass) in a given volume of a substance.

(b) $\text{Density} = \frac{Mass}{Volume} = \frac{80g}{100\ cm^3} = 0.8g/cm^3$

(c) Floating objects.

(d) The average density of the boat is less than that of water while the average density of the nail is more than that of water.

(e) As the water fills the boat, its average density increases. As soon as the average density of the boat filled with water exceeds the density of water, the boat sinks.

2 (a) The volume of the metal is equal to the volume of the water it displaced.

(b) Volume at point marked 'A' on the cylinder = volume of metal
$V = length \times breadth \times height = 4cm \times 3cm \times 2cm = 24\ \text{cm}^3$

(c) Mass$= 48\ \text{g}, \text{Volume} = 24\ \text{cm}^3$.
$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{48\ \text{g}}{24\text{cm}^3} = 2\ \text{g/cm}^3$

(d) The metal block could not float on water because its density ($2\ \text{g/cm}^3$) is more than the density of water ($1\ \text{g/cm}^3$). It will sink.

(e) By making it hollow or making a boat out of it, thereby increasing the volume. As the volume increases, the average density reduces because the air in the hollow space considered part of the boat. When the density becomes less than that of water ($1\ \text{g/cm}^3$), it will float.

3 (a) $\text{Volume} = \text{length} \times \text{breadth} \times \text{height} = 10\text{cm} \times 5\text{cm} \times 15\text{cm} = 750\ \text{cm}^3$

(b) Increase in height$= 18\ \text{cm} - 15\ \text{cm} = 3\ \text{cm}$

(c) $\text{Volume of stone} = \text{length} \times \text{breadth} \times \text{height} = 10\text{cm} \times 5\text{cm} \times \mathbf{3\ cm} = 150\ \text{cm}^3$

(d) Density of the stone = $\frac{\text{Mass}}{\text{Volume}} = \frac{300\ \text{g}}{150\ \text{cm}^3} = 2\ \text{g/cm}^3$

(e) Density of a different solid = $\frac{\text{Mass}}{\text{Volume}} = \frac{50\ \text{g}}{150\ \text{cm}^3} = 0.33\text{g/cm}^3$. This solid would float on the water because its density is less than that of water which is $1\ \text{g/cm}^3$.

4 (a) 12 g (shown on the scale)

(b) Volume of liquid 'H' = $4\ \text{cm}^3$ (shown on measuring cylinder).

(c) Density of liquid 'H'= $\frac{\text{Mass}}{\text{Volume}} = \frac{12\ \text{g}}{4\text{cm}^3} = 3\ \text{g/cm}^3$

(d) Liquid 'H' is denser than water (pure water has density of **1 g/cm³**), so it will collect at the bottom of the container and the water will float on it.

(e) To make sure that the measurement is accurate.

10.1 HEAT TRANSFER & EXPANSION OF SUBSTANCES – PAPER 1

1 C

2 C

3 A

4 A – metal X expands more than metal Y. K (0.30) expands more than M (0.09).

5 C – the metal on the outer side of the strip (curve) expands more than the one on the inner side.

6 D

7 B – Y is a vacuum that prevents heat transfer by radiation.

10.2 HEAT TRANSFER & EXPANSION OF SUBSTANCES – PAPER 2

1 (a) (i) Conduction

(ii) To prevent heat transfer by radiation.

(b) (i) Wax 1

(ii) Because Mercury is a good conductor of heat.

(iii) Water is a bad conductor of heat.

(c) Convection.

2 (a) Aluminum cup.

(b) It is a good conductor

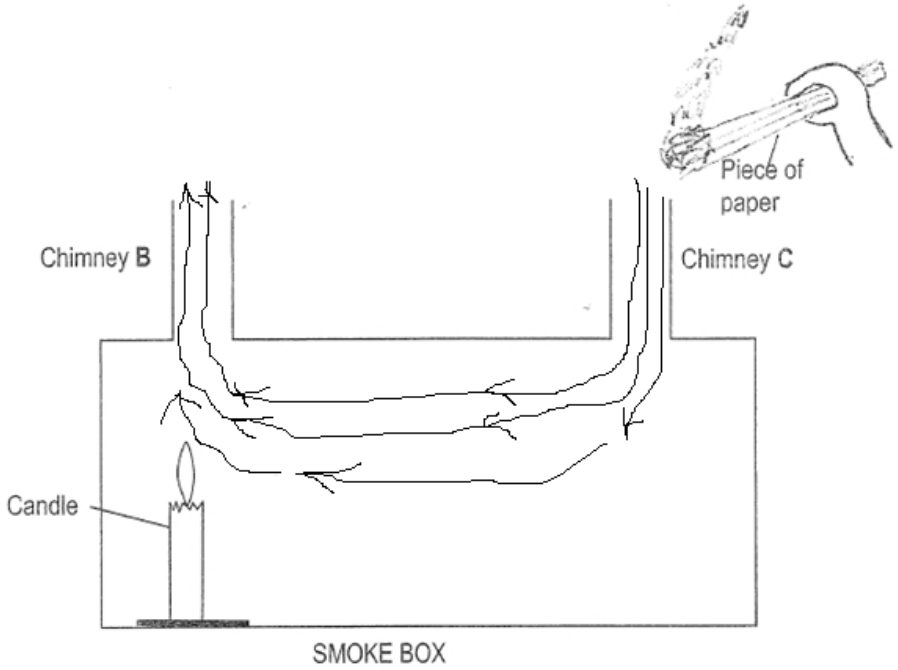
(c) (i) Aluminum cup

(ii) Ceramic cup

(d) Good conductors of heat get heated quickly and also lose their heat quickly, but bad conductors slowly get heated and also lose their heat slowly (or retain their heat longer).

(e) Vacuum flask or thermos flask.

3 (a)



(b) To set up a convection current by heating the air at one side of the smoke box

(c) Convection

(d) Air

(e) Sea breeze. (During the day the surface of the land is warmed by the sun and the air above it rises. The water (sea) takes time to warm up and so the air above it is cooler, therefore taking longer to rise. The air above the sea therefore moves towards the land to replace the hot air which has risen.

(f) Conduction (in solids). Radiation (in a vacuum).

11.1 REFLECTION AND REFRACTION OF LIGHT – PAPER 1

1 A

2 A – Angle of incidence is the angle between the incident ray and the normal. Angle of refraction is the angle between the refracted ray and the normal.

3 **A** – When a ray is moving from a less dense medium to a denser medium, the refracted ray will bend towards the normal. When a ray is moving from a denser medium to a less dense medium, the refracted ray will bend away from the normal.

11.2 REFLECTION AND REFRACTION OF LIGHT – PAPER 2

1 (a) The normal. (The perpendicular lines)

(b) Refraction. It is the bending of light (or change of direction) when it passes from one medium to another.

(c) The rays of light bend away from line B because the glass block is denser than water

(d) The ray will bend toward the normal because water is denser than air.

NB – when a ray is moving from a less dense medium to a denser medium, the refracted ray will bend towards the normal. When a ray is moving from a denser medium to a less dense medium, the refracted ray will bend away from the normal.

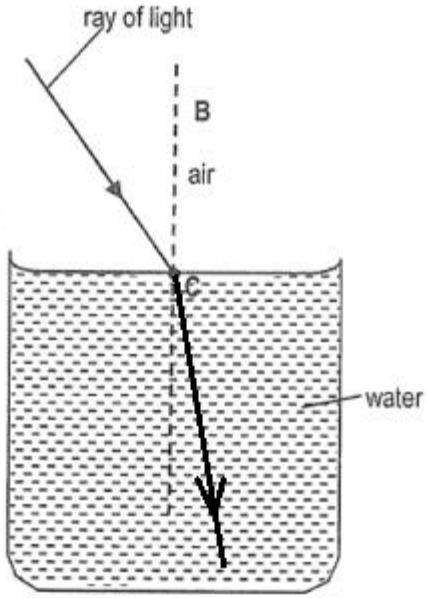


Figure 2.0

(e) The density is the highest in glass. (Followed by water, then air)

12.1 COMPOSITION OF AIR – PAPER 1

1 B

2 D

3 B

4 **C** – The acid reacts with the carbonate solution to produce carbon dioxide to extinguish fire.

5 **D** – Oxy-acetylene flame. It can reach a temperatures of about 3000 ^{0}C.

12.2 COMPOSITION OF AIR – PAPER 2

1 (a) (i) P – Oxygen and Q - Nitrogen.

(ii) P (Oxygen) is used in welding and cutting metals; manufacturing steel from iron; breathing for patients and premature babies in hospitals; combustion of fuel in vehicle and rocket engines.

(b) R (Carbon dioxide)

(c) The amount of water vapour in the air varies with weather conditions (humidity, rainfall, temperature).

2 (a) Carbon dioxide.

(b) It is denser (heavier) than air. It does not support burning.

(c) Gas Y (Carbon dioxide) is denser (heavier) than air, so it would not escape from the jar.

(d) Carbon dioxide turns lime water milky (when passed through lime water, the lime water turns milky).

(e) It is used to make carbonated drinks like coca-cola. (The carbon dioxide is dissolved in the drinks under pressure and gives the drinks a pleasant taste.
It is used in baking. (The yeast or baking powder produces carbon dioxide which makes the dough to rise.)
Solid carbon dioxide or dry ice is used to cool food and drinks.
It is used by plants during photosynthesis.

GRADE 9 TOPICS

13.1 CIRCULATORY SYSTEM – PAPER 1

1 B

2 B – Human blood is red in colour, ranging from **bright red when oxygenated** to a very dark, almost blackish-red when deoxygenated. The only vein that carries oxygenated blood is the pulmonary vein which carries blood from the lungs to the heart.

3 B

4 **D** – **Vena cava** has a thin wall, carries deoxygenated blood and has valves. Pulmonary vein carries oxygenated blood from the lungs. Pulmonary artery and Aorta are arteries. Arteries have no valves. Arteries have thick walls to withstand high blood pressure.

5 **A** – Left Atrium and Left Ventricle have oxygenated blood from the lungs ready to be pumped to all parts of the body through the Aorta.

6 A

7 **D** – The left ventricle pumps blood to all parts of the body (but not to the lungs)

13.2 CIRCULATORY SYSTEM – PAPER 2

1 (a) Tricuspid valve

(b (i) Z and V are arteries (carry blood away from the heart.)

(ii) U and W are veins (carry blood to the heart).

(c) Lungs (air sacs).

(d) From the **left auricle**, oxygenated blood will flow to the **left ventricle via the bicuspid valve**, then move to all parts of the body through blood vessel **Z** (**Aorta**). Deoxygenated blood from all parts of the body will then pass through blood vessel **U** (**Vena cava**) back to the heart in the **right auricle**, then into the right **ventricle via part Y (Tricuspid valve).** From there, it passes through vessel **V** (**pulmonary artery**) to the **lungs**. Oxygenated blood from the lungs passes through vessel W (pulmonary vein) back to the heart in the left auricle and the whole process repeats itself.

14.1 RESPIRATORY SYSTEM – PAPER 1

1 A'8

2 $$Percentage\ of\ used\ oxygen = \frac{Inhaled - exhaled\ air}{Inhaled\ air} \times 100\%$$
$$= \frac{20 - 14}{20} \times 100\% = \frac{6}{20} \times 100\% = 30\%$$

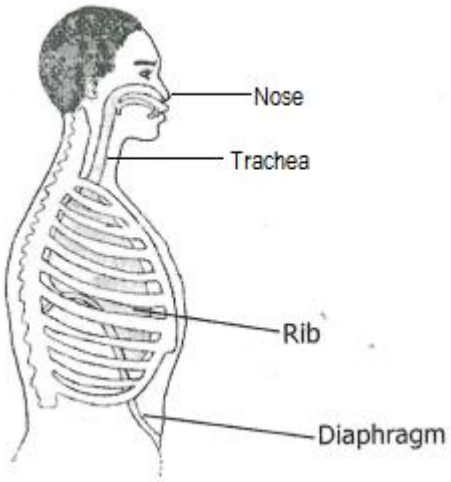
3 D

4 B

5 A

14.2 RESPIRATORY SYSTEM – PAPER 2

1 (a)



(b) (i) Expiration.

(ii) The diaphragm is curved upwards

(c) [**During inspiration** the intercostal muscles (between ribs) contract, ribs move upwards and outwards, the diaphragm is straightened and flattened (moves downwards), the volume of the chest increases, the air pressure in the chest decreases and air is drawn in.]

2 (a) **E** – diaphragm. **F** – air sac or alveolus

(b) **G**(ribs) – to protect the lungs and the heart.

(c) (i) The lungs are filled with air and become bigger. (Air is drawn in).

(ii) **G (ribs)** move upwards and outwards.

[**During inspiration** the intercostal muscles (between ribs) contract, ribs move upwards and outwards, the diaphragm is straightened and flattened (moves downwards), the volume of the chest increases, the air pressure in the chest decreases and air is drawn in.]

[**During expiration** the intercostal muscles (between ribs) relax, ribs move downwards and inwards, the diaphragm curves upwards, the volume of the chest decreases, the air pressure in the chest cavity increases and air is pushed out.]

(d) Lung capacity is the amount of air we can breathe out after filling them completely.

15.1 SEXUALLY TRANSMITTED INFECTIONS (STIS) – PAPER 1

1 C 2 B 3 D

16.1 CYCLES IN THE BIOSPHERE – PAPER 1

1 D 2 D

16.2 CYCLES IN THE BIOSPHERE – PAPER 2

1 (a) Nitrogen fixation by lightning

(b) Bacteria

(c) Legumes

(d) - animal dropping and urine; decomposition after death

(e) Applying compost manure; applying inorganic fertilisers; planting legumes

17.1 WATER MANAGEMENT – PAPER 1

1 D

2 A

17.2 WATER MANAGEMENT – PAPER 2

1 (a) M

(b) The water from the well, J, is contaminated with water from the pit latrine.

(c) The water cycle keeps re-cycling the water.

(d) Sand and gravel act as a filter to clean up water as it seeps into the ground

(e) Groundwater

(f) Surface water. Under river flow

18.1 CONSERVATION OF ANIMALS AND PLANTS – PAPER 1

1 D

2 D

3 C

4 B

5 D

6 A

7 B

8 A

9 A

10 A

18.2 CONSERVATION OF ANIMALS AND PLANTS – PAPER 2

1 (a) (i) Farmers practice cross breeding to produce better offspring from animals or plants with desirable qualities by bringing together two organisms of different animal breeds or plant varieties.

(ii) In-breeding is the producing of pure plant varieties from hybrids by self-pollination.

(b) (i) Early maturity. High production of eggs.

(ii) Drought resistance. Early maturing.

(c) Artificial insemination is the process by which semen (sperms) is collected from the desired male and deposited into the desired female in order impregnate it.

The advantage is that it is easier and cheaper to import semen than to import male animals and create condition for mating. The method minimises the spreading of veneral diseases. The disadvantage is that it requires well-trained operations and special equipment.

2 (a) Rhino, elephant, giraffe, zebra etc.

(b) Animals in game parks are protected. People are not allowed to hunt without license.

(c) Do not give food to animals. Do not throw litter in the park. Use designated footpaths or drives. Killing or disturbing any kind of wildlife is forbidden. Do not light fires, except in designated places. Do not remove any vegetation from the park.

(d) Camping sites and lodges. Footpaths and drives.

(e) It attracts local and foreign tourists who bring in revenue for the country. Creation of employment.

19.1 PHOTOSYNTHESIS – PAPER 1

1 B

2 A – Sodium hydroxide (Soda lime) absorbs the carbon dioxide in Plant B. Therefore, photosynthesis doesn't occur. No starch in the leaf of Plant B.

3 A

4 D

5 **D** – The guard cells help to regulate the rate of transpiration by opening or closing of stomata.

6 **D** – Oxygen is a product of photosynthesis. Its identity test is that it relights a glowing splint. Photosynthesis is process of producing sugar (glucose) and oxygen by green plants (plants with chlorophyll) using carbon dioxide and water in the presence of light.

7 **A** – The palisade layer has many chloroplasts where photosynthesis occur.

8 **D** – Soda lime absorbs the carbon dioxide. Therefore, photosynthesis doesn't occur. No starch in the leaf.

9 **A** – Region **A** didn't turn blue/black because it did**n't have starch** since it was not exposed to light.

10 **A** – Sodium hydroxide absorbs carbon dioxide from the air.

19.2 PHOTOSYNTHESIS – PAPER 2

1 (a) (i) To remove carbon dioxide

(ii) To supply carbon dioxide gas to the plant.

(b) Potted plant V. (There is carbon dioxide. Potted plant U has no carbon dioxide.)

(c) Iodine solution.

(d) - Boiling the leaf in water to kill the protoplasm. This stops all activities in the leaf which might interfere with the experiment.
- The leaf is placed in boiling alcohol in order to remove chlorophyll so that colour changes can easily be seen.

2 (a) Photosynthesis
(b) Pond water has dissolved carbon dioxide required for photosynthesis to occur.
(c) Experiment R is a control, to check whether the gas came from water or the pond weed.
(d) Oxygen
(e) Oxygen relights a glowing splint.
(f)

$$\text{Carbon dioxide + water} \xrightarrow[\text{Chlorophyll}]{\text{light}} \text{sugar + oxygen}$$

Note – Photosynthesis is the process of producing sugar and oxygen by green plants (plants with chlorophyll) using carbon dioxide and water in the presence of light.)

3 (a) Because there was no light.
(b) Sodium hydroxide absorbs carbon dioxide.
(c) To prevent light from reaching the plant.
(d) (i) Starch will be absent in experiment D (there is no carbon dioxide)
(ii) In E, the leaf will turn blue/black.
In F the leaf will be stained brown (the colour of iodine solution) because there is no starch due to the absence of light.

4 (a) Starch
(b) Iodine solution
(c)
(i) Boiling the leaf in water kills the protoplasm. This stops all activities in the leaf which might interfere with the experiment.
(ii) The leaf is placed in boiling alcohol in order to remove chlorophyll so that colour changes can easily be seen.
(d) Because alcohol is flammable (can burn).
(e) The leaf is dipped in hot water to make it soft after it has been made brittle by alcohol
Additional Notes – When drops of iodine solution added to the leaf after Stage IV, the leaf turns blue/black if starch is present. It is stained brown if there is no starch.

20.1 TRANSPIRATION – PAPER 1

1 A

2 A – Xylem vessels transport water and mineral ions from the root.

3 C

4 C – Xylem vessels carry water and mineral salts from the roots to the leaves. Phloem cells carry food from the leaves to all parts of the plant.

20.2 TRANSPIRATION – PAPER 2

1 (a) The air-tight seal is used to prevent water from evaporating or leaking from the tubing, making the experiment inaccurate.

(b) The potometer measures the rate of uptake of water.

(c) Blue cobalt chloride paper or anhydrous copper sulphate.

(d) Water turns blue cobalt chloride paper pink. It also turn anhydrous copper sulphate from white to blue.

(e) - Uptake of water and mineral nutrients from the roots to the rest of the plant.

- Water is needed for photosynthesis in the plant.

- Loss of water vapour though the stomata cools the plant and prevents overheating.

- keeps plant cells turgid, which helps keep plants standing upright.

(f) - thickened waxy cuticle; sunken stomata; reduced leaf size; hairy leaves

2 (a) **B**. Water will be lost into the atmosphere

(b) Transpiration (the process of losing water in form of water vapour by plants into the atmosphere).

(c) Liquid **O** is water. It turns blue cobalt chloride paper pink. It also turn anhydrous copper sulphate from white to blue.

(d) Transparent polythene bag prevents the water vapour from the soil in pots from escaping.

(e) Source of **O** – leaves
Source of **P** – soil in water pot

21.1 CHEMICAL REACTION – PAPER 1

1 A

2 B

3 B

4 **D** – Potassium, Sodium, Lithium, Calcium, Magnesium, – react with water to produce hydrogen gas.
Aluminium, Zinc, Iron – react with **steam** to produce hydrogen gas.
Tin, Lead, Copper, Silver, Gold, Platinum – **DO NOT** react with water nor steam.

5 **A** – An acid enables water to conduct electricity. Pure water is a bad conductor.

6 **A** – When a mixture of potassium chlorate and magnesium dioxide is heated, oxygen is produced. The other product is potassium chloride. Magnesium dioxide remains unchanged. It is a catalyst.

7 **D** – Downward delivery or displacement of air is for gases that are denser than air.

8 **D** – The test for water. Water turns blue cobalt chloride paper pink. It also turn anhydrous copper sulphate from white to blue.

9 **B** – Magnesium reacts faster with steam than with cold water.

10 **B** – Potassium, Sodium, Lithium, Calcium, Magnesium, – react with water (**in the shown order**)
Aluminium, Zinc, Iron – react with steam
Tin, Lead, Copper, Silver , Gold, Platinum – **DO NOT react with water** nor steam.

11 **D** – Calcium carbonate (marble chips) + hydrochloric acid → Calcium chloride + water + carbon dioxide

21.2 CHEMICAL REACTION – PAPER 2

1 (a) A chemical reaction is the changing of one or more substances into different substances.

(b) (i) 3

(ii) 2 and 4

(iii) 1

(c) 2 and 4.

2 (a) **J** – Potassium chlorate. When a mixture of potassium chlorate and magnesium dioxide is heated, oxygen is produced. The other product is potassium chloride. Magnesium dioxide remains unchanged. It is a catalyst.

b) Potassium chlorate → potassium chloride + oxygen

(c) Downward displacement of water.

(d) To speed up the reaction. It is a catalyst.

(e) The first bottle is not pure oxygen, but a mixture of oxygen and air.

(f) Oxygen is used in welding and cutting. It is used in burning fuels, steel production.
(Other uses include: used in hospitals to aid patients with breathing problems, deep sea divers and mountain climbers)

3 (a) The burning will become brighter. (Oxygen supports burning)

(b) Carbon dioxide. (It does not support burning)

(c) Oxygen is heavier (denser) than air.

(d) Photosynthesis. (The process of producing sugar and oxygen by green plants (plants with chlorophyll) using carbon dioxide and water in the presence of light.)

(e) **Manganese dioxide** will catalyse the release of oxygen when added to hydrogen peroxide. Oxygen gas will relight a glowing splint.

(f) Oxygen is used in welding and cutting. It is used in burning fuels, steel production.
(Other uses include: used in hospitals to aid patients with breathing problems, deep sea divers and mountain climbers)

4 (a) Apparatus labelled J – Thistle funnel.

(b) Gas I – carbon dioxide.

(c) Lime water turns milky when carbon dioxide passes through it.

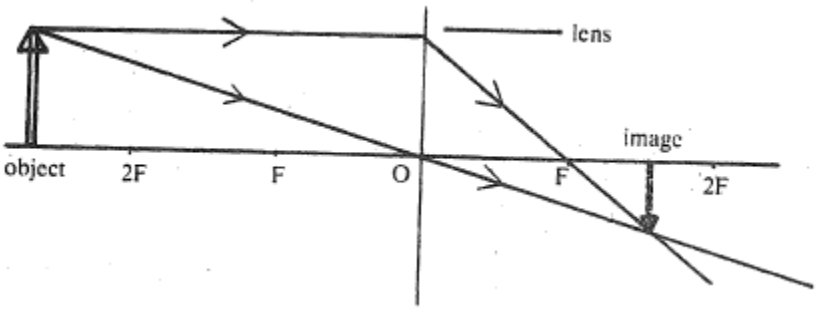
(d) Calcium carbonate + hydrochloric acid → calcium chloride + water + carbon dioxide.

(e) Carbon dioxide is denser than air. This makes downward delivery possible.

(f) Carbon dioxide is used in fire extinguishers, making carbonated soft drinks, used in baking (makes the dough to rise). It is used in cooling drinks and food (when cooled to form a solid called dry ice). It is used by plants in photosynthesis.

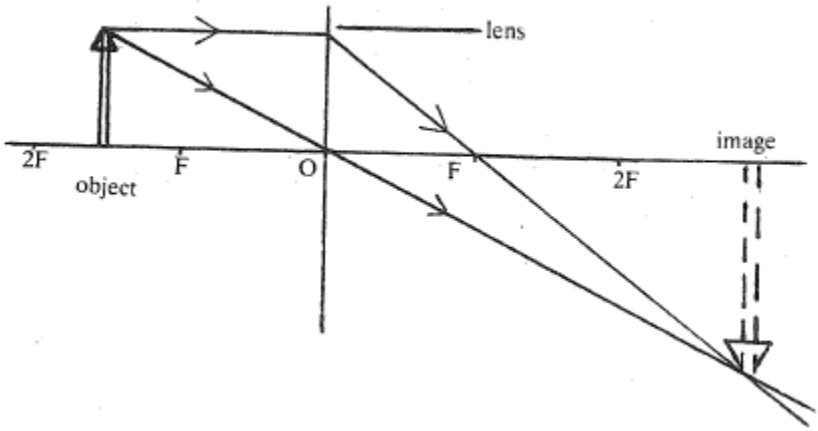
22.1 LIGHT AND ITS NATURE – PAPER 1

1 C – When the object is **on the left of 2F** – the image is real, upside down and diminished
When the object is **at 2F** – the image is real, upside down and same size as object.
When the object is **between 2F and F** – the image is real, upside down and magnified.
When the object is **between F and the lens** – virtual, magnified and upright.



2 C

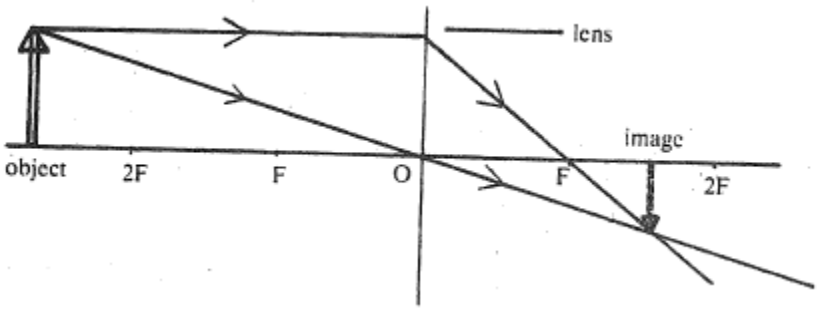
3 C – When the object is **on the left of 2F** – the image is real, upside down and diminished
When the object is **at 2F** – the image is real, upside down and same size as object.
When the object is **between 2F and F** – the image is real, upside down and magnified.
When the object is **between F and the lens** – virtual, magnified and upright.



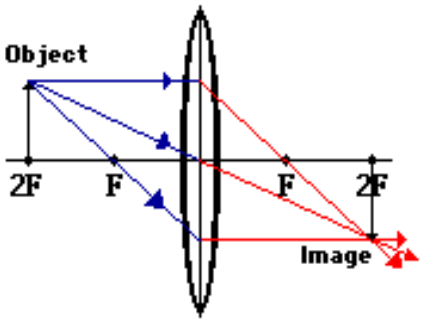
4 A – The fourth is green. (**ROY-G-BIV** for **R**ed, **O**range, **Y**ellow, **G**reen, **B**lue, **I**ndigo, **V**iolet). Violet is same side as the base of the triangular prism. 1 is violet, 4 is green, 5 is yellow. Green is always in the middle!

5 B – a red shirt looks red because the shirt has absorbed the other colours of the spectrum. Red light is the only light that is reflected from the shirt. If only blue light is shone onto a red shirt, the shirt would appear black, because the blue would be absorbed and there would be no red light to be reflected. White objects appear white because they reflect all colours. Black objects absorb all colours so no light is reflected.

6 **C –** When the object is **on the left of 2F** – the image is real, upside down, diminished, located between F and 2F.

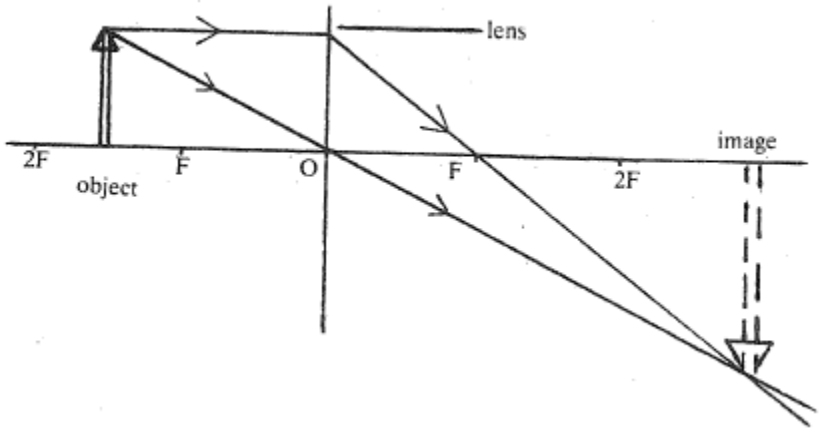


When the object is **at 2F** – the image is real, upside down, same size as object, located at 2F.

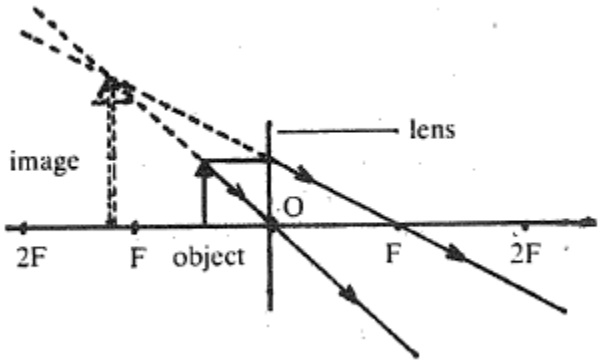


Ray Diagram for Object Located at 2F

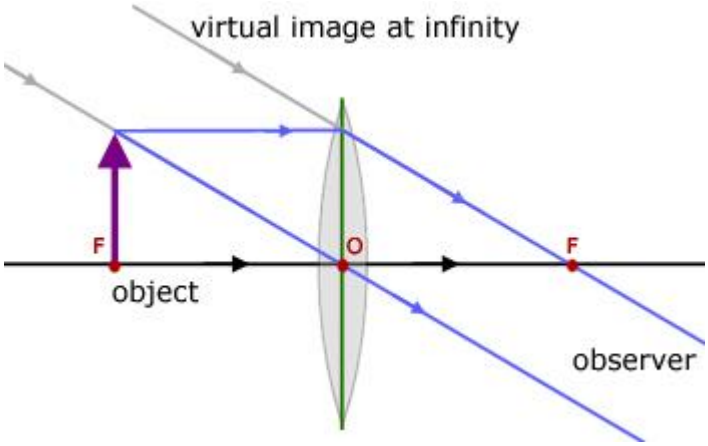
When the object is **between 2F and F** – the image is real, upside down, magnified, located on the right of 2F.



When the object is **between F and the lens** – the image is virtual, magnified, upright, on same side of the lens as the object.



When the object is **at F** – virtual image at infinity (in other words, the image is **not formed)**



7 **A –** White light is a mixture of all colors, in roughly equal proportions. White objects look white because they reflect back all the visible colours of light that shine on them - so the light still looks white to us. Colored objects, on the other hand, reflect back only some of the colours; the rest they absorb. For example, if white light shines on a red ball, the ball reflects back mostly red light, and so we see red. Most of the greens and blues that are part of white light are absorbed by the ball so we cannot see them. Likewise, a blue book is reflecting the blue part of the white light spectrum. The red and green parts are absorbed by the book.
What happens when red light shines on a red ball? It continues to reflect the red light, and so it is still red -- but a white ball would also look red in red light, because it reflects all colors. If instead we shine blue light on a red ball, it will look dark, because it does not reflect blue light. It cannot look red unless there is red light coming to it from the light source. And it cannot look blue because the red ball absorbs blue light. So when we ask what color an object is, the answer is not simple - it depends on what color light we are using to see the object.

8 A

9 **B** – Angle of refraction is angle between the refracted ray and the normal

10 **B** – The fourth is green. (**ROY-G-BIV** for **R**ed, **O**range, **Y**ellow, **G**reen, **B**lue, **I**ndigo, **V**iolet). Violet is same side as the base of the triangular prism.

11 **B** – When the object is **on the left of 2F** – the image is real, upside down and diminished
When the object is **at 2F** – the image is real, upside down and same size as object.
When the object is **between 2F and F** – the image is real, upside down and magnified.
When the object is **between F and the lens** – virtual, magnified and upright.

22.2 LIGHT AND ITS NATURE – PAPER 2

1 (a) (i) Lens G – Convex lens
(ii) Lens H – Concave lens
(b) (i) Spaces between lines on lens G are wider than spaces on the paper.
(ii) Spaces between lines on lens H are narrower than spaces on the paper.
(c) The image is diminished.
(d) The paper would burn.

23.1 ELECTRIC CURRENT AND VOLTAGE IN CIRCUIT – PAPER 1

1 C

2 A $Current = \frac{Voltage}{Resistance} = \frac{10}{2+3} = \frac{10}{5} = 2A$

3 B

4 A

5 C – the parallel connection (2) will have less resistance and more current than the series (1) connection.

6 **A** – The filament has high resistance to produce heat and light. (Usually tungsten or nichrome)

7 C

8 B

9 **D** – A fuse is placed on the live (brown) wire.

10 **B** – There are 3 cells, so total voltage 3×2 volts $=$ **6 volts**. There are 4 bulbs, each has resistance of 3 ohms. Therefore, total resistance $= 4 \times 3 =$ **12 ohms**

11 **A** – The lead should be a bad conductor of electricity (insulator). Metal is a good conductor.

12 B

13 D – $1st\ setting$: $Current = \frac{V}{R} = \frac{12}{3} = 4\ amps.$

$2nd\ setting$: $Voltage = I \times R = 5 \times 2 = 10\ volts$

14 B

15 **D** – When bulbs are connected in parallel, more current flows where there is less resistance. Since **R** is equal, all bulbs will have same current flowing through them.

16 B

17 **C** – The higher the resistance the more heat and light. High melting point prevents the filament from melting

18 **A** – Increasing the voltage and the number of turns in the coil makes the magnet stronger.

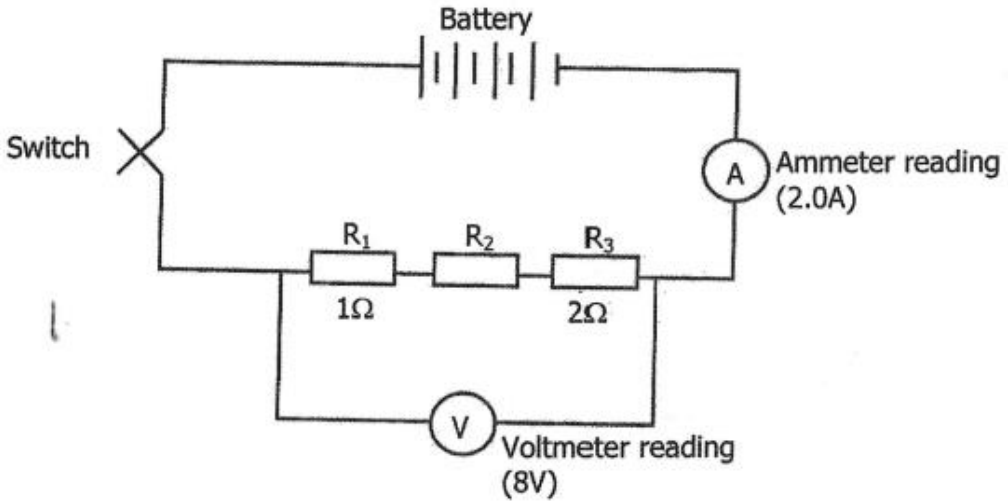
19 D – $Total\ Resistance = \frac{V}{I} = \frac{1.5 \times 4}{2} = \frac{6}{2} = 3\ \Omega.$

$Resistance\ for\ B1 = \frac{Total\ resistance}{2} = \frac{3}{2} = 1.5\ \Omega$

23.2 ELECTRIC CURRENT AND VOLTAGE IN CIRCUIT – PAPER 2

1 ELECTRICITY

Carefully study the electric circuit diagram below. [2016.P2.Q9]



(a) 4 cells. (A pair of one long (positive) and one short (negative) vertical lines represent one cell.)

(b) $Resistance = \frac{Voltage}{Current}$

$$R_1 + R_2 + R_3 = \frac{8}{2}$$

$$1 + R_2 + 2 = 4$$

$$R_2 = 4 - 2 - 1 = 1\ ohm$$

(c) $Voltage\ of\ 1\ cell = \frac{Total\ voltage}{Number\ of\ cells} = \frac{8}{4} = 2\ V$

(d) $Current = \frac{Voltage}{Resistance} = \frac{8}{R_1 + R_2} = \frac{8}{1+1} = \frac{8}{2} = 4A$

(e) $Power = Voltage \times Current = VI = 8 \times 4 = 32W$

2 (a) **S** is the ammeter (measures current). Current$= \frac{\text{Voltage}}{\text{Resistance}} = \frac{28\text{V}}{5\text{ ohms}+3\text{ ohms}} \frac{28}{8} = \mathbf{3.5\ A}$.

T is the voltmeter (measures voltage or potential difference across resistors).
Reading $= \mathbf{28\ V}$

(b) **T measures** potential difference or voltage across resistors.

(c) Power = Voltage × Current $= 28\text{V} \times 3.5\text{A} = \mathbf{98\ watts} = \mathbf{98\ W}$

(d) Current$= \frac{\text{Voltage}}{\text{Resistance}}$ → Voltage $=$ Current × Resistance
Voltage across the 3 ohm resistor$=$ Current × Resistance $= 3.5 \times 3 = 10.5$ V

(e) A fuse protects an electrical appliance by melting and breaking the circuit when current exceeds the fuse rating.

3 (a) Heating effect.

(b) (i) Tungsten

(ii) It has high resistance and high melting point

(c) Electric kettle, pressing iron.

(d) Element A produces heat while the element in a bulb produces more of light

(e) Heating effect

4 (a) The resistors R1 and R2 are arranged in series.

(b) For resistors in series, add their resistance.

Total resistance $= \mathrm{R1} + \mathrm{R2} = 1.5 \text{ ohms} + 1.5 \text{ ohms} = 3 \text{ ohms}$

(c) (i) $\text{Voltage} = \text{Current} \times \text{Resistance} = 4 \times 3 = 12 \text{ V}$

(ii) There are 4 cells in the battery. Total voltage is 12 V.

Therefore, voltage across one cell $= \frac{12}{4} = 3 \text{ V}$

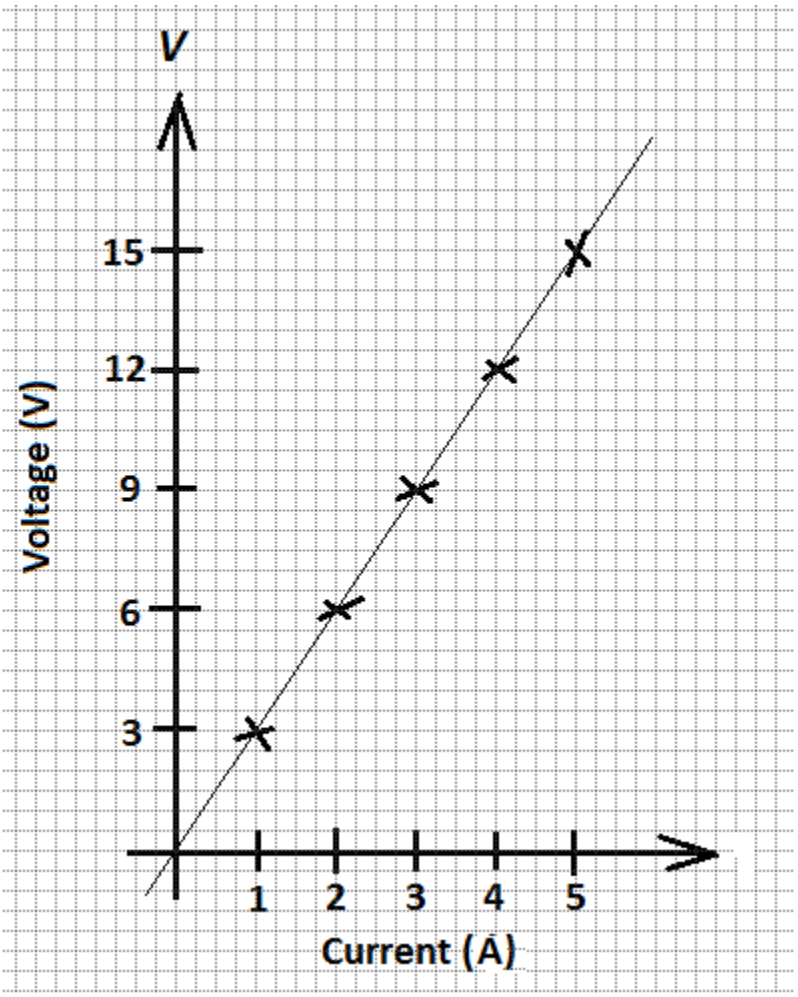
(d) $\text{Power} = \text{Voltage} \times \text{Current} \ (\mathrm{P} = \mathrm{V} \times \mathrm{I})$

$\mathrm{I} = 4\mathrm{A}, \ \mathrm{V} = \mathrm{I} \times \mathrm{R} = 4 \times 1.5 = 6\mathrm{V}$

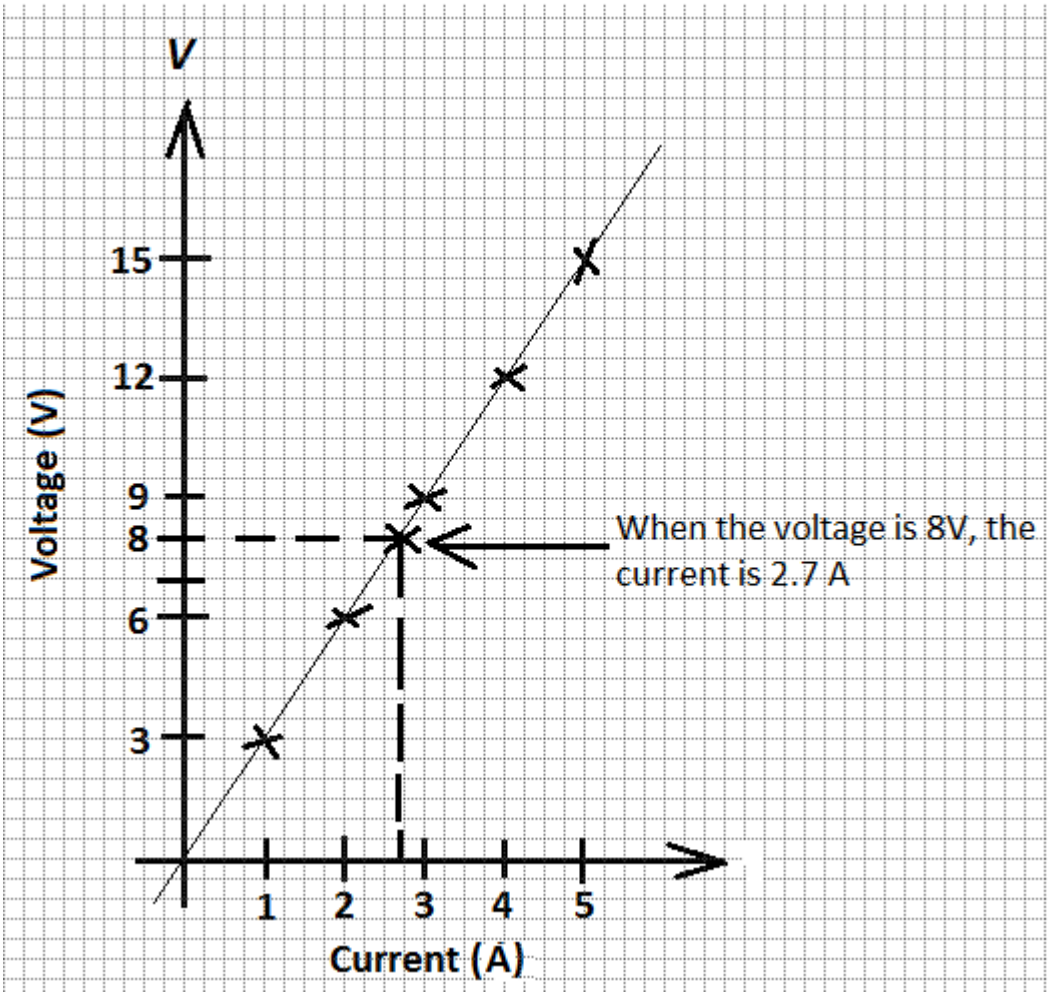
$\mathrm{P} = \mathrm{V} \times \mathrm{I} = 6 \times 4 = 24 \text{ W}$

5 (a) **Ohm's Law** states that the current, ***I***, through a conductor is proportional to the voltage, ***V***, between the two ends of the conductor provided the temperature remains the same.

(b)

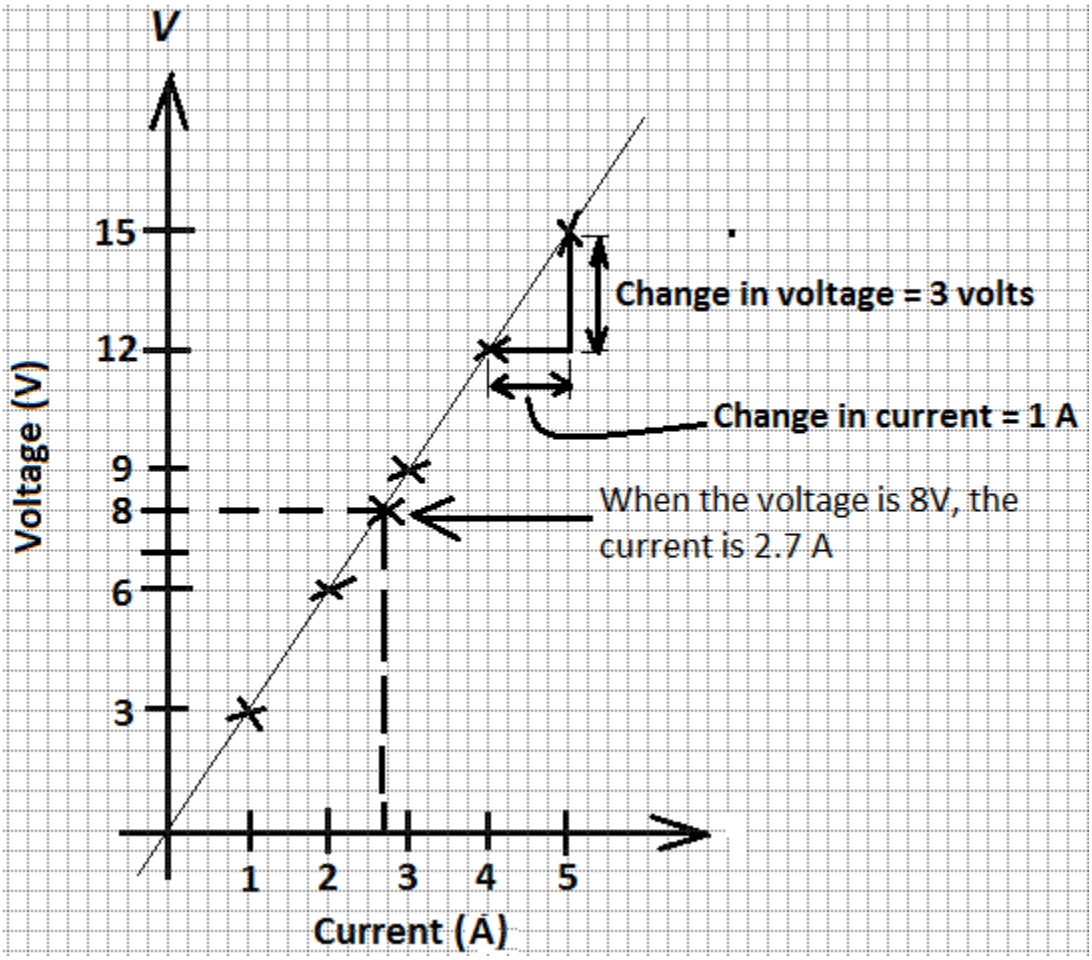


(c) Draw a horizontal line from 8V on the y-axis to the graph. When it meets the graph, draw a vertical line from that point to the x-axis. The value on the x-axis is the current, approximately 2.7 A.



(d) The resistance of the appliance used in the experiment to obtain the results in the table is equal to the gradient of the graph (straight line). Choose any two points on the graph and find the gradient. (For example, (5,15) and (4,12).)

$$\text{Resistance} = \text{Gradient} = \frac{\text{Change in Voltage}}{\text{Change in Current}} = \frac{15-12}{5-4} = \frac{3}{1} = \mathbf{3\ ohms}$$



(e) Power = Voltage X Current or **P** = **VI**= $\mathbf{15}\boldsymbol{V} \times \mathbf{5}\boldsymbol{A} = \mathbf{75\ W}$

24.1 PRESSURE – PAPER 1

1 D – the smallest surface area exerts the greatest pressure.

2 $Pressure = \frac{Force}{Area} = \frac{600N}{6m^2} = 100\ N/m^2$

24.2 PRESSURE – PAPER 2

1 (a) Pressure is a force per unit area that is applied at right angles to the surface of an object.

(b) (i) Area of the base of the cuboid $= l \times b = 3m \times 2m = 6m^2$

(ii) $Pressure = \frac{Force}{Area} = \frac{90N}{6m^2} = 15\ N/m^2$

(c) Pressure would reduce. The smaller the surface area the greater the pressure exerted by the surface..

25.1 ENERGY AND ITS CONSERVATION – PAPER 1

1 B

2 D

3 B

4 **A** – Natural gas, coal and oil are diminishing resources (they can finish)

5 **B** – Destructive distillation is the process of heating coal in the absence of air in order to separate coal into coke, coal gas (natural gas), pitch etc

6 B

25.2 ENERGY AND ITS CONSERVATION – PAPER 2

1 (a) Coal is a diminishing resource.

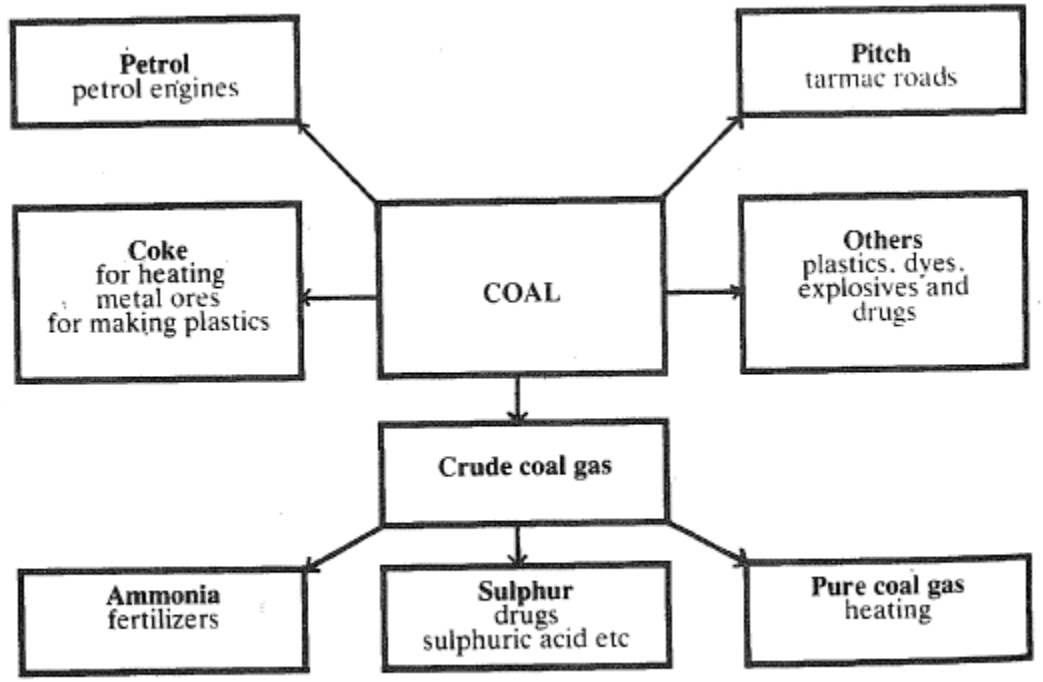
(b) **Destructive distillation** is the process of heating coal in the absence of air in order to separate coal into coke and other substances.

(c) (i) **Ammonia** is used to make fertilisers to increase food production.

(ii) **Pitch** can contribute to land pollution if not used or disposed properly.

(d) Pitch can be used to make tarred roads, thereby improving transport and communication.

(e) Unleaded petrol **does not emit harmful gase**s into the environment.



Products of destructive distillation of coal

2 (a) Hydro-electric power station

(b) **K** represents the **turbines**

(c) Water pressure is higher at the bottom of the dam than at the top

(d) **N**. Water will be channeled down the tunnel with more force.

(e) (i) Hydro-electric power station does not cause pollution to the environment.
It is cheaper to maintain.

(ii) Hydro-electric power station is very expensive to build.

Note – L represent the dynamo that actually produces electricity.

26.1 COMMUNICATION – PAPER 1

1 B

2 D

3 C

4 D

5 C

6 B

7 C

8 **A** – Tracking disc sends and receives signals from satellites in space.

9 **C** – A microphone changes sound into electrical signals during recording.

10 A

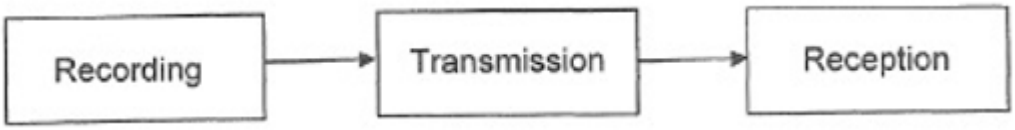
11 A

12 **D** – The microphone changes sound energy into electrical energy.

26.2 COMMUNICATION – PAPER 2

1 (a) Television set.

(b)



(c) Signals from the satellite will not be tracked and will not be received.

(d) Repeater

(e) The repeater is a device that picks up a radio signal, amplifies it and then sends it in a particular direction. This enables the signal to be strong (good quality) and be able to reach many homes.

2 (a) **R** – Microphone. **Q** – Transmitter.

(b) (i) **R** (Microphone) – coverts sound into electrical signals.

(ii) The amplifier makes the signals stronger.

(c) (i) **Repeater station** – it picks up radio signals from a transmitter, amplifies them and send them in a particular direction.

(ii) **Communication satellites** – spacecraft that receives and transmits messages as it orbits the Earth.